

Strategic Trade Infrastructure: Understanding China's Belt and Road Initiative*

Pablo Fajgelbaum[†], Amit Khandelwal[‡], Theodore Naff^{*}, and Edouard Schaal[§]

[†]UCLA and NBER

[‡]Yale and NBER

^{*}UCLA

[§]CREI, ICREA, UPF and BSE

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Abstract

Since 2000, China has invested \$259 billion in 535 transportation infrastructure projects across 144 countries. Are these investments economically justified for China? What are the trade and welfare impacts for host and third countries? First, we examine the economic and geopolitical drivers of Chinese overseas investments and estimate their impacts on port-level infrastructure and trade. Second, in a standard two-country trade model extended with a transport sector we characterize the welfare impacts of foreign infrastructure investments and show how they arise in equilibrium among competing countries. Third, in a multi-country framework with a multi-modal transport network with congestion in transport hubs, we measure the impacts of the observed BRI investments, back out country-specific preferences of China that rationalize them, and compute optimal economic investments of China and other major investors. Computationally, we tackle the high dimensionality and complex non-linearities using piecewise linear approximations that allow us to apply standard optimal-transport methods.

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*E-mail: pfajgelbaum@econ.ucla.edu, amit.khandelwal@yale.edu, theodore.naff@ucla.edu, eschaal@crei.cat.

1 Introduction

Starting in the early 2000’s, and accelerating through the 2013 launching of the Belt and Road Initiative (BRI), China has made large-scale overseas infrastructure investments across the world. These investments focus on transportation infrastructure that facilitates international trade in low- and middle-income countries, including seaports across South Asia, the Middle East, and West Africa and rail and road corridors through Central Asia. China has also pursued similar investments in some high-income countries throughout the world. While the BRI also encompasses investments in other sectors such as energy, pipelines, and telecommunications, its transportation component represents an unprecedented effort by a single country to reshape the global trade network. In terms of sheer value, these transportation investments amount to approximately \$259 billion, roughly twice the present value of the U.S. Marshall Plan.

The motivations behind China’s involvement in financing foreign infrastructure projects is subject to debate. Much of the narrative spoused by China and host countries supporting the BRI centers on the economic benefits. In this view, the investments will facilitate China’s access to both import sources and exports markets, promoting economic development in host countries and fostering deeper integration via future trade and investment agreements. Critical views instead argue that the economic impacts are small or negative and that China’s financing is a vehicle for political influence. In this paper, we inform this discussion by answering several questions. First, are China’s BRI investments economically profitable for China, and how are host and third countries affected? Second, what investments would be economically optimal from China’s perspective, how do they compare with observed investments, and what factors can explain the difference?

We answer these questions combining new empirical evidence, theory, and quantitative methods. First, we assemble a dataset at the port level linking detailed information on investments by China and other major financiers (such as the World Bank and Japan), port infrastructure, container shipments, and vessel rotation. We provide stylized facts on the determinants of the infrastructure investments of China and the other investors, and on their impacts on port-level infrastructure and container trade. Second, we use a simple two-country neoclassical trade model extended with a transportation sector to study trade infrastructure investments among two competing countries. The model demonstrates conditions under which a foreign country benefits from investing in another country’s international trade infrastructure. Third, for the quantification, we extend the theory to a multi-country framework where trade occurs on a multi-modal transport network with congestion at transport hubs. The large dimension and endogeneity of node-level congestion presents computational challenges that we tackle with piecewise-linear approximation allowing us to apply standard optimal transport methods.

Section 2 describes the data. China’s investment data come from AidData’s China’s Global Loans Dataset and the Council on Foreign Relations. After our cleaning procedures, the database captures 535 transportation infrastructure projects with total committed value of \$259 billion across 144 countries between 2000 and 2023. We also compile transportation projects financed by bilateral (U.S. and Japan) and multilateral agencies (World Bank and the IFC). For seaports, we link the

investments to several other sources on infrastructure and trade and over time: port-level attributes (such as cargo and channel depths) from the the World Port Index, infrastructure (such as cranes and berths) from Lloyd’s Port of the World, port-to-port container vessel routes from the Maritime Data Systems and Technology database, port-to-port shipments of twenty-foot equivalent units (TEUs) from Containerized Trade Statistics (CTS), and geophysical data from various sources to construct a measure of dredging costs.¹

Section 3 presents the descriptive evidence. Our goal is two-fold. First, we show that economic factors, alongside geopolitical variables, are important correlates of the Chinese investment allocation across countries and across ports, and that these correlates differ from those associated with other investors. Second, we establish that China’s investments materially change port infrastructure and trade. For both questions, we compare the drivers and the effects of China’s investments with those of the other large financiers of trade infrastructure. We provide four key findings. First, since 2000, the geographic footprint of China’s investments has been broader than other investors, spanning more countries, more ports, and a more varied set of countries, including both poor and rich, across the globe. Second, across countries, and relative to other sources of investment, China directed more financing to countries that trade more, both overall and with China in particular, and to countries less politically aligned with the U.S. Third, across ports, like other investors, China invests more in ports with certain characteristics that correlate with trade (such as harbor size). However, China is more responsive to several measures of depth (a strong predictor of a port trading capacity) and to dredging costs (a key determinant of returns to investment). Fourth, China’s investments are correlated with sizable changes in port depth, suggesting that the loans enable dredging (which we validate from newspaper reports). Moreover, China’s investments increase throughput in terms of both capacity of vessels rotating through a port and TEU’s (containers).

The reduced form evidence thus establishes that China’s investments seem to respond to both economic and non-economic factors in ways that seem specific to China, and that these investments materially affect port infrastructure and trade. Next, we move to quantifying the economic impacts and computing optimal investments. To study these questions, we need a tractable model of foreign trade infrastructure investments. A recent literature in spatial economics, including among others [Fajgelbaum and Schaal \(2020\)](#), [Allen and Arkolakis \(2022\)](#), and [Bordeu \(2023\)](#), quantifies investments in transport infrastructure but lacks three crucial ingredients to study the BRI theoretically or quantitatively. First, in these papers, the investment decision directly affects price wedges; however, China’s investments in transportation hubs is itself an activity that generates economic returns. For example, investing in the port of a source country for its imports allows China to enjoy *both* lower import prices and a investment return. Second, in these models, no agent besides a planner can invest in transport infrastructure; however, private investors can also finance infrastructure projects. Theoretically, it is unclear why China may benefit from distorting market investments, but incorporating private investors allows us to establish when the strategic

¹Future versions of this project will include investments from additional bilateral and multilateral agencies as well as additional infrastructure data for airports and land investments.

incentives of foreign investors push towards subsidizing infrastructure. Third, these existing models only quantify planners investing within their own economy. In contrast, the BRI improved connections beyond China’s jurisdiction. With these forces—economic returns from investing in transport infrastructure, private financing of infrastructure, and investments outside a planner’s jurisdiction—it is not obvious that foreign infrastructure investments are economically desirable for an investing country, barring (plausibly important) asymmetries such as technological superiority or cheaper access to financing.

To clarify the welfare impacts of foreign infrastructure investments and whether they arise in equilibrium, Section 4 characterizes an otherwise standard neoclassical two-country trade model extended with these features. A “foreign investor” and a “host country” trade through a transport hub owned by the host. The transport hub demands infrastructure, supplies services to shipping companies, and is congested as function of trade volume. Better infrastructure lowers congestion, both governments and the private sector can supply infrastructure, and governments can tax or subsidized private investors, foreign investment, and trade. We show that foreign infrastructure investments and foreign subsidies to private investors naturally arise in equilibrium as a result of how infrastructure affects the prices of traded goods, the returns to infrastructure, and shipping costs.

Starting from a poor level of infrastructure in the host, the foreign country (China) obtains high returns from investing and gains from better prices on traded goods (thanks to lower shipping costs), but loses because of diminishing returns to infrastructure, which it exports. Around no foreign investment, the decreasing returns to infrastructure have no first-order welfare impact; therefore, the foreign country necessarily gains from investing. In contrast, when the host country invests, it benefits from lower infrastructure returns and from better terms-of-trade on goods, but the latter are offset by lower fees charged by the port. As a result, the foreign country has stronger incentives to invest than the host: even if countries have identical marginal costs of financing infrastructure, in equilibrium the foreign country always invests and typically subsidizes its private investors, but the host country may be crowded out. Moreover, because foreign investment creates net surplus for the world, the host country also gains if it can tax foreign investment or pay a subsidized rate for infrastructure—a feature consistent with the terms in BRI data. These results go through with optimal tariffs; in that case, infrastructure does not have welfare impacts through changes in prices of traded goods, but still affects returns to infrastructure and shipping costs directly. Section 5 extends this setting to many countries trading along a multi-modal global transport network with congestion. Each country consists of a set of nodes representing a trade hub such as a port, airport, rail junction, or land crossing. Trade costs on a link depend on congestion at the nodes it connects, itself a function of the volume of shipments passing through the node and infrastructure. As in the baseline model, transport operators supply shipping services and use infrastructure as an input. Solving this model is challenging due to the non-linearities arising from congestion at nodes and the scale of the problem. These features make it unfeasible to use existing solution methods incorporating optimal transport in general equilibrium, such as [Fajgelbaum and Schaal \(2020\)](#).

To make the model computationally tractable, we adapt an approach from operations research on intermodal transportation relying on approximations with piecewise linear methods (Crainic and Kim, 2007; Croxton et al., 2007). These approximations allows us to globally solve the investment problem using a standard optimal transport approach.

Section 6 presents the preliminary quantitative implementation. This section is still very preliminary, and we show it to illustrate how we can implement the model and to give a sense of the overall approach we are taking to do so.

2 Data

We assemble a project-level database of China’s overseas infrastructure investments and pair it with several port-level, country-level, and trade datasets. This section describes the data sources. Appendix A describes our data processing steps in more detail.²

2.1 UN/LOCODES

We use the U.N. Code for Trade and Transport Locations (UN/LOCODE) as the geographic key linking investments to infrastructure points. The grey dots in Panel A of Figure 1 show the universe of seaport, airport, road terminal, and rail terminal locodes across the globe.³

2.2 China’s Overseas Investments

Institutional Background China’s overseas investment portfolio is primarily state-driven. The financing for the Belt and Road stemmed from China’s surplus USD foreign exchange reserves. The majority of developing country infrastructure loans were issued by state-owned development banks, the largest being the Export Import Bank and China Development Bank, and by the Silk Road Fund, although other entities including state policy banks, state-owned commercial banks, and sovereign funds were also involved. While China’s loans take different forms and are issued by different entities over the course of the BRI period, the Chinese state plays a central role in financing and directing the portfolio. In addition to debt-based financing, Chinese state-owned companies are also actively involved in infrastructure as contractors and as equity partners. The majority of our data comes from AidData and records loans or grants from these Chinese entities, with a smaller number of projects recorded in the CFR data also showing equity investments.

AidData CLG-Global 1.0 We use AidData’s China’s Global Loans and Grants dataset (CLG-Global 1.0), which compiles 33,593 Chinese overseas loans in 200 countries committed between 2000 and 2023, with a total committed value of \$2.2 trillion (2023 USD). The dataset spans investments

²Future versions of this paper hope to bring in more characteristics of airports and land points to better understand the driver of those projects, but currently our coverage of port-level information is most complete.

³The road locodes essentially follow population density and are the least precise of the infrastructure points as they capture depots, trucking hubs, or even towns where may roads converge.

from 1,242 Chinese agencies across 25 sectors (agriculture, construction, energy, transport, education, mining, health, etc). Transport is the third largest sector of China’s investments, accounting for \$312 billion in total investments over this period.

AidData constructs their loan panel through the Tracking Underreported Financial Flows methodology, which triangulates project-level reports from news media, official documents, annual reports of Chinese state-owned banks and enterprises, IMF Article IV consultations, financial statements, and oversight institutions of recipient governments. The dataset captures debt financing, overseas development grants, and syndicated loans (including those not led by China). The database does not capture purely equity-financed projects, though equity may be a component in the financing of the projects they capture. AidData may miss investments done through off-balance-sheet investments and, although the data attempts to capture information on contractual terms and renegotiations, this information is not systematically available for all projects.⁴

We restrict the analysis to infrastructure projects which can be assigned a locode (i.e., projects linked to a specific seaport, airport, road, or rail segment). We exclude projects with no implementation date, but retain projects not yet completed. After this filter, AidData contains 495 infrastructure projects in 134 countries, totaling \$245 billion.

Council on Foreign Relations (CFR) We supplement AidData with the Council on Foreign Relations datasets "Tracking China’s Control of Overseas Ports" and "Tracking China’s Control of Overseas Airports." This source tracks both debt- and equity-financed seaport and airport investments by Chinese state-owned enterprises and complements AidData’s debt-focused coverage. We apply the same inclusion and exclusion criteria and de-duplicate against AidData to bring any additional seaports and airports investments into the project.

Combined Dataset with China Overseas Infrastructure Investment Nodes Combining AidData and CFR, and breaking the universe out by transport mode, yields 178 seaport projects, 87 airport projects, and 350 land (road and rail) projects. Total committed values are \$103.7 billion (seaports), \$24.2 billion (airports), and \$130.7 billion (land). Collectively, there are 535 infrastructure projects in 144 countries, totaling \$259 billion. Seaport investments account for 40% of the total commitment.⁵

⁴AidData provides loan-level detail on the interest rate and the Chinese ownership share for a subset of investments, and we have so far cleaned these for seaports. Of China’s seaport investments with recorded contractual terms (53 projects), the average reported interest rate is 3.3% (standard deviation 1.8%). This interest rate is inclusive of the grant element on the loan. Interest rates could be either fixed or floating (we are currently processing the fields that capture floating rates and their market benchmarks such as LIBOR and SOFR), and the rate reflects terms at origination only. In terms of equity stakes at the time of investment, 110 locodes have no Chinese ownership in their direct receiving agencies when the agreement is made. This means the equity ownership of the projects is typically non-Chinese at the time of signing. This motivates our modeling assumption that seaport financing agreements are primarily based on debt rather than on equity financing from Chinese companies. Conditional on non-zero Chinese ownership at time of investment, the average share is 52%.

⁵Figure C.1 plots the timeline of Chinese commitments by implementation year. Although the Belt and Road Initiative was formally launched in 2013, and the mid 2010s was a period of substantial lending, it is evident that China’s overseas investments were also active prior to the formal announcement. Since 2020, there has been a slowdown in loans to infrastructure projects.

Table 1 (top row, "China") summarizes the investments by mode. As a benchmark to these numbers, the World Bank committed \$50 billion in transportation projects in the last decade.

2.3 Investments by Other Major Sources

To benchmark China’s investments, we assemble parallel project-level data from four bilateral and multilateral sources covering 2000-2025: (i) U.S. International Development Finance Corporation (DFC); (ii) Japan International Cooperation Agency (JICA); (iii) World Bank IBRD/IDA; (iv) International Finance Corporation (IFC).⁶ Each source publishes project-level data through its respective portals, and we assign the locode using the same procedure applied above. These four institutions span a spectrum of mandates (bilateral development finance for DFC and JICA; multilateral lending for IBRD/IDA; co-investment with private-sector for IFC) and provide benchmarks for the patterns we document below. Rows 2-5 of Table 1 summarizes the investment projects by source.

Table 1: Overseas Infrastructure Investments Dataset

source	seaports			airports			land		
	locodes	countries	value (\$b)	locodes	countries	value (\$b)	locodes	countries	value (\$b)
China	178	90	103.7	87	62	24.2	350	106	130.7
IBRD/IDA	54	34	9.0	31	27	5.1	—	—	—
IFC	79	38	8.3	21	19	2.4	—	—	—
JICA	21	17	12.3	15	12	8.8	—	—	—
DFC	10	10	2.5	4	4	0.7	—	—	—

Notes: This table reports the locode-level investments across transport modes and sources. (Data collection on non-China land investments is in progress.)

2.4 Port Characteristics

We obtain the panel of port-level characteristics from the World Port Index (WPI), maintained by the U.S. National Geospatial-Intelligence Agency, for 14 years between 2000-25.⁷ WPI collects a range of characteristics for 3943 ports across 189 countries.

We focus on a subset of characteristics that are asked consistently throughout the sample period. For each port, we observe its locode, harbor size, harbor type, shelter afforded, tidal range,

⁶Future versions of this project will include investments from Asian Infrastructure Investment Bank (AIIB), Asian Development Bank (ADB), African Development Bank (AFDB), European Bank for Reconstruction and Development (ERBD), European Investment Bank (EIB), and Organization of the Petroleum Exporting Countries (OPEC).

⁷We have obtained data for 1996, 1998, 2000, 2001, 2002, 2003, 2004, 2005, 2006, 2007, 2008, 2009, 2010, 2011, 2012, 2013, 2014, 2015, 2016, 2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024, 2025. We are trying to track down additional years of WPI.

and three depth measures: cargo pier depth, channel depth, and anchorage depth.⁸ Below, we examine how these characteristics correlate with measures of port-level trade and how they change with investments. We use these responses to measure the infrastructure growth from Chinese investments.

We complement WPI with Lloyd’s Ports of the World (POW) for the years 2004-2011, which provides information on number of cranes, number of berths and berth length for 2994 ports across 191 countries.⁹

2.5 Vessel Rotations and Container Shipments

We build the network of container shipping routes using MDS Transmodel’s (MDST) Container-ship Databank. These data, which we observe for the month of May in 2006, 2011, 2024, record the sequence of port calls that a vessel rotates through on its regular service. For example, in 2006, Maersk Line TP5 rotated four vessels on a weekly basis between Los Angeles, Oakland, Shanghai, Ningbo, and returning to Los Angeles; the average capacity of each vessel was 4503 TEUs. We observe a total of 4,773 service rotations which report the rotation of port calls (encoded by UN/LOCODE), the number of vessels deployed, the average TEU (container) capacity per vessel, and the annual sailing frequency. This dataset allows us to build an annualized number of vessels and total capacity that flows out of 1100 locodes in 182 countries. We use port-to-port container shipments from Container Trade Statistics in the month of May for 2011, 2012, 2023, and 2024. This database captures the global trade in containers, with 80% of the data coming directly from the large global carriers and the remaining 20% from other sources. We observe shipments across 564 locodes in 160 countries. We use these data to measure responses of port level shipments to investments and to build the global network of directly connected ports used in the quantification¹⁰

2.6 Geophysical Data

We build a measure for the costs of dredging at each locode. For this, we obtain geophysical attributes of the WPI ports from several gridded datasets: ocean depth and seabed slope (GEBCO);

⁸WPI provides four values of harbor size: large, medium, small, very small; 13.2% of ports are classified as large or medium. This size is *not* based on area alone; instead, WPI classifies harbor size based on “several applicable factors, including area, facilities, and wharf space”, and is not “not based on area alone or on any other single factor.” There are eight possible harbor types: three coastal types (natural, breakwater, tide gates), three river types (natural, basic, tide gates), canal/lake, and open roadstead. A tidal gate is a barrier that manages the effects of tidal swings, and a breakwater is a structure built to reduce wave energy entering the port harbor; the breakwater/tidal gates flag is switched on for 24.0% of ports. Shelter afforded is the degree of protection from wind, sea, and heavy swell, which WPI classifies as excellent, good, fair, poor, or none; 54.5% are assigned excellent or good. The tidal range is the vertical distance between the highest high tide and the lowest low tide. The cargo pier depth is the largest depth across piers where vessels dock. Channel depth is controlling depth of the waterway that leads into the port. Anchorage depth the depth of the designated waiting area for vessels.

⁹We are attempting to obtain more recent data on cranes and berth lengths for ports. Currently, these data are only used to measure infrastructure stocks pre-BRI investments.

¹⁰Figure 5 displays this network. The figure shows only one link for each connected country-pair, with the nodes for each link corresponding to the largest port-pair by trade volume used by the country-pair. In future versions of this project we anticipate adding information on ships transporting bulk cargo or natural gas.

water turbidity (MODIS-Aqua); sediment thickness (GlobSed); distance to the nearest river mouth (GloRiC); seafloor sediment lithology (Dutkiewicz et al., 2015); and coastal geomorphology (Dürr et al., 2011). These attributes will be used to construct a predicted cost of dredging at each port; see discussion below.

2.7 Country Characteristics

We compile country-level characteristics from several sources. To inspect correlates foreign investments, we obtain the value of natural resource stock from the World Bank’s Adjusted Net Savings dataset and countries’ ideal point based on U.N. voting records compiled by Bailey et al. (2017), where higher values indicate closer alignment to a U.S.-led international order. For the quantification, we use GDP and population from World Development Indicators (WDI), bilateral FOB exports from BACI, road network length from Global Roads Inventory Project (GRIP), and rail network length from ESRI.

3 Facts

We organize the descriptive evidence around four questions: What is the geographic reach of China’s Investments? How do the investments target country and port characteristics? How do they impact infrastructure and trade? And, how do these determinants and responses compare between China and other sources? We focus this analysis on seaport investments where we have made the most progress on data processing.

3.1 Fact 1: China’s Investments Have a Broader Geographic Coverage

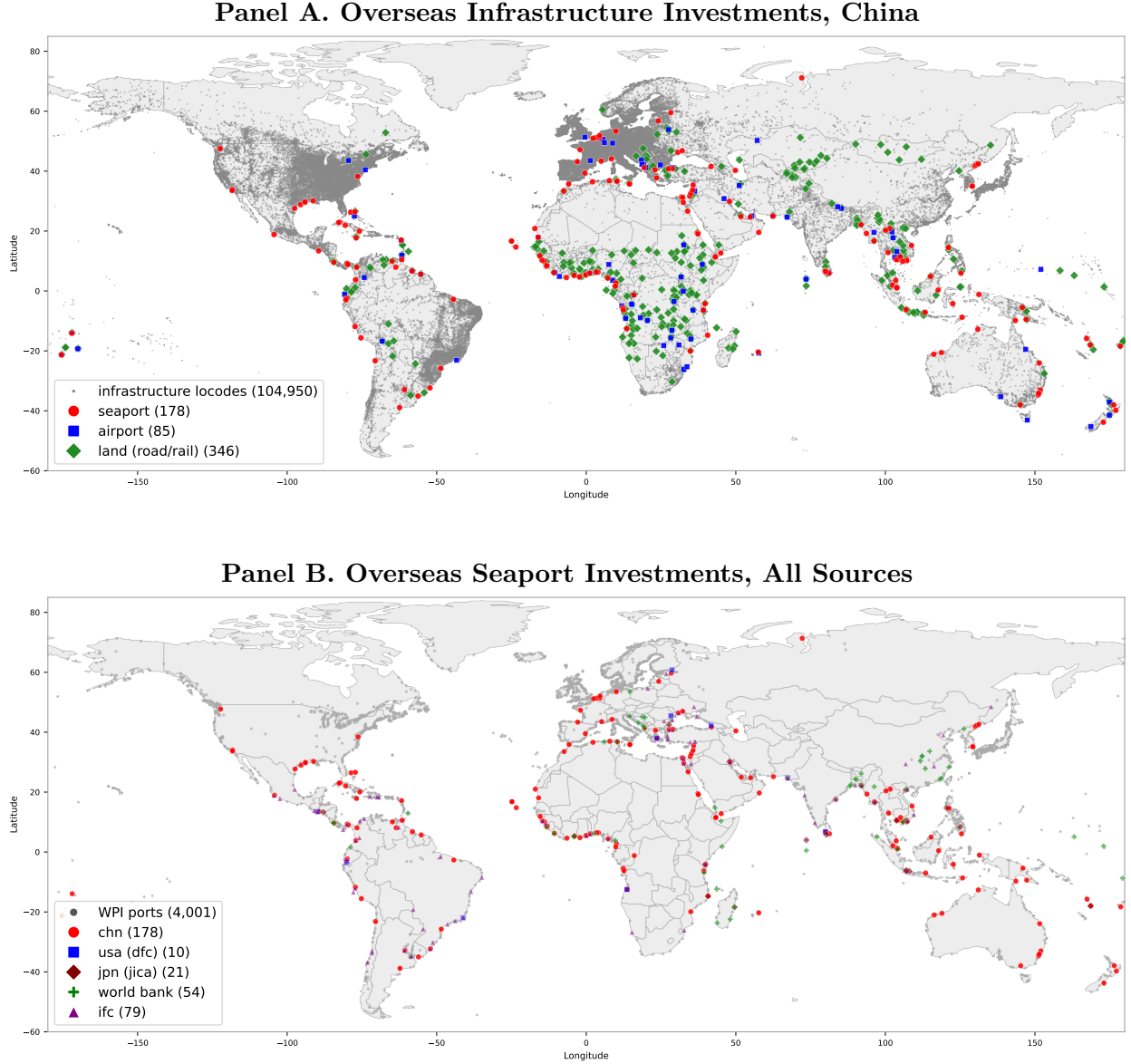
Table 1 illustrates that China has invested in nearly twice as many countries as the World Bank or IFC; and that the receiving country from China is, on average, richer: the average per capita income of countries receiving Chinese seaport investments is \$13827, compared to \$7371, \$2469, \$4405, and \$5134 for the U.S., Japan, World Bank, and IFC, respectively.

Panel A of Figure 1 shows the location of all Chinese overseas infrastructure investments by mode. China’s footprint is global, with a dense concentration of seaports in Western Africa as well as land projects throughout Sub-Saharan Africa and Southeast Asia. China has also financed airports and seaports in Europe and the U.S. The only large populous countries that have not received any infrastructure investments from China are Japan, India, and South Africa.¹¹ Panel B plots the location of all ports in WPI and overlays seaport investments from all five sources. Chinese investments span a wider geographic range than any of the other individual investors. As expected, the other lenders have not financed any ports in U.S., Europe or Australia. However, JICA, IFC, and the World Bank have financed ports in India.¹²

¹¹Appendix Table C.1 reports the 15 largest infrastructure projects financed by China.

¹²Appendix Figure C.2 shows that most ports do not receive overlapping financing from these five sources, although China and the IFC have co-invested in 16 ports, and China and the World Bank in 9 ports. There are 2 ports that

Figure 1: Overseas Infrastructure Investments, 2000-25



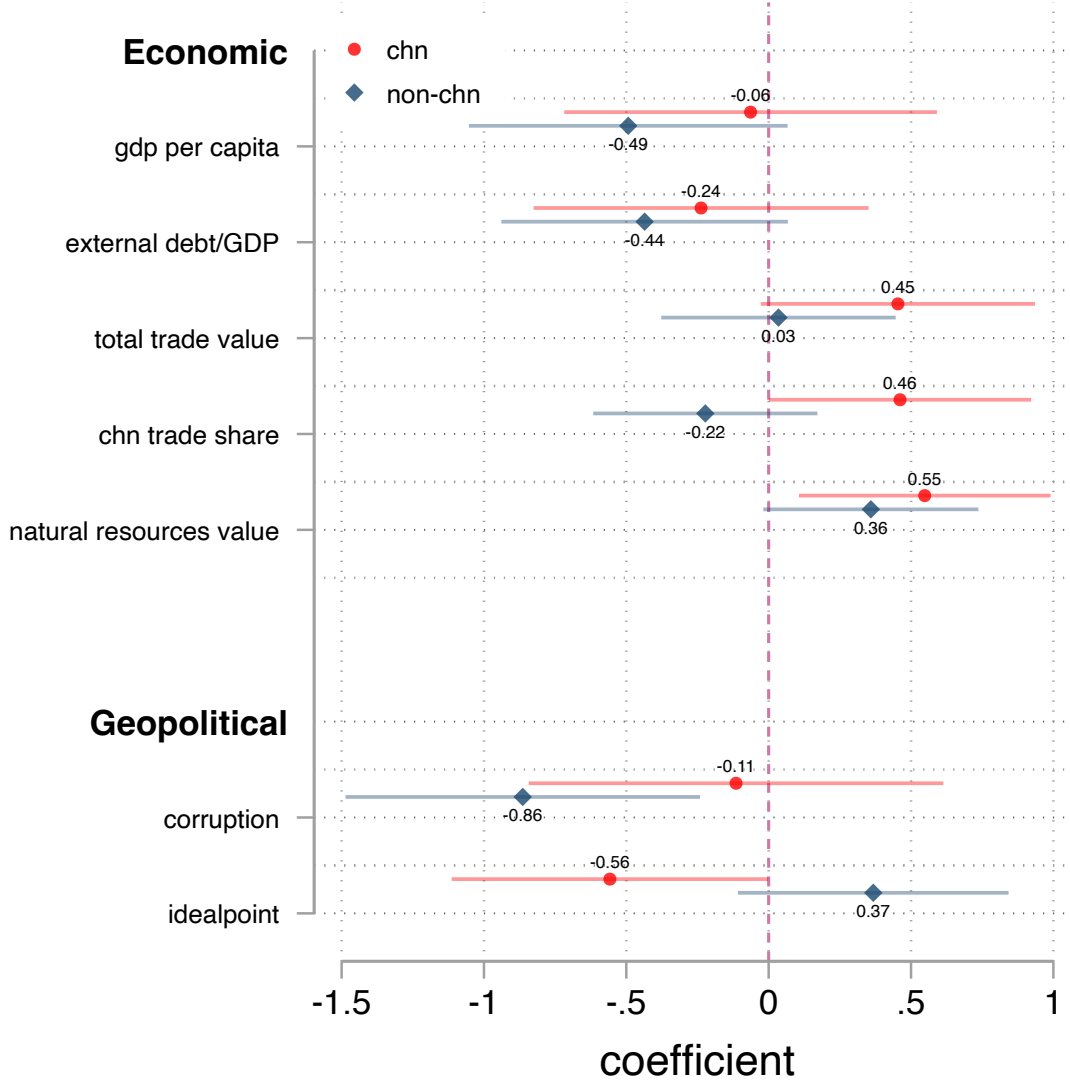
Notes: Panel A plots the locations of China's overseas investments in seaports, airports, and land infrastructure (roads and railways). Panel B plots the locations of overseas seaport investments across the five sources.

3.2 Fact 2: China Targets Countries that Trade More and are Less U.S. Aligned

We next ask how China's seaport investments correlate with broad country-level characteristics, and how those correlations compare with those of the non-Chinese investors. For each lender

have received financing from both China and the U.S. DFC.

Figure 2: Investments and Country Correlates, China vs Non-China



Notes: Figure plots the coefficients on the country correlates in (1) for China and non-China's investments. All variables are standardized; if a covariate is missing, the standardized value is set to zero and a missing-value indicator is included. Total number of observations in each regression is 209. The number of treated countries are: CHN 89; non-CHN 72. 90/10 errors bars.

$s \in \{CHN, USA, JPN, WB, IFC\}$, we run separate OLS regressions:

$$\text{asinh}(\text{value}_{is}) = \alpha_s + \beta_s Z_i + \varepsilon_{is} \quad (1)$$

where value_{is} is the cumulative committed value of investor s 's seaport investments in country i over 2000–25 (inverse hyperbolic sine to include zeros), and Z_i is a vector of country characteristics averaged over 1995–1999. The choice of characteristics seeks to include variables that plausibly capture broad rationales for investment. We group covariates into two blocks; economic variables

that include GDPPC, external debt to GDP ratio, trade (total imports plus exports, and the share of total trade with China), and the value of natural resource stocks, and two geopolitical variables (an index of corruption and the ideal point where higher values indicate closer alignment to the U.S.). All variables are standardized, and if a covariate is missing, we set the standardized value to zero and include a missing-value indicator.

To facilitate comparisons with China, we group the non-Chinese lenders into a single non-CHN aggregate by summing the values across the sources before applying the inverse hyperbolic sine transformation. We report results the coefficients for China and non-China in Figure 2, with the disaggregated lenders reported in Appendix C.3.

The figure suggests some differences in the countries that receive seaport investments between the two groups. On average, non-China lenders target slightly poorer countries, consistent with the unconditional averages reported above. There is some evidence that non-China lenders direct financing towards countries with lower debt to GDP ratios (conditional on other attributes), but the coefficient is not statistically different from zero. China targets countries that trade more, and particularly those who traded more with China in the 1990s. And, relative to the non-China lenders, China targets countries with more natural resources (though there is overlap in the confidence bars between the two). In terms of two geopolitical variables, the non-China lenders are more averse to corruption than China, and the coefficient ideal point variable is also quite different: China directs lending to countries characterized as not aligned with the U.S., while the opposite occurs for non-China lenders.

Overall, the regressions suggest that China’s has directed seaport lending to countries differently according to these particular covariates. They further suggest that while geopolitical factors are correlated with investment, economic factors are also important correlates.

3.3 Fact 3: China Targets Ports with High Trade Potential and Lower Dredging Costs

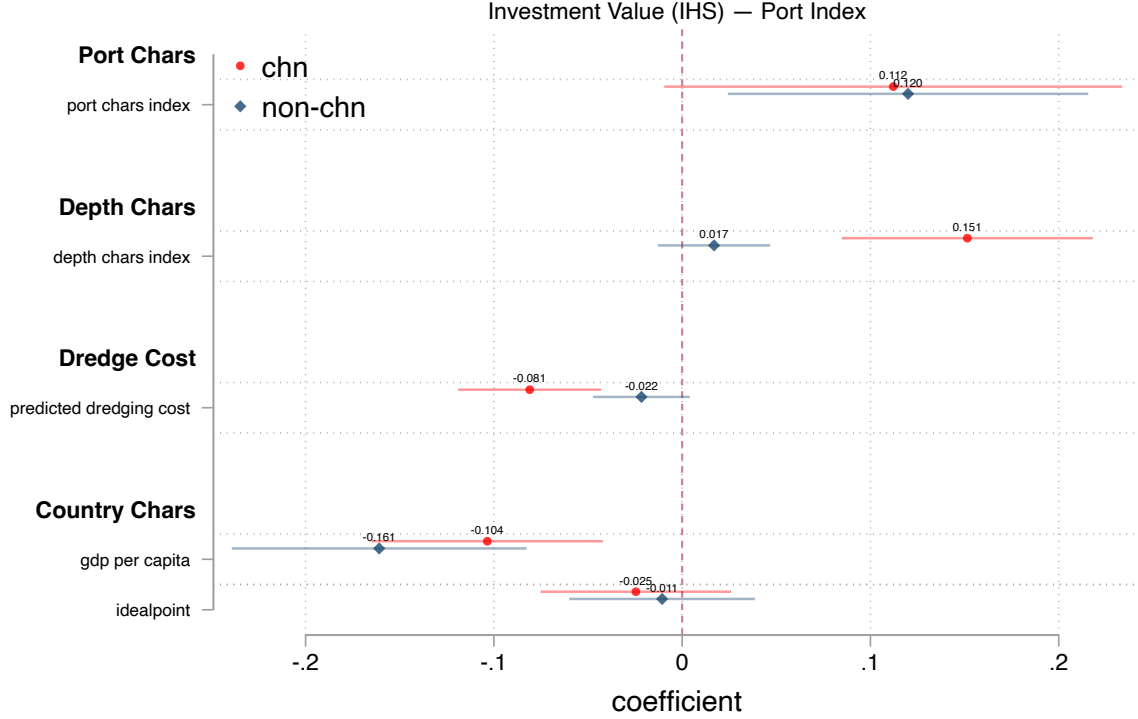
We next examine correlates of lenders’ financing at the port level. We again seek to understand if China has directed financing towards ports with different *ex ante* (before 2005) port-level characteristics than the other lenders. To do so, we run analogous regressions at the port level:

$$\text{asinh}(\text{value}_{ps}) = \alpha_s + \beta_s Z_i + \gamma_s X_p + \varepsilon_{ps} \quad (2)$$

where p denotes a seaport in the WPI database. X_p contains seven possible port attributes: binary indicators for harbor size (=1 if the harbor is medium or large), whether the harbor has a breakwater or tidal gates, and shelter afforded (=1 if shelter is rated good or excellent), and four continuous measures tracking tidal range and three depth characteristics—cargo pier depth, channel depth, and anchorage depth. We also include the predicted cost of dredging at the port based on geophysical characteristics discussed in Section 2.6.¹³ As before, we run these specifications separately by

¹³To do so, we leverage data on historical dredging contracts from the U.S. Army Corps of Engineers (USACE). The USACE dataset records detailed dredging contracts for U.S.-based projects from 1990-2026, including the actual

Figure 3: Initial Port Characteristics and Investments, China vs Non-China



Notes: Figure plots the coefficients on the port characteristics in (2) for China and non-China's investments. "Port chars index" is the average of harbor size (0/1), breakwater/tidal gates (0/1), shelter (0/1), and standardized tidal range. "Depth chars index" is the average of standardized depths of cargo pier, channel, and anchorage. Total number of observations in each regression is 3,531. The number of treated ports by source are: CHN 152; non-CHN 96. Standard errors clustered by country; 90/10 error bars shown.

investor, where the port characteristics are averaged over 2000-05. We also include two initial country characteristics from the regressions above, GDPPC and the ideal point, to see how these point estimates change at the port level. The continuous port characteristics are standardized, and standard errors clustered by country.

We do not have any particular prior on which of these attributes positively or negatively co-vary with trade volumes. Instead, we can use the port-level trade data in the baseline period to examine these correlates. So, before showing how investments correlate with attributes, we regress three measures of port-level trade on these eight characteristics (three separate multi-variate regressions): port capacity and number of vessels (in 2006), and total TEU throughput (total TEU imports plus exports in 2011/12, the earliest period we have the data for). The left panel of Appendix Figure C.4 reports these results. We see that a large ports are associated with much larger metrics of trade; for

dredging cost per cubic yard. We deflate nominal to a common base year using the Corps of Engineers' Civil Works Cost Index (CWCCIS). We then extract geophysical covariates at those dredging sites, and train a random-forest regression (500 trees) on the USACE sample of 8,010 contracts, projecting cost per cubic yard on the geophysical features. On a 20% held-out test set, the random-forest model achieves an out-of-sample R-squared 0.296. We then apply the trained model to each port in the WPI panel using the same geophysical covariates evaluated at the port's geo-coded location, yielding a port-specific index of predicted dredging cost.

example, a larger port is associated with $0.425sd$ more vessels, $0.201sd$ more capacity, and $0.413sd$ more TEU throughput. Ports with tidal gates or breakwater do not systemically have more or less trade than ports without these structures. Ports classified with better shelter are associated with $0.070sd$ more capacity and $0.120sd$ more vessels but not more TEU throughput. The tidal range negatively correlates with trade volume.¹⁴ The pier and channel depth correlate positively with output, while the anchorage depth (conditional on the other covariates) correlates negatively. The predicted dredging cost is negatively correlated with trade.

To simplify subsequent exposition, we collapse port attributes into two indexes—a port characteristics index, which is the average of three indicators—harbor size, breakwater/tidal gates, good shelter, and the standardized tidal range; and, a “depth” index which is the average of the three (standardized) depth measures. We then re-run the regressions of port-level trade on the two indexes, the predicted dredging cost, and the country correlates in the right panel of Appendix Figure C.4. The regressions show that both indexes correlate positively with trade and the dredging cost correlates negatively.

With a better understanding of how port characteristics map to trade, we now turn to (2) to examine how lenders’ investments correlate with port-level attributes, where we use the indexes as the attributes. Figure 3 shows the results for China and non-China. We see different patterns of lending between the two. While both China and non-China target ports with higher port index characteristics, China systemically seeks ports with higher initial depth compared to non-China lenders. Moreover, China is more sensitive to dredging cost than non-China: a one standard deviation increase in dredging costs reduces China’s investment by 8.1%; on the other hand, non-China lenders appear insensitive to this metric.¹⁵ Appendix Figure C.5 shows that these differences are even starker if we compare China against any individual lender.

A pattern that emerges is that China selects ports with structural characteristics that predict higher trade flows relatively more than non-China lenders. This pattern then motivates our next question: do these investments, in turn, also change port characteristics and, subsequently, trade?

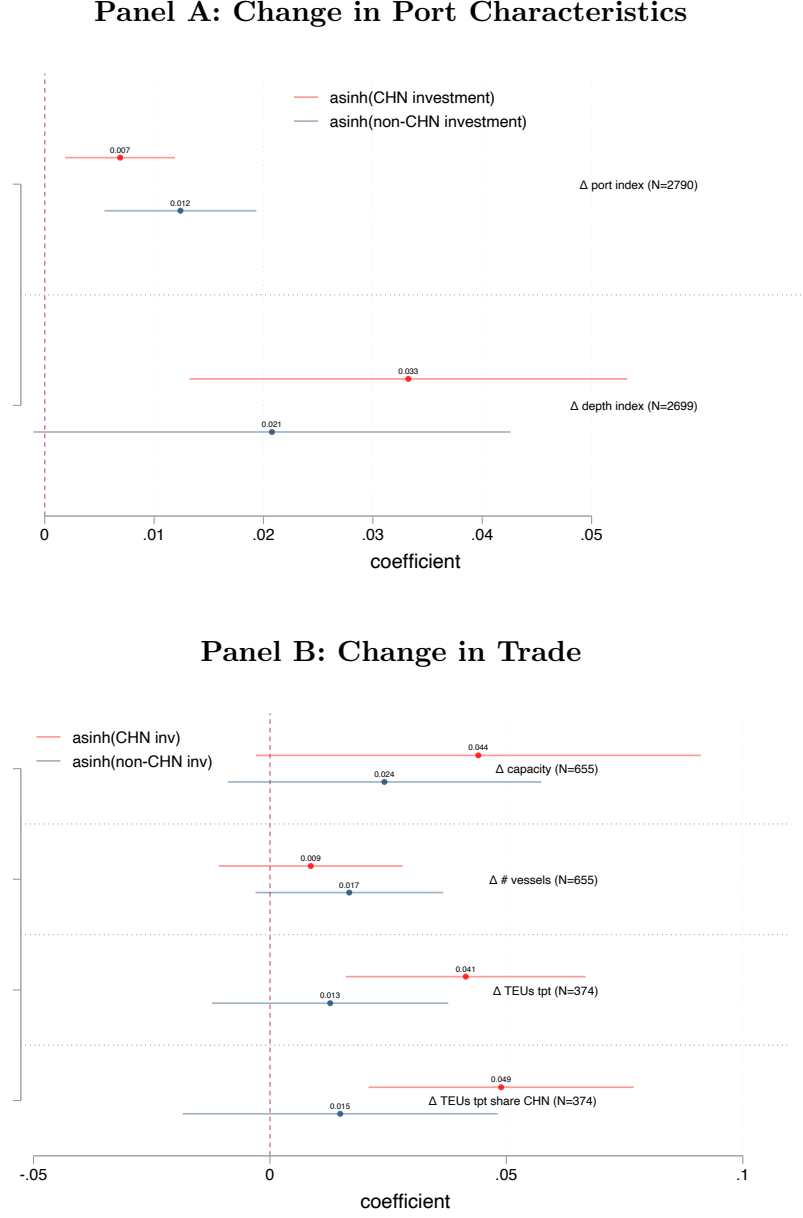
3.4 Fact 4: China’s Investments Raise Infrastructure and Trade (Similarly to Other Sources)

We now examine if ports that receive investment subsequently experience changes in their physical characteristics and trade volumes. For each port we construct long differences of the standardized characteristics and trade outcomes between a early period (2000-05 for port characteristics, 2006 for capacity and vessels, and 2011/12 for TEU flows) and a late period (2021-25 across all outcomes), then regress these long differences on the (inverse hyperbolic sine) cumulative

¹⁴This finding differs from [Brancaccio et al. \(2024\)](#) who find that a larger tidal range is associated with higher port productivity. The difference could come from that study examining U.S. ports, while we study ports around the world, or that we have additional controls in these specifications, or because they examine port productivity while we examine output.

¹⁵The patterns are the same if we instead use a binary indicator in (2) instead of $\text{asinh}(\text{value})$.

Figure 4: Investment and Changes in Port Characteristics and Trade



Notes: Top panel reports coefficients for changes in port indexes on China's and non-China's (inverse hyperbolic sine) investment. The bottom reports coefficients for changes in port trade outcomes on each investment. Standard errors clustered by country; 90/10 errors bars noted.

investment value received at the port:

$$\Delta X_p = \alpha_s + \beta_s \text{asinh}(\text{value}_{ps}) + \varepsilon_{ps} \quad (3)$$

As before, variables are standardized, and standard errors are clustered by country.

Of the port attributes analyzed, the dredging cost is time invariant and so we do not analyze its change over time. As classified by WPI, the harbor size is not necessarily time invariant, as it is based on several factors including area, facilities, and wharf space (see footnote 8). The pier, channel, and anchorage depths can change with investment, particularly through dredging. The tidal range is also, in principle, time invariant, although there is some evidence that tidal ranges could change due to dredging (Ralston et al. 2019). Other “above-ground” port characteristics correlated with trade volumes, such as number of berths, berth length, and types of cranes, could also, of course, be financed by investments, but at present we do not have these port-level attributes in the recent period.

The top panel of Figure 4 reports the regressions of the port characteristics and port depth indexes on China’s investment. We see no impact of investments on port characteristics, but we do see that China’s investment increases the depth index; a 1% increase in investment is associated with a $0.033sd$ increase in the depth index. This finding is suggestive that China’s investments change port infrastructure in part through increasing the depth at the port.¹⁶

We validate the mechanism of dredging causing changes in port characteristics by examining newspaper reports at the seaport locodes that China has invested at. We find 55 cases of confirmed dredging. Appendix Table C.4 shows that dredging is not correlated with changes in harbor size, tidal gates/breakwater, good shelter, and the tidal range. Ports with dredging do experience a large increases in pier depth ($0.487sd$) and anchorage depth ($0.430sd$), although no differential change in channel depth (relative to ports that received Chinese investment but have no confirmed reports of dredging).

We next examine how China’s investments are correlated with changes in port-level trade. The bottom panel of Figure 4 examines how port-level trade responds to investment. We see that China’s investments correlates positively with an increase in total port capacity—a 1% increase in investment increases port capacity by $0.044sd$. There is a positive correlation with the change in number of vessels, although the point estimate is quite noisy. Taken together, these suggest that China’s investments are increasing the sizes of vessels that a port can handle, though not necessarily the number of vessels. Turning toward actual trade volumes, we find that China’s investments is correlated with an increase in overall TEU throughput, as well as the share of that throughput to China.¹⁷

We re-do the analysis examining how non-China investment changes port characteristics and trade; this is also reported in Figure 4. We see that non-China investments change the same sets of port characteristics as with China’s investment (and at similar magnitudes). This suggests

¹⁶Panel A of Table C.2 reports the response of each underlying variable for these indexes. As expected, there is no response of the underlying variables in the port characteristics index. Among the three depth variables, we see that China’s investments is correlated with an increase in the pier and channel depth: a 1% increase in investment raises the pier depth by $0.053sd$ and the channel depth by $0.025sd$, which is suggestive that part of the investments are financing dredging costs. The point estimate on anchorage depth is a positive but not statistically significant.

¹⁷Panel B of Appendix Table C.2 shows that TEU imports, rather than exports, drives the throughput response. This is consistent with ports importing containers from China and our data being unable to capture exports of bulk cargo.

that there is not something unique about China’s investment, but rather port receiving investment experiences changes in some structural characteristics. The right-side, however, shows the impacts of non-China investment on trade are not sharp (although the magnitudes would be statistically similar to impacts from China’s investments).

3.5 Robustness

We assess robustness of these key findings along two dimensions by examining if the patterns hold when examining patterns across non-OECD countries only, and examine patterns among the large ports in WPI.

China has financed ports in advanced economies, including the U.S., E.U., and Australia, and so we examine how sensitive the findings are to the exclusion of these countries. Appendix Figure C.7 demonstrates that the correlates of country-level port investments for China and non-China sources are similar to the full sample. Among non-OECD countries, China targets richer countries relative to non-China lenders (although not statistically different from each other) and countries that are less aligned to the U.S. Interestingly, they do not target countries with more trade, although do direct financing towards countries with pre-existing larger share of trade with China and more natural resources.

Assessing to which ports China directs financing requires taking a stand on the potential set of ports it may consider. The baseline analysis includes all WPI ports, but its possible China only considered a subset of these ports for possible investment. We re-run (3) on these ports that WPI initially classifies as large (465 ports of the 3943 total ports in WPI), with the assumption that larger ports are more commercially viable (which is supported by the left panel of Figure C.4). Appendix Table C.5 examines the change in port indexes finds similar patterns are the full sample. The right panel examines changes in trade volumes, also showing qualitatively similar patterns as the full sample but noisier given the fewer number of observations.

4 Foreign Infrastructure Investments: Baseline Theory

4.1 Setup

Consumers and Producers We set up a simple model to study theoretically a country’s incentives to invest in international transport infrastructure. We consider an economy with two countries, an importer (country M) and an exporter (country X). The exporter produces a differentiated commodity but does not consume it, and the importer consumes it but does not produce it. Both countries are endowed with, and consume, an outside good that is freely traded. Both countries may set trade taxes, a per unit import tax $t_{Q,M}$ in the importer and an export tax $t_{Q,X}$ in the exporter. Agents are competitive.

The countries are indexed by M and X depending on who exports the differentiated good. However, the key distinction is what country owns the profits Π_T of the transport hub, defined

below. We call the country owning the profits the “host” and the other country the potential “foreign investor”.

Consumers in the importing country M have quasilinear preferences $U(Q) + C_M^O$, where $U(Q)$ is a concave utility function over the quantity imported of the differentiated good, and C_M^O is consumption of the outside good. Therefore, indirect utility in the importer is:

$$V_M = \max_Q U(Q) - p_M Q + \bar{O}_M + \mathcal{T}_M, \quad (4)$$

where p_M is the import price for Q , $\bar{O}_M$ is the endowment of the outside good, and $\mathcal{T}_M$ are additional sources of consumer income (government transfers and returns to the transport sector) defined below. Producers in the exporting country X have a convex cost function $\Psi(Q)$ in units of the outside good. As the consumers in X only buy the numeraire good, their indirect utility equals their net income:

$$V_X = \max_Q p_X Q - \Psi(Q) + \bar{O}_X + \mathcal{T}_X, \quad (5)$$

where p_X is the producer price for Q , $\bar{O}_X$ is the endowment of the outside good, and $\mathcal{T}_X$ are the additional sources of income. Standard consumer and firm optimization gives the first order conditions:

$$p_M = U'(Q), \quad (6)$$

$$p_X = \Psi'(Q). \quad (7)$$

Shipping and Ports Competitive shipping companies buy the good from exporters, pay the (endogenous) user fees τ per unit shipped to operators of transport infrastructure, spend their own (exogenous) shipping expenses $\bar{\tau}$, and sell to importers. Free entry into shipping gives the no-arbitrage condition:¹⁸

$$p_M \leq p_X + \tau + \bar{\tau} + t_Q, \quad = \text{ if } Q > 0, \quad (8)$$

where

$$t \equiv t_{Q,M} + t_{Q,X} \quad (9)$$

is the sum of import and export taxes.

Shipping companies use services provided by transportation hubs such as ports, airports, or railway companies (“ports” in short). We assume there is a single port (the quantitative framework includes transport hubs across modes and countries). The port uses transport infrastructure I (physical infrastructure such as cranes or services such as dredging) and has a total cost $T(Q, I)$ (in units of the outside good) to process output Q . For example, it costs an $T(Q, I)$ dollars to process Q containers when the port has installed I cranes. Ports charge a fee τ per unit of Q and

¹⁸It does not matter who pays the taxes. We can equivalently assume t_M is paid by consumers in M and t_X by producers in X .

pay a user cost r per unit of I to maximize profits:

$$\Pi_T = \max_{Q,I} \tau Q - T(Q, I) - rI. \quad (10)$$

The port's profits are owned by its host country. We define a binary variable to index whether a given country j is the host:

$$\chi_j = \begin{cases} 1 & \text{if } j \text{ is host country,} \\ 0 & \text{if } j \text{ is foreign country.} \end{cases} \quad (11)$$

The first order conditions give the inverse supply of port services and inverse demand of infrastructure:

$$\tau = T_Q(Q, I), \quad (12)$$

$$r = -T_I(Q, I). \quad (13)$$

We make neoclassical assumptions on port technologies: T is increasing and convex in Q (increasing marginal cost due to port congestion) and decreasing and convex in I (decreasing returns to transport infrastructure). In addition, $T_{QI} < 0$ (infrastructure reduces the marginal cost of processing trade). We further impose $T(Q, I)$ is convex ($T_{QQ}T_{II} > T_{QI}^2$), ensuring an interior solution.¹⁹

Infrastructure Supply Both national governments and private agents can invest to develop international transport infrastructure. Each country $j \in \{M, X\}$ can transform κ_j units of outside goods into one unit of infrastructure I_j . We can interpret κ in two ways. A first interpretation is that κ is the marginal cost of producing a unit of infrastructure and I_j is measured in units of infrastructure. A second interpretation is that κ_j is a financial friction and I_j is measured in units of wealth invested.²⁰

A government of country j that invests $I_{G,j}$ charges a lump-sum tax $\kappa_j I_{G,j}$ to its residents to finance infrastructure and receives a return $r I_{G,j}$ that is transferred back to its citizens. Atomistic private firms in country j can use the same technology as the government to develop private infrastructure $I_{P,j}$. Private developers may also pay a tax $t_{P,j}$ to their government (a subsidy if negative). In the case of an international investment, private developers may face a foreign

¹⁹This condition is satisfied in the parametrization.

²⁰For example, under the first interpretation, κ is the dollar cost of building a crane and I_j is the number of cranes. In this interpretation, a country transforms a dollar of wealth (in units of the outside good) into a dollar invested at a 1-1 rate, and then earns a return $R = r/\kappa$ on each dollar invested (and therefore a rate r on each crane built). Under the second interpretation, a country transforms a dollar of wealth into a dollar invested at rate $1/\kappa$, earning a rate r per dollar invested. In the quantification, we directly observe dollars invested by China and we measure the rate κ at which dollars translate into infrastructure I_j proxied by cranes or by kilometers of road and rail.

investment tax t_I . Hence, private infrastructure developers in country j have profits

$$\Pi_{P,j} = I_{P,j} (r - \kappa_j - t_{P,j} - (1 - \chi_j) t_I) \leq 0 \quad (14)$$

with equality if $I_{P,j} > 0$. Letting $I_j = I_{G,j} + I_{P,j}$ be the total infrastructure developed by country j by either the private or the public sector, the global supply of infrastructure to the transport hub is

$$I = I_X + I_M. \quad (15)$$

We now define the transfers $\mathcal{T}_j$ entering in (4) and (5). The net transfers to final consumers in country $j = M, X$ are:

$$\mathcal{T}_j = \underbrace{\chi_j \Pi_T}_{\text{Transport Profits}} + \underbrace{(r - \kappa_j) I_j}_{\text{Returns to Infrastructure}} + \underbrace{t_{Q,j} Q}_{\text{Trade Tax}} + \underbrace{t_I [\chi_j I_{-j} - (1 - \chi_j) I_j]}_{\text{Foreign Investment Tax}} \text{ for } j = X, M. \quad (16)$$

Residents of country j own the port's profits (if the country is host) and the returns to the investment (regardless of whether public or private sector implement it). They also earn the revenue from the trade tax, and either pay for investment tax (if they export infrastructure) or earn the proceeds (if they are the host country).

Equilibrium Given Government Policies To recap, the equilibrium depends on a set of government policies, $\Gamma_j \equiv \{I_{G,j}, t_{P,j}, t_{Q,j}\}$, plus a foreign investment tax t_I by the foreign country. Given these policies, a market equilibrium consists of a quantity traded Q^* , a return to infrastructure r^* , a port fee τ^* , prices $\{p_j^*\}$, and private sector investments $\{I_{P,j}^*\}$ such that agents optimize according to (6), (7), (8), (12), (13), and (14). Combining these conditions, we obtain:

$$\underbrace{U'(Q^*)}_{=p_M^*} = \underbrace{\Psi'(Q^*)}_{=p_X^*} + \underbrace{T_Q(Q^*, I^*)}_{=\tau^*} + t_Q + \bar{\tau}, \quad (17)$$

where total infrastructure $I^* = \sum_{j=M,X} I_{G,j} + I_{P,j}^*$ is determined by free entry of private investors:

$$\underbrace{-T_I(Q^*, I^*)}_{=r^*} - t_{P,j} - (1 - \chi_j) t_I \leq \kappa_j \text{ for } j = X, M, \quad (18)$$

with equality if $I_{P,j}^* > 0$. (17) and (18) jointly determine $\{Q^*, I^*\}$. (17) is a standard no-arbitrage condition: when there is trade in final goods, the price at the importer's port equals the price at the exporter's plus shipping costs and trade taxes. (18) is the free entry condition of private investors. Private investors in j are crowded out if the returns from investment r^* net of taxes are below the marginal cost κ_j .

4.2 Impacts of Infrastructure on Prices and Welfare

Around an equilibrium, marginally shifting public investments only changes the global infrastructure stock I^* if private investors are absent; otherwise, public capital crowds private investments one to one, either in or out, without affecting equilibrium outcomes. However, a country can always tax or subsidize private investors, or undertake discrete public investments that fully crowd out the private sector, and in this way shrink or raise the global stock.²¹ Therefore, we can consider the welfare effects of country j raising the global infrastructure stock I^* by changing its own (public and private) investment I_j through some combination of public investments I_j and domestic taxes $t_{P,j}$. Part (i) considers impacts on the market outcomes entering in (17), and part (ii) shows effects on welfare (V_M and V_X in (4) and (5)).

Proposition 1. (*Effects of infrastructure investments on prices and welfare*). Suppose that, around an initial equilibrium, country j implements (via subsidies or public investment) an increase in equilibrium infrastructure $dI^* = dI_j$. Then,

(i) trade grows ($\frac{dQ^*}{dI^*} > 0$), import prices fall ($\frac{dp_M^*}{dI^*} < 0$), export prices increase ($\frac{dp_X^*}{dI^*} > 0$), port fees fall ($\frac{d\tau^*}{dI^*} < 0$), and returns to infrastructure fall ($\frac{dr^*}{dI^*} < 0$); and

(ii) letting f be the foreign investor and h be the host country, then

$$\frac{dV_f}{dI_f} = (r^* - \kappa_f - t_I) + Q^* \left| \frac{dp_f^*}{dI^*} \right| + I_f^* \frac{dr^*}{dI^*} + t_{Q,f} \frac{dQ^*}{dI^*} \quad (19)$$

and

$$\begin{aligned} \frac{dV_h}{dI_h} = (r^* - \kappa_h) + Q^* \underbrace{\left(\left| \frac{dp_h^*}{dI^*} \right| + \frac{d\tau^*}{dI^*} \right)}_{= - \left| \frac{dp_f^*}{dI^*} \right|} - I_f^* \frac{dr^*}{dI^*} + t_{Q,h} \frac{dQ^*}{dI^*}; \text{ and} \end{aligned} \quad (20)$$

(iii) if import and export taxes $t_{Q,h}$ are optimally set, then

$$\frac{dV_f}{dI_f} = (r^* - \kappa_f - t_I) - Q^* \frac{\partial \tau^*}{\partial I} + I_f^* \frac{\partial r^*}{\partial I} \quad (21)$$

and

$$\frac{dV_h}{dI_h} = (r^* - \kappa_h) - I_f^* \frac{\partial r^*}{\partial I}. \quad (22)$$

Consider first the market outcomes in part (i). On impact, at an initial level of trade Q^* , when infrastructure increases then congestion port fees τ^* fall ($T_{QI} < 0$). As a result, the price gap between importer and export shrinks; trade Q^* grows, p_M^* falls, and p_X^* grows. As trade increases, congestion raises ($T_{QQ} > 0$) but not enough to offset the direct effect of infrastructure. The effect

²¹Around an equilibrium with private investors from their own country, country j can either provide domestic subsidies to private investors to grow the capital stock or increase public investments $I_{G,j}$ to a point that the private sector is crowded out. Similarly, around an equilibrium with private investors from the foreign country, a country can fully crowd out foreign investors by using subsidies to domestic investors or public investment. A country cannot destroy the publicly invested infrastructure of the foreign country, so there is a lower bound to the global infrastructure a country can choose. See Section (17) and (A.1) for a more formal discussion.

on returns r^* is more subtle. On impact, assuming decreasing returns to infrastructure ($T_{II} < 0$), the direct effect is to lower the returns r^* . However, because infrastructure reduces congestion ($T_{QI} < 0$), as the quantity traded the marginal returns to infrastructure grow (and therefore r^*) grow. This indirect effect resulting in higher r^* is dominated by the direct effect that lowers r^* if T is globally convex (a condition bounding the absolute value of T_{IQ}); we discuss below what would happen in the case when T is not globally convex.

Conditions (19) and (20) in part (ii) show why host and foreign countries have different incentives to invest. These conditions distinguish between foreign investor and host country, rather than between importer and exporter. First, there are economic rents $r^* - \kappa_j$: with low enough initial infrastructure (high enough r^*), the investment is profitable. Naturally, the investment incentives are stronger for the more efficient country (with lower κ_j). Second, there are strategic terms-of-trade considerations. For both countries, there are positive welfare effects from the price of traded goods: the price falls at the importer and increases at the exporter ($|dp_f^*/dI^*| > 0$ and $|dp_h^*/dI^*| > 0$). However, the host country also exports shipping services and the loss in port fees exceeds the price gain for consumers or producers so that, overall, the host country loses from the price impact on final goods.²² Third, the decreasing returns to infrastructure ($dr^*/dI^* < 0$) imply a loss for the foreign country (an exporter of infrastructure) and a gain for the host country (an importer). With trade taxes, both countries also experience a fiscal externality $t_j dQ^*$.

Part (iii) modifies part (ii) for the case in which countries optimally set trade taxes. Optimization over trade taxes removes the welfare impacts from infrastructure investments operating through the manipulation of prices of traded goods. To see why, note that the changes in infrastructure affect all prices (p_M^* , p_X^* , τ^* , and r^*) both indirectly, through the quantities traded Q^* , and—in the case of τ^* and r^* —also directly through the port technology $T(Q, I)$.²³ When the tariffs are optimally set, the indirect effects vanish from welfare impacts through standard envelope arguments. Hence, the case with optimal tariffs is relevant because it demonstrates that the strategic incentives to lower trade costs by the foreign country persist even when its investment decision ignores the welfare gains through terms-of-trade manipulation. Indeed, comparing (21) and (22) with (19) and (20) shows that both countries still take into account the direct impact on infrastructure returns $I_f(\partial r^*/\partial I)$, while only the foreign country cares about impacts on shipping costs, $(\partial r^*/\partial I)$.

²²I.e., $|dp_h^*/dI^*| + d\tau^*/dI^* = -|dp_f^*/dI^*| < 0$. Specifically, if the host country is the exporter, better infrastructure raises the terms-of-trade on exports, $dp_X^* > 0$, but lowers port fees, $d\tau^* < 0$, with the combined effect $dp_X^* + d\tau^* = dp_M^* < 0$. In turn, if the host country is the importer, better infrastructure raises the terms-of-trade on imports, $-dp_M^* > 0$, but lowers port fees, $d\tau^* < 0$, with the combined effect $-dp_M^* + d\tau^* = -dp_X^* < 0$.

²³Formally, $\frac{dr^*}{dI} = \frac{\partial r^*}{\partial I} dI + \frac{\partial r^*}{\partial Q} dQ$ and $\frac{d\tau^*}{dI} = \frac{\partial \tau^*}{\partial I} dI + \frac{\partial \tau^*}{\partial Q} dQ$ where τ^* and r^* come from (12) and (13).

4.3 Equilibrium Infrastructure with Free Trade and Foreign Investment

We move on to understanding equilibrium investments.²⁴ We define the Nash Equilibrium in investments as the infrastructure of host and foreign country, I_h^* and I_f^* such that:

$$I_j^* = \arg \max_{I_j} V_j(I_j, I_{-j}^*) \quad (23)$$

subject to $\{Q^*, p_M^*, p_X^*, \tau^*, r^*\}$ given by (17) and total infrastructure $I^* = I_j + I_{-j}^*$.²⁵ We first conduct the analysis assuming free trade and investment ($t_{Q,j} = t_I = 0$). As we just discussed, countries have different incentives to invest due to marginal costs and strategic terms of trade manipulation. Naturally, without strategic incentives, only the country with lowest κ_j will invest. To study the implications of strategic investments, we now assume $\kappa_X = \kappa_M = \kappa$.

To understand the Nash equilibrium, consider first a situation without foreign investment ($I_f = 0$). Around that point, the foreign country does not experience any terms-of-trade loss from investment, but the host country does. Therefore, the foreign country gains more than the host from investing, $dV_f/dI_f > dV_h/dI_h$. So, if the host country is optimally investing ($dV_h/dI_h = 0$) then also the foreign country will want to invest. As a result, in a Nash equilibrium the foreign country must invest.

Consider next the host's optimal investment when $I_f = 0$. Because more infrastructure worsens its terms of trade on goods, the host country taxes its private sector. Letting r^{*a} be the equilibrium return to infrastructure in that case, from (20) we obtain the first order condition:

$$\frac{dV_h}{dI_h} = (r^{*a} - \kappa) - Q^* \left| \frac{dp_f^*}{dI^*} \right| = 0, \quad (24)$$

which gives a rate r^{*a} a rate above the marginal cost. From Proposition (1), r^* falls with infrastructure, so the host country taxes the private sector, $t_{P,h}^* > 0$. Around that point, from (19), the welfare effect of foreign is:

$$\frac{dV_f}{dI_f} = \left| \frac{dp_f^*}{dI^*} \right| Q^* + (r^{*a} - \kappa). \quad (25)$$

Because $r^{*a} > \kappa$, without any intervention from the foreign government the private investors will enter and raise the infrastructure stock in the host until $r^* = \kappa$. At that point, (25) gives $dV_f/dI_f = \left| dp_f^*/dI^* \right| Q^* > 0$: by investing, the foreign country can improve the terms of trade at zero cost. Because r^* falls with infrastructure, this means the foreign country over-invests compared

²⁴Our quantification below considers the impacts of actual and optimal BRI investments taking as given the infrastructure stock of each host country before the BRI and, as such, does not incorporate a Nash equilibrium. However, we still inspect the Nash equilibrium theoretically in the simple model to demonstrate that, in our framework, foreign investments also arise in equilibrium even when every country can invest and have the same marginal costs of funds.

²⁵As we have argued, each country can manipulate entry or exist of the private sector via taxes. Therefore, the Nash Equilibrium can be defined directly in terms of the total (private and public) investments of each country, with condition (18) that determines the private sector entry adjusting in the background depending on the policies used by countries to reach a desired level of total (private and public) investment.

to what its private sector would choose undistorted. Hence, around the infrastructure chosen by the host country in the absence of foreign investment, the foreign country subsidizes its private investors.

The way infrastructure is used to manipulate terms of trade of final goods can be compared with how trade taxes are typically used. Here, absent foreign investments, the host country manipulates infrastructure investments downwards to shrink trade in final goods, similarly to how tariffs are used (because this baseline model has only two countries, this effect is particularly strong as there are no substitute ports, while in the quantitative model this effect is weaker because traders can re-route through other hubs). In contrast, the foreign country subsidizes exports of infrastructure because, around a point with $r^* = \kappa$, it allows it to improve its terms of trade of the goods it imports at no fiscal cost. Section 4.5 shows that, when countries also implement optimal taxes, the strategic incentives to distort prices of final goods through infrastructure vanish, but the incentives to distort r^* and τ^* remain.²⁶

As we have argued, in a Nash equilibrium the foreign country must invest. However, the host country may be fully crowded out when the strategic incentives of the foreign country are strong enough. It follows from (19) that, with the foreign country investing,

$$r^* = \kappa - \left(\left| \frac{dp_f^*}{dI^*} \right| Q^* + I_f^* \frac{dr^*}{dI^*} \right). \quad (26)$$

From (20), the host will lose from entering if

$$- \left| \frac{dp_f^*}{dI^*} \right| Q^* < I_f^* \frac{dr^*}{dI^*} \quad (27)$$

at the (I_f^*, Q^*) that solve 17 and 26. In sum, when foreign-country prices p_f^* are very responsive to infrastructure, or when the returns to infrastructure r^* are not, the foreign country might crowd out the host. In turn, if 27 fails at the (I_f^*, Q^*) that solves 17 and 26, in equilibrium both countries must invest.

Last, we consider the efficiency properties of the equilibrium. Because of the outside good, utility is transferrable between countries; therefore, a global planner chooses trade and investment to maximize total surplus.²⁷ In the Pareto efficient allocation of infrastructure and trade, only the most efficient country invests. The efficient allocation is achieved by the market in the absence of government intervention, and in it the returns to infrastructure equal the marginal cost in the

²⁶This discussion also highlights the key role of modeling economic returns r . A model where each country can directly invest in lowering the trade costs τ without earning any economic return, such as Fajgelbaum and Schaal (2020) or Allen and Arkolakis (2022), would have different properties. In the data, most of the Chinese BRI investments have a direct economic return in the form of loan repayments. In some cases, China has offered concessional grants with little economic return. We incorporate information on the rates of returns of Chinese loans in our quantification.

²⁷The planner chooses trade Q , investment by country $\{I_M, I_X\}$ and consumption of the outside good by country $\{C_M^O, C_X^O\}$ to maximize utility of one country given a level of utility of the other. Section (20) and (A.2) derives this result from the planner's problem.

cheapest source:

$$r^{PE} = \min \{ \kappa_X, \kappa_M \}. \quad (28)$$

Our previous discussion implies that, if both countries invest, then the allocation of trade and infrastructure is efficient; in that case, 20 and 19 readily imply that $r^* = \kappa = r^{PE}$. However, when the strategic incentives of the foreign investor are strong enough and the host is crowded out, 20 and 26 and 27 imply $r^* < \kappa = r^{PE}$: there is over-investment of infrastructure in the market allocation.

We summarize the results so far in the following proposition.

Proposition 2. (*Equilibrium Investments with Free Trade*) Assume both countries have the same costs to develop infrastructure ($\kappa_M = \kappa_X = \kappa$) and there is free trade and foreign investment ($t_{Q,h} = t_{Q,f} = t_I = 0$). Then, in equilibrium, the foreign country necessarily invests in infrastructure and, depending on the parameters, it may be the only country investing. When both countries invest, the equilibrium global infrastructure is Pareto efficient ($r^* = \kappa$) and governments do not tax or subsidize private developers ($t_{P,j}^* = 0$). When the host country is crowded out, global infrastructure is over-supplied ($r^* < \kappa$) and the foreign government subsidizes its private developers ($t_{P,f}^* < 0$).

The main takeaway from this proposition is that strategic considerations pressure equilibrium outcomes towards foreign investment. In a *laissez-faire* equilibrium without technological differences ($\kappa_j = \kappa$), the source of infrastructure investments is undetermined in both the market and the efficient allocation. However, when strategic considerations come into play, foreign countries necessarily invest. Because the strategic motivations heavily rely on terms-of-trade effects that can be manipulated via trade and investment taxes, we now move to studying the equilibrium investments when countries can simultaneously use these other instruments.

4.4 Equilibrium Investments with a Foreign Investment Tax

Around the equilibrium in Proposition 2, 27 implies $dV_h/dI_f < 0$: in an equilibrium where the host country is crowded out, it would gain from restricting foreign investment.²⁸ However, the host country could still gain by letting foreign investment in and taxing it. To see why, consider again the welfare impacts in the absence of trade or investment policies. Around the autarky returns to infrastructure r^{*a} that solve 24, entry of foreign investors lowers welfare in the host country through terms-of-trade effects,

$$\frac{dV_h}{dI_f} = -Q^* \left| \frac{dp_f^*}{dI^*} \right| < 0, \quad (29)$$

but, because the initial equilibrium features under-investment, it raises global welfare by bringing infrastructure closer to the Pareto efficient allocation,

$$\frac{d(V_h + V_f)}{dI_f} = r^{*a} - \kappa > 0. \quad (30)$$

²⁸In the equilibrium without crowding out, $dV_h/dI_f = 0$: the host country is indifferent with respect to marginal changes in foreign investment.

Because global surplus increases, the host country is better off by letting the foreign country invest and taxing it (or by investing more itself and asking the foreign country for a transfer to do so).

To formalize this intuition, we maintain the assumption of free trade ($t_{Q,j} = 0$) but now let the host country choose both infrastructure and investment tax,

$$(I_h^*, t_I^*) = \arg \max_{I_h, t_I} V_j(I_h, t_I, I_f^*) \quad (31)$$

subject to the same constraints as in 23, plus an additional relationship $I_f^*(t_I)$ for how the host country perceives foreign investments to change t_I .²⁹ The foreign country takes t_I as given and continues to solve 23. With this foreign investment tax, the results from Proposition 23 and 2 are now extended as follows.

Proposition 3. (*Equilibrium Investments with Foreign Investment Tax*) Assume both countries have the same costs to develop infrastructure ($\kappa_M = \kappa_X = \kappa$), there is free trade ($t_{Q,h} = t_{Q,f} = 0$), and the foreign country sets an optimal foreign investment tax ($t_I = t_I^*$). Then, in equilibrium, the foreign country necessarily invests in infrastructure and, depending on the parameters, it may be the only country investing. The equilibrium features under-investment compared to the efficient allocation ($r^* > \kappa$), and the foreign country subsidizes private developers ($t_{P,f}^* < 0$).

With an investment tax, we still find that the foreign country necessarily invests and may even be the only country investing. However, Proposition 23 and 2 changes in one substantive way: because of the investment tax, the equilibrium now features under-investment compared to the global planner ($r^* > \kappa$). Despite infrastructure r^* above marginal cost κ , private investors in the foreign country would enter at a loss in the absence of government support. As a result, the foreign country subsidizes private developers, $t_{P,f}^* \equiv r^* - \kappa - \tau_I^* < 0$, so they are willing to enter.

4.5 Equilibrium Investments with Trade Policy

We extend the previous results to a context where countries can implement trade taxes. We define the Nash Equilibrium in investments and taxes such that:

$$(I_j^*, t_{Q,j}^*) = \arg \max_{I_j, t_{Q,j}} V_j(I_j, t_{Q,j}, I_{-j}^*, t_{Q,j}^*) \quad (32)$$

subject to $\{Q^*, p_M^*, p_X^*, \tau^*, r^*\}$ given by (17) and total infrastructure $I^* = I_j + I_{-j}^*$. That is, countries change infrastructure and tariffs simultaneously. The next proposition updates the equilibrium features from Proposition 2 for the case with optimal tariffs using the results from part (iii) of Proposition 1.

²⁹We assume that $I_f^*(t_I)$ is strictly decreasing. In a region with entry of foreign investors and no entry of domestic investors $I_f^*(t_I)$ results from from 18, $T_I(Q^*, I^*(t_I)) - t_{P,f} - t_I = \kappa$. Otherwise, it results from the foreign country's first-order condition in 23; i.e., such that $\frac{\partial V_f(I_f^*(t_I), I_h^*; t_I)}{\partial I_f} = 0$. However, the results in this section only assume the host country perceives a decreasing $I_f^*(t_I)$, regardless of the micro-foundation.

Proposition 4. (*Equilibrium Investments with Trade Taxes*) Assume both countries have the same costs to develop infrastructure ($\kappa_M = \kappa_X = \kappa$), countries set optimal trade taxes ($t_X = t_X^*$ and $t_M = t_M^*$), and there is no foreign investment tax ($t_I = 0$). Then, in equilibrium, the foreign country necessarily invests in infrastructure and subsidizes private developers ($t_{P,f}^* < 0$) and, depending on the parameters, it may be the only country investing.

Importantly, the key asymmetry in the incentives to invest around autarky we have discussed in the context of free trade in Section 4.3: around autarky in infrastructure, the foreign country has stronger incentives to invest than the host country. Moreover, because the adverse terms-of-trade effects on prices of traded goods are eliminated for the host country, under infrastructure autarky that country chooses a conditionally efficient level of infrastructure ($r^{*a} = \kappa$). Around that point, the host country is indifferent (to a first order) with respect to the foreign country investing, while the foreign country gains from entering because it still experiences a gain from lower trade costs ($\partial\tau/\partial I < 0$). Therefore, the equilibrium features $r^* < \kappa$, and private developers are subsidized in the foreign country.

4.6 Taking Stock

In this section we have asked: under what conditions does a country obtain an economic benefit from investing in foreign infrastructure? In part the answer is straightforward a foreign country can benefit from investing if the initial stock of infrastructure is low (a low I_h with associated high r^*) or if it is more efficient than the host (a lower κ). However, even if countries share the same technologies (in which case the source of investments is indeterminate in the efficient allocation), the equilibrium always features foreign investment. Moreover, across the various cases we considered (free trade, optimal foreign investment tax, and optimal trade taxes) the foreign country typically intervenes to subsidize private investors.³⁰ These subsidies may shift the global allocation of infrastructure away from efficiency in the case of free trade, but closer to efficiency if the host country sets investment taxes or both countries set trade taxes. Importantly, these incentives persist even if countries implement terms-of-trade manipulation via trade taxes. Because the foreign investments raise global efficiency, both countries are better off from the investments if the host can tax the invest or obtain a subsidized rate. Next, we move to a quantitative setup to measure these forces.

5 Quantitative Framework

We develop a multi-country Armington trade model embedded in a multimodal transport network with endogenous transport costs and strategic infrastructure investment. This framework generalizes and nests the simple model presented in the Section 4. The key extensions are as follows. First, now there are J countries, each producing its own differentiated good, and every

³⁰The only case in which the foreign country does not intervene is under no trade or investment taxes, and within the range of parameters such that terms-of-trade effects are weak and both countries invest in equilibrium. This is also the only case in which the market allocation delivers the Pareto efficient allocation.

country demands the output from every country. Production functions in every country use domestic labor and the outside good and therefore have curvature (allowing output to expand and contract with trade) with country-specific TFP shifters. On the demand side, each country has a vector of demand shocks over imports by origin.

Second, now trade happens over a multi-modal transport network. Each country consists of a set of nodes (representing seaports, airports, land crossings, and rail junctions). Each node s operates a transport technology $T_s(Q_s, I_s)$, as in the baseline model. Now each trade flow congests the origin and destination nodes (instead of crossing through only one gate as in the baseline model). The throughput Q_s adding congestion to the transportation node s combines all the volume shipped through that node, including shipments en route to third countries. The network also includes trans-shipment costs when between different modes within a country, while international connections are within-mode.

5.1 Setup

Countries and Preferences The world economy consists of a set of countries $\mathcal{J} = \{1, \dots, J\}$, with China indexed by $j = CH$. There are $J + 1$ goods: J differentiated varieties, each produced by a different country, and a homogeneous outside good. Each country quasi-linear utility over differentiated products C_j and the outside good C_j^O :

$$V_j = U_j(C_j) + C_j^O, \quad (33)$$

C_j is a CES aggregator of the differentiated varieties from country $n = 1, \dots, J$,

$$C_j = \left(\sum_{n=1}^J d_{nj}^{\frac{1}{\sigma}} C_{nj}^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}}, \text{ with } \sigma > 0, \quad (34)$$

with d_{nj} denoting an origin-destination-specific preference shifter. Country j faces the budget constraint

$$C_j^O + \sum_{n=1}^J p_{nj}^C C_{nj} \leq \Pi_{Y,j} + \Pi_{T,j} + \Pi_{I,j} + \bar{O}_j + \sum_n t_{nj}^M C_{nj} + \sum_n t_{jn}^X C_{jn}, \quad (35)$$

where $p_{nj}^C = p_{nj} + t_{nj}^M + t_{nj}^X$ is the consumer price, p_{nj} is the pre-tax price of good n upon arrival in country j (inclusive of transport costs), and t_{nj}^M and t_{nj}^X denote, respectively, the import tax set by importer j on exporter n and the export tax set by the exporter n on the importer j . On the income side, $\Pi_{Y,j}$ represents profits from production of final goods, $\Pi_{T,j}$ are profits from the transport sector, and $\Pi_{I,j}$ are profits from domestic and foreign infrastructure investments. The term $\bar{O}_j$ denotes country j 's endowment of the outside good, and the last two terms are the tariff revenue.³¹

³¹ As shown in the baseline model, any desired level of infrastructure investment from a country can be implemented with a mix of private-sector subsidies and public investments by that country. Moreover, the welfare implications

Consumer optimizations yields the standard conditions

$$U'_j(C_j) = P_j^C \quad (36)$$

$$C_{nj} = d_{nj} \left(\frac{p_{nj}^C}{P_j^C} \right)^{-\sigma} C_j \quad (37)$$

where $P_j^C = \left(\sum_{n=1}^J d_{nj} (p_{nj}^C)^{1-\sigma} \right)^{\frac{1}{1-\sigma}}$ is the consumer price index.

Production Country j produces its differentiated good using as input the outside good

$$\Pi_{Y,j} = \max_{Y_j} p_{jj} Y_j - \Psi_j(Y_j), \quad (38)$$

with inverse supply

$$p_{jj} = \Psi'_j(Y_j). \quad (39)$$

Geography Each country j is represented by a set of nodes $\mathcal{S}(j)$, which includes a *central* node $s_0(j)$, where production and consumption take place, as well as transport nodes (e.g., seaports or rail hubs). Therefore, the pre-tax price of food from n in j is the price at the node $s_0(j)$, $p_{nj} = p_{ns_0(j)}$. The global set of nodes $\mathcal{S} = \cup_{j \in \mathcal{J}} \mathcal{S}_j$ is arranged on an undirected graph $(\mathcal{S}, \mathcal{E})$ where the $\mathcal{E} \subset \mathcal{S}^2$ describe feasible transport links. We denote $\mathcal{N}(s)$ the set of neighbors of node s .

Transport technology Goods are shipped across the transport network. Let $Q_{nss'}$ the quantity of good n transported over the link $(s, s') \in \mathcal{E}$ and define total flow through the node s as $Q_{ns} = \sum_{s'} (Q_{nss'} + Q_{ns's})$. We denote by p_{ns} the pre-tax price of good n at node s upon leaving the node. Abusing notation slightly, p_{nj} denotes the price of good n at the central node of country j , i.e., $p_{nj} \equiv p_{ns_0(j)}$.

As in the simple model, shipping is carried out by competitive firms. On link (s, s') , shipping companies purchase good n at node s at price p_{ns} , incur port fees τ_{ns} and $\tau_{ns'}$ at origin and destination, and pay an exogenous transport cost $\bar{\tau}_{nss'}$. They then sell the good at node s' at price $p_{ns'}$. Free entry yields the no-arbitrage condition:

$$p_{ns'} \leq p_{ns} + \bar{\tau}_{nss'} + \tau_{ns} + \tau_{ns'}, \text{ with equality if } Q_{nss'} > 0. \quad (40)$$

Shipping goods requires the services provided by infrastructure operators (“ports”) at a each node. Handling goods is costly, with cost function $T_s(Q_s, I_s)$, measured in units of the outside good,

depend on the total investment resulting from a given policy mix regardless of how they are implemented. Therefore, the quantitative model omits explicitly modeling the private sector. At any equilibrium of the model, a given mix of private and public investments can be implemented by allowing for private investors and properly choosing a subsidy to that sector

where

$$Q_s = \sum_{n=1}^J \omega_n Q_{ns} \quad (41)$$

is the total volume of goods transiting through node s and I_s is the quantity of infrastructure. As in the simple model, T_s is increasing and convex in Q_s (congestion), but decreasing and convex in I_s (i.e., infrastructure reduces costs with diminishing returns). Infrastructure I_s is rented at user cost r_s from either domestic or foreign suppliers. Ports operate in competitive markets and charge a per-unit fee τ_{ns} to handle good n . Port operators solve the problem:

$$\pi_{Ts} = \max_{\{Q_{ns}\}, I_s} \sum_n \tau_{ns} Q_{ns} - T_s(Q_s, I_s) - r_s I_s \quad (42)$$

where $Q_s = \sum_n \omega_n Q_{ns}$ and $Q_{ns} = \sum_{s' \in \mathcal{N}(s)} (Q_{nss'} + Q_{ns's})$. Optimization by port operators yields the supply of transport service and the demand of infrastructure,

$$\tau_{ns} = \omega_n \frac{\partial T_s(Q_s, I_s)}{\partial Q_s}, \quad (43)$$

$$r_s = - \frac{\partial T_s(Q_s, I_s)}{\partial I_s}. \quad (44)$$

We assume that port operators are owned by the host country, so that total transport profits are $\Pi_{Tj} = \sum_{s \in \mathcal{S}(j)} \pi_{Ts}$.

Infrastructure Total profits from infrastructure investment is

$$\Pi_{I,j} = \underbrace{\sum_{s \in \mathcal{S}(j)} (r_s - \kappa_{js}) I_{js}}_{\text{domestic investments}} + \underbrace{\sum_{s \in \mathcal{S} \setminus \mathcal{S}(j)} (r_s - \kappa_{js}) I_{js}}_{\text{foreign investments}}. \quad (45)$$

Country j supplies infrastructure I_s^j at node s at per unit cost κ_{js} , measured in terms of the outside good. The total supply of infrastructure at node s is

$$I_s = \sum_{j \in \mathcal{J}} I_{js}. \quad (46)$$

Resource constraint For each good $n \in \mathcal{J}$, feasibility requires that for each country j and each node $s \in \mathcal{S}(j)$:

$$1_{s=s_0(j)} C_{nj} + \sum_{s' \in \mathcal{N}(s)} Q_{nss'} \leq 1_{s=s_0(j)} 1_{n=j} Y_n + \sum_{s' \in \mathcal{N}(s)} Q_{ns's}. \quad (47)$$

Condition (47) states that, at each node, total outflows (consumption and shipments) cannot exceed total inflows (production and incoming shipments). Consumption C_{nj} occurs only at the central node $s_0(j)$, while production Y_n occurs only at the central node of the producing country.

Equilibrium We define an equilibrium as follows.

Definition 1. An equilibrium given infrastructure $\{I_{js}\}_{j \in \mathcal{J}, s \in \mathcal{S}}$ is a set of prices $\{p_{ns}, \tau_{ns}\}_{j \in \mathcal{J}, s \in \mathcal{S}}$, returns to infrastructure $\{r_s\}_{s \in \mathcal{S}}$, consumption $\{C_{nj}\}_{(n,j) \in \mathcal{J} \times \mathcal{J}}$, output $\{Y_n\}_{j \in \mathcal{J}, s \in \mathcal{S}}$, and gross trade flows through the network $\{Q_{nss'}\}_{j \in \mathcal{J}, (s,s') \in \mathcal{E}}$ such that:

- In every country $j \in \mathcal{J}$, households maximize utility, i.e. (36) and (37) hold.
- In every country $j \in \mathcal{J}$, firms maximize profits, i.e. (53) holds.
- In every node $s \in \mathcal{S}$, infrastructure operators maximize profits, i.e. (43) and (44) hold.
- For every link $(s, s') \in \mathcal{E}$ and every good $n \in \mathcal{J}$, prices satisfy no-arbitrage (40).
- For every good $n \in \mathcal{J}$, country $j \in \mathcal{J}$ and node $s \in \mathcal{S}(j)$, the resource constraint (47) is satisfied.
- At every node $s \in \mathcal{S}$, the infrastructure market clears, i.e. (46) holds

5.2 Key Welfare Features versus the Baseline Model

This quantitative framework nests the baseline model we have studied in Section 4. It also includes new elements for welfare impacts. Now, infrastructure investments happen in a network and China can invest in a portfolio of ports. Countries are impacted by infrastructure both because the prices of their consumption and output change and because traffic changes through the ports they own. For example, a country whose exports are typically routed through a given foreign port will typically benefit, on impact, when more investment happens in that port: the lower fees along the route to a final destination will lower the import price and increase demand. Similarly, if a country's imports from some country travel through a port that received investments, the country will benefit from investments happening in that port.

We can summarize these forces by recomputing the welfare gradient of each country in this case. Given any vector of bilateral (country-node) investments in the world dI_{js} in the world, including both foreign and domestic investments, the generalized version of (19) and (20) is:

$$\begin{aligned}
dV_j = & \underbrace{(r_s - \kappa_{js}) dI_{js}}_{\text{Return to Investment}} + Y_j \underbrace{(dp_{jj} + \overline{d\tau_j})}_{\text{ToT on final goods}} - \underbrace{\sum_{n=1}^J C_{nj} dp_{nj} + \sum_{s \in \mathcal{S} \setminus \mathcal{S}(j)} I_{sj} dr_s - \sum_{s \in \mathcal{S}(j)} (I_s - I_{js}) dr_s}_{\text{ToT on } I} \\
& + \underbrace{\sum_{s \in \mathcal{S}(j)} \sum_{n \neq j} Q_{ns} d\tau_{ns}}_{\text{Foreign Demand for Own Ports}}, \tag{48}
\end{aligned}$$

where $\overline{d\tau_j} \equiv \sum_{s \in \mathcal{S}(j)} Q_{js} d\tau_{js} / Y_j$ is the change in transport fees charged by country j on its own goods. All the terms in the first line of 48 extend terms that already appeared in 19 and 20. The term in the second line is zero in the baseline model, and it shows that here countries also care

about their ports shipping goods from other countries. Countries benefit from investments that redirect trade to their ports, increasing congestion and fees τ_{ns} .

We highlight that dV_j is the welfare change of any country j when any set of investments (domestic or international) take place in the world. Hence, these effects are present for China, host countries of Chinese investments, and third countries not receiving any investments. When China invests, in any country j other than China the first term in the first line of 48 (return to investment) is zero, but the remaining terms will generically not be and can lead to either positive or negative welfare impacts depending on the ensuing vector of price changes.

5.3 Numerical Resolution

The numerical resolution of our quantitative model is challenging for several reasons. First, the problem is extremely high-dimensional: the large number of endogenous variables rules out the use of standard optimization routines that rely on computing Hessians. Given the size of the network and number of countries described in the next section, the number of endogenous flow variables $Q_{nss'}$ with their associated non-negativity multipliers approaches 1 million. Computing and storing a Hessian for a problem of this scale would require computing and storing matrices of multiple terabytes of memory. Second, congestion at each node is highly nonlinear in the throughput of all goods flowing through node, making it unfeasible to reduce dimension by expressing outcomes as function of prices at nodes. Third, it is not possible to cast the optimization problem over infrastructure as the decision of a single global planner. In other words, the allocation of infrastructure does not maximize global surplus because Chinese planners internalize price effects that are zero-sum for the world and therefore ignored by a global planner. These features imply we cannot use the methods from Fajgelbaum and Schaal (2020), who formulate the equilibrium determination, routing, and optimal network as a single-agent optimization problem.

To address these challenges, we develop a novel numerical strategy that combines a standard iterative procedure over prices in an outer loop with a separate inner loop that solves the optimal flow and routing problem. We show that the optimal flows in our framework coincides with the solution to a standard convex-cost flow problem, which is well-known in the operations research literature. A key step in our approach is the use of a fast piecewise-linear approximation method (Croxtan et al., 2007; Crainic and Kim, 2007), which transforms the optimal flow and routing problem into a standard linear programming (LP) problem. The key idea of the approach is to approximate the $T_s(Q_s)$ function mapping throughput to trade costs at a node with multiple capacity-constrained links with different levels of (constant) marginal costs. The resulting LP can then be solved efficiently using simplex-type algorithms and optimization solvers such as Gurobi. Although this reformulation introduces additional variables and constraints relative to the original problem, it avoids the repeated computation of large Hessian matrices at each iteration and substantially improves computational tractability. Appendix B.2 provides a detailed description of the numerical algorithm.

This approach to micro-founding and solving for the transport sector is quite flexible and can

be readily generalized to accommodate a wide range of congestion effects, including congestion at the port vs. link level, both within and across goods, with or without external effects. Section B.4 in the Appendix demonstrates how our numerical approach can be extended to incorporate congestion externalities as well as iceberg transport costs.

6 Quantitative Results [IN PROGRESS]

This section is still very preliminary, and we show it to illustrate how we can implement the model and to give a sense of the overall approach we are taking.

6.1 Quantification Approach

We bring the model to the data as follows. First, in Section 6.2, we match the model to the pre-BRI equilibrium in the data, using averages from 2000-2012. In that equilibrium, we take as given the observed distribution of infrastructure by country and mode, and choose the model parameters to match observed distributions of trade and GDP. Given a set of external elasticities, our calibration exactly matches the global distribution of trade and GDP, and we verify as an over-identifying check that the model makes reasonable prediction for gross capacity flows, measured in the data by vessel container capacity in each route.

Then, we simulate the BRI investments in the model. For this, we need to translate the Chinese investments (measured in USD) infrastructure growth. In Section 6.3, we rely on the empirical analysis from Section 3, which show that investments affected port infrastructure, to measure the marginal cost of infrastructure in each country and measure the changes in infrastructure caused by the BRI. Section 6.4 simulates these shocks into our pre-BRI calibrated economy and measures the changes in trade and welfare across the world.

Finally, we compare the observed BRI investments with those that would have been implemented by a Chinese planner that only cares about maximizing income. In Section 6.5, we back infrastructure costs that rationalize the observed BRI investments as the outcome of a planning problem by Chinese authorities. We contrast these costs with those we measure directly in the data in the previous step by observing how invested BRI dollars actually translate into infrastructure. The gaps between these cost measures derive non-economic value attached by Chinese planners to each country. We conclude in Section 6.6 by simulating the optimal investments that a Chinese planner would have implemented under economic costs only.

6.2 Calibration to Pre-BRI Equilibrium

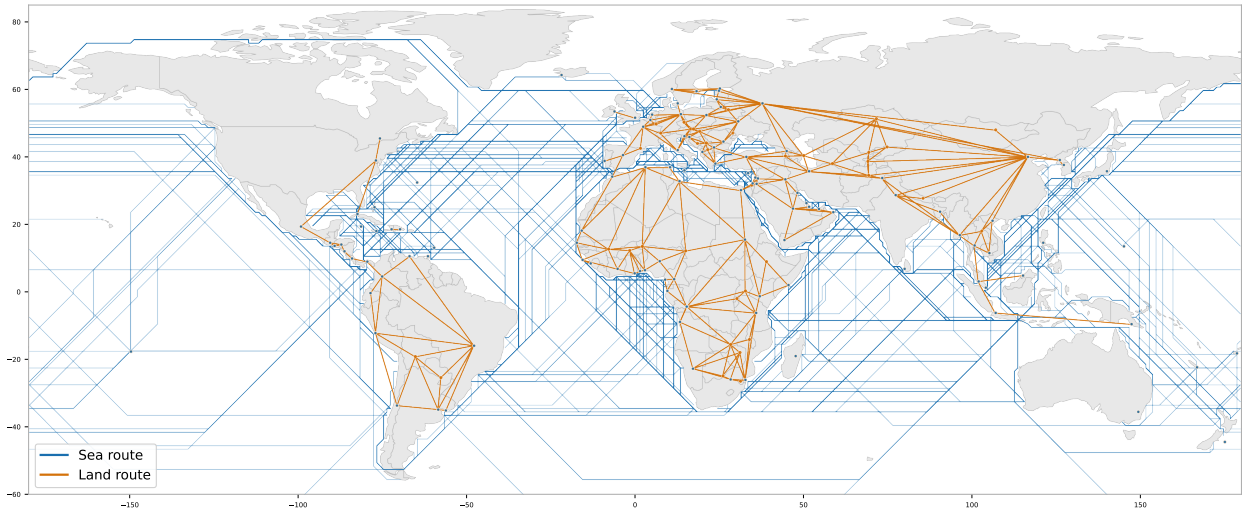
Network Our preliminary quantification focuses on land and maritime transport.³² We include the 160 largest countries. Each country $j \in \mathcal{J}$ is represented by a set $S(j)$ consisting of a land node and a sea node (if the country is not landlocked). To build the network, we assume land

³²We are currently working on adding the air network.

nodes belonging to two countries are connected if the countries share a land border. The kilometer distance between connected land nodes is computed using spherical distance between the capitals of the connected countries. The sea nodes of two countries are connected if, in our MDST data, we observe at least one container ship making consecutive stops in ports belonging to these two countries. The kilometer distance between connected sea nodes is calculated using the shortest path on a 5-degree sea raster between the two ports handling the highest bilateral container traffic.³³

These assumptions yield a land network with 554 edges and a sea network with 1,655 edges (Figure 5). We allow for transshipment between the land and sea nodes within each country. The resulting network is fully connected (with any node reachable from any node). We assume that the land node serves as the country’s central node, where both national production and consumption take. We denote by $m(s) \in \{land, sea\}$ the transport mode of node $s \in \mathcal{S}$.

Figure 5: Land and Sea Network



Notes: Maps plots the network of land and sea routes. Land routes connect capitals through straight line distance. Sea routes are built from the port-to-port vessels calls in the MDST data.

Functional Forms We parametrize the model to match data corresponding to averages from 2000 to 2012. First, we need to assume functional forms for transport costs. From (40), the trade cost for good n between connected nodes s and s' consist of an exogenous component and port fees at origin and destination:

$$\tau_{nss'} = \bar{\tau}_{nss'} + \tau_{ns} + \tau_{ns'}. \quad (49)$$

³³Therefore, even though we include a single “sea node”, countries ship via sea using origin and destination ports at the actual locations of the largest ports handling the most bilateral traffic. We plan to include a more detailed geography within larger countries.

We parametrize the exogenous components follows:

$$\bar{\tau}_{nss'} = \begin{cases} \omega_n \exp(-\zeta_{m(s)} \text{dist}(s, s')) & \text{if } m(s) = m(s') \\ \omega_n \bar{\tau}_{TS} & \text{if } m(s) \neq m(s') \end{cases},$$

The term ω_n is a good-specific shifter capturing the weight or volume per unit of goods from n , $\zeta_{m(s)}$ is a transport mode-specific distance elasticity for international shipments, and $\text{dist}(s, s')$ the kilometer distance between the two nodes. We assume a common transshipment cost $\bar{\tau}_{TS}$ between modes.

We assume a power function for the transport costs of nodes entering in (42):

$$T_s(Q_s, I_s) = \bar{T}_{m(s)} Q_s^{1+\beta_{m(s)}} I_s^{-\gamma_{m(s)}}, \quad (50)$$

From (43), the port fees are:

$$\tau_{ns} = (1 + \beta_{m(s)}) \omega_n \bar{T}_{m(s)} Q_s^{1+\beta_{m(s)}} I_s^{-\gamma_{m(s)}}. \quad (51)$$

The parameters $\bar{T}_{m(s)}$, $\beta_{m(s)}$ and $\gamma_{m(s)}$ denote, respectively, mode-specific cost shifters, congestion elasticities, and infrastructure elasticities. The infrastructure index I_s proxies for the transport infrastructure in each country.

Externally Set Parameters To parametrize our model, we externally set a few parameters. Utility in country j , with population L_j , is given by

$$U_j(C_j) = \frac{C_j^{1-\rho}}{1-\rho}. \quad (52)$$

We set the curvature parameter to $\rho = 2$, consistent with standard values in the macroeconomic literature, implying an inter-temporal elasticity of substitution (IES) of 0.5. The cost function in country j is derived from a Cobb-Douglas production function $Y_j = A_j M_j^\alpha L_j^{1-\alpha}$ with share α on the outside good and $1 - \alpha$ on a residual fixed factor, which we assume to equal the national labor endowment, and TFP shifter A_j implying

$$\Psi_j(Y_j) = \frac{\alpha^{\frac{\alpha-1}{\alpha}}}{L_j} \left(\frac{Y_j}{Z_j} \right)^{\frac{1}{\alpha}}. \quad (53)$$

We set $\alpha = 0.33$ to match a standard labor share of 67%. In line with standard estimates in the trade literature, we set the CES elasticity of substitution in (34) to $\sigma = 3$, which lies within the range of estimates in [Head and Mayer \(2014\)](#). We currently set all tariffs to 0, i.e., $t_{nj}^M = t_{nj}^X = 0$.

We follow [Fajgelbaum and Schaal \(2020\)](#) and parametrize the land congestion and infrastructure parameters, β_{land} and γ_{land} , to match estimates from [Couture et al. \(2018\)](#) on the elasticity of travel time to kilometers of roads and vehicle-miles, which give $\beta_{land} = 0.13$ and $\gamma_{land} = 0.10$.

For the sea congestion and infrastructure parameter, we use the Cobb-Douglas estimates of port production functions in the US by [Brancaccio et al. \(2024\)](#), which translated to our functional form in (50) yield $\beta_{sea} = 0.79$ and $\gamma_{sea} = 0.31$.³⁴

Calibrated Parameters The remaining parameters are: demand shifter d_{nj} , TFP shifters A_n , good-specific weight shifters ω_n , mode-specific distance elasticity $\zeta_{m(s)}$, mode-specific shifters $\bar{T}_{m(s)}$, and trans-shipment cost $\bar{\tau}_{TS}$. We calibrate these parameters to match FOB trade, GDP and the rate of return on infrastructure loans by land and sea observed in the BRI data. In the model, the FOB value of trade from origin n to destination j is given by

$$X_{nj}^{FOB} = p_{nn} d_{nj} C_j \left(\frac{p_{nn} + \tau_{nj}^{LD}(\Theta)}{P_j^C} \right)^{-\sigma}, \quad (54)$$

where $\tau_{nj}^{LD}(\Theta)$ is the “long-distance” transport cost to ship good n from n to j on the optimal route on the transport network, itself a function of the trade-cost parameters $\{\zeta_{m(s)}, \bar{T}_{m(s)}, \bar{\tau}_{TS}, \omega_n\}$. We take logs, choose units by imposing the normalization $p_{nn} = 1$ in the initial equilibrium, and decompose the demand shifters $d_{nj} = O_n D_j \varepsilon_{nj}$ into an origin, a destination term and a residual.³⁵ Then, we obtain the estimating equation:

$$\ln X_{nj}^{FOB} = \psi_n + \psi^j - \sigma \ln(1 + \tau_{nj}^{LD}) + \ln \varepsilon_{nj}, \quad (55)$$

where the fixed effects are $\psi_n \equiv \ln O_n$ and $\psi^j \equiv \ln C_j (P_j^C)^\sigma D_j$.

Our estimation iterates between two steps. First, for each candidate parameter vector $\Theta \equiv \{\zeta_{m(s)}, \bar{\tau}_{TS}, \omega_n\}$, we set $\bar{T} \equiv \{\bar{T}_{land}, \bar{T}_{sea}\}$ such that the average r_s/κ_s in the model matches the average rate of return R_m^{BRI} per dollar across all (mode-specific) loans observed in the BRI data:

$$\mathbb{E}[r_s/\kappa_s | m(s) = m] = R_m^{BRI}, \quad m \in \{land, sea\}. \quad (56)$$

³⁴[Brancaccio et al. \(2024\)](#) estimate a service time per ship using (in their notation) the functional form $TIME(\frac{L}{s}, \frac{c}{s}) = \omega \left(\left(\frac{L}{s} \right)^{\alpha_{cd}} \left(\frac{c}{s} \right)^{\beta_{cd}} \right)^{-1}$, where L is workers, c is cranes, s is ships, and ω is a TFP shifter of the port. To use their estimates, we assume: $Q \propto s$ (shipments Q in our model are proportional to ships s); $I \propto s$ (infrastructure I is proportional to the number of cranes); and $T \propto Q * TIME$ (total cost of a of a port is proportional to total time spent processing all ships) processing time. We further assume that labor is optimized out by ports given infrastructure. That is, for any Q and I , ports in our model solve $\max_{Q,I} \tau Q - \left\{ \min_L \omega \frac{L^{\alpha_{cd} + \beta_{cd}}}{L^{\alpha_{cd}} I^{\beta_{cd}}} + wL \right\} - rI = \max_{Q,I} \tau Q - \bar{T}_{sea} Q^{\beta_{sea}} I^{\gamma_{sea}} - rI$, where the last problem coincides with (50) with $\bar{T} \equiv \omega w^{\frac{\alpha_{cd}}{\alpha_{cd} + 1}} (\alpha_{cd}^{-1} + 1) \alpha_{cd}^{\frac{1}{\alpha_{cd} + 1}}$, $\beta_{sea} \equiv \frac{\alpha_{cd} + \beta_{cd}}{1 + \alpha_{cd}}$ and $\gamma_{sea} \equiv \frac{\beta_{cd}}{1 + \alpha_{cd}}$.

³⁵This normalization amounts to an arbitrary choice of units for each good n , with a corresponding rescaling of $\{A_n, \omega_n, O_n\}$ and quantities of good n . In the counterfactuals, p_{nn} adjusts to clear markets. Calculating the long-distance cost $\tau_{nj}^{LD}(\Theta)$ requires solving for the optimal flows on the network under congestion for each combination of parameters Θ . Having chosen the units of each good n , we back out the quantities C_{nj} from the FOB flows directly ($p_{nn} C_{nj} = X_{nj}^{FOB}$) as well as output from GDP ($p_{nn} Y_n = GDP_n$). Knowing consumption and production, for estimation we can skip the calculation of the fixed point over producer prices described in Section B.2 and directly solve for the optimal flows $Q_{nss'}$ and the resulting long-distance $\tau_{nj}^{LD}(\Theta)$ using our piecewise linear for each draw of Θ .

Then, given $\bar{T}$, we estimate (55) by nonlinear least squares, computing the long-distance transport costs $\tau_{nj}^{LD}(\{\Theta; \bar{T}\})$ and choosing the Θ that minimizes the squared residuals $MSE(\Theta; \bar{T}) = \sum_{n,j} (\ln \varepsilon_{nj})^2 / J^2$. In sum, our estimation procedure solves the minimization problem

$$\min_{\Theta, \bar{T}_{m(s)}} MSE(\Theta; \bar{T}) \text{ subject to (56).}$$

The demand shifters O_n and D_j are recovered from the estimated fixed effects ψ_n and ψ_j , and the residual component ε_{nj} of the demand shock corresponds to the residual in (55), that exactly matches gravity. Finally, we infer the price-adjusted country-level productivity $p_{jj}A_j$ by equating each country's production of the differentiated good to GDP. From (39) and (53),

$$GDP_j = \alpha^{\frac{\alpha}{1-\alpha}} (p_{jj}A_j)^{\frac{1}{1-\alpha}} L_j. \quad (57)$$

The calibrated parameters are reported in Table 2. Trade flows and GDP are exactly matched, as we have just described. For model validation, Figure 6 compares model-implied gross flows (Q_s) through each country's seaports with observed capacity-adjusted vessels flowing through each country in the MDST data, untargeted by our calibration. The resulting share of port fees in total trade costs is 22.39%, and the share of sea trade in world trade is 44.75%.

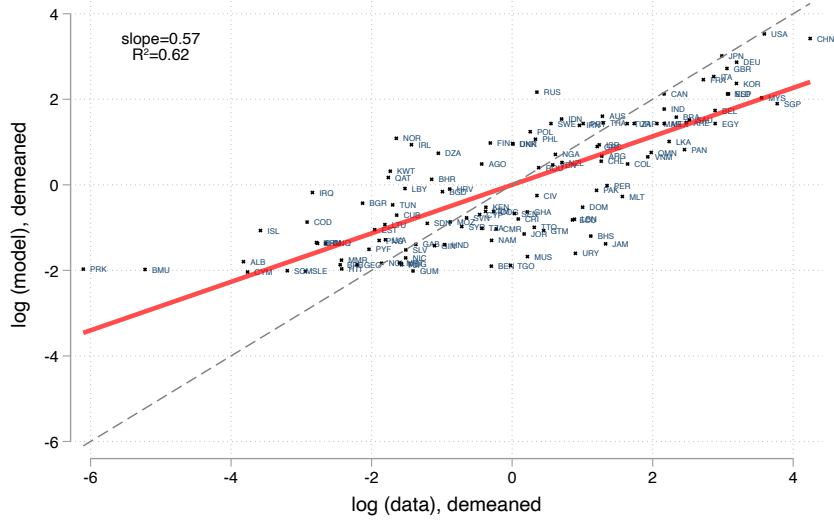
Table 2: Model Parameters

Parameter		Source	Value
Externally set			
Elasticity of production	α	66% labor share	0.33
CES elasticity	σ	Head and Mayer (2014)	3.00
Curvature of utility	ρ	IES of 0.5	2.00
Congestion elasticity	β_{land}	Couture et al. (2018)	0.13
	β_{sea}	Brancaccio et al. (2024)	0.79
Infrastructure elasticity	γ_{land}	Couture et al. (2018)	0.10
	γ_{sea}	Brancaccio et al. (2024)	0.31
Estimated			
Distance elasticity	ζ_{land}		1.99
	ζ_{sea}		0.24
Port coefficient	$\overline{T}_{land}$		6.22
	$\overline{T}_{sea}$		0.10
Transshipment cost	$\overline{\tau}_{TS}$		0

6.3 Measuring Investment Costs and the BRI Infrastructure Shock

To simulate the impact of BRI, we need to translate China's financing into real infrastructure changes at project sites. I.e., we need to measure the marginal cost $\kappa_{CH,s}$ that translates dollars invested into units of infrastructure I_s at node s of our model. As we have mentioned, in the pre-BRI equilibrium we proxy I_s with the stock of cranes observed in each sea node and with kilometers

Figure 6: Gross Flows, Model and Data (untargeted)



Notes: Gross flows compare model predictions for Q_s at country-level sea nodes against average capacity-adjusted vessels from MDST flowing through each country in our data for the years the MDST is available within that period. The calibration fits GDP and trade flows exactly, and does not target the capacity flows.

of rail and road in each land node. By using cranes and kilometers of roads, we can use externally estimated elasticities of transport costs to these measures of infrastructure.

Both sea and land infrastructure are multi-dimensional and depend on other complementary inputs (port of infrastructure also depends on berth length; land infrastructure also depends on rail stations), so we implicitly assume that the complementary investments shift in fixed proportions with the number of cranes and kilometers of roads, making these good proxies. Therefore, we need node-specific conversion rates from dollars invested by China to cranes or kilometers of road and rail resulting from these investments. This is challenging since we do not directly observe the marginal cost of infrastructure and because, in addition to funding dredging operations or cranes, financing may go towards funding non-infrastructure activity, such as trade facilitation, software systems, or training. For seaports, we rely on the reduced-form analysis from Section 3.4 to empirically map financing into realized infrastructure changes; see Appendix Section B.5 details the steps of this procedure. We obtain an average total cost of expanding a port by one crane (including the costs of all the complementary investments such as dredging or berth expansion for which the BRI loan is used) of \$371m.³⁶

For land and road investments, we do not have the proxies for changes in infrastructure over time as we do for seaports. However, for 289 of the total 350 land investments (spanning 91 countries), field notes in AidData contain information on the total kilometers of road or rail constructed. Since this is not available for all projects, we run an analogous regressions of total kilometers on

³⁶Brancaccio et al. (2024) estimates, within the U.S., a total cost of \$115m to expand total capacity of a port by one ship.

investment interacted with GDPPC on the projects for which we do observe this information. Our estimates imply an average 0.18 kilometers of road and 0.07 kilometers of rail per USD 1 million investment from China.³⁷ We can then impute the kilometers of road or rail built from China's investments in each country. Scaling this increase by the initial stock of road and rail kilometers yields the implied percentage increase in land infrastructure.

As a result of these steps, we obtain $\kappa_{CH,s}$ for each each land and sea node in the data, as well as a shock $\hat{I}_s \equiv \frac{\Delta I_s}{I_s^{pre=BRI}}$ in number of cranes or kilometers of road and rail due to BRI investments. Figure 7 plots these changes.

6.4 Welfare Impacts of the BRI

We begin by evaluating the cross-country effects of the BRI investments. Figure 8 reports the total welfare change in each country following the implementation of the BRI, expressed as a percentage of initial GDP. There is a strong positive relationship between the investment size (as share of initial GDP) or the percentage growth in infrastructure and the total welfare change. Sri Lanka is the largest beneficiary. Across regions, African countries experience the largest welfare gains; Sierra Leone, Togo, Burkina Faso, Mauritania, and Guinea in West Africa, and Ethiopia and East Africa (Ethiopia, Somalia). East Asian countries also benefit, particularly those in the Mekong region. At the same time, receiving substantial Chinese investment does not necessarily translate into welfare gains, reflecting the interaction of multiple offsetting forces. Countries such as Belarus, Sudan, and Cuba experience only gains despite large investments (as a share of their GDP). Furthermore, others may even face negative effects, with Panama the largest net loser.³⁸ Finally, most of China's major trading partners, including the United States, Western Europe, and Japan, appear to be only marginally affected by the BRI. China itself experiences a modest loss, suggestive that non-economic drivers may be important.

To better understand these effects, Figure 9 decomposes welfare changes across countries into four components. Starting from the definition of country level welfare, V_j in (33), and letting ΔX denote the difference in a variable between the calibrated pre-BRI equilibrium and the equilibrium with BRI investments, we can write:

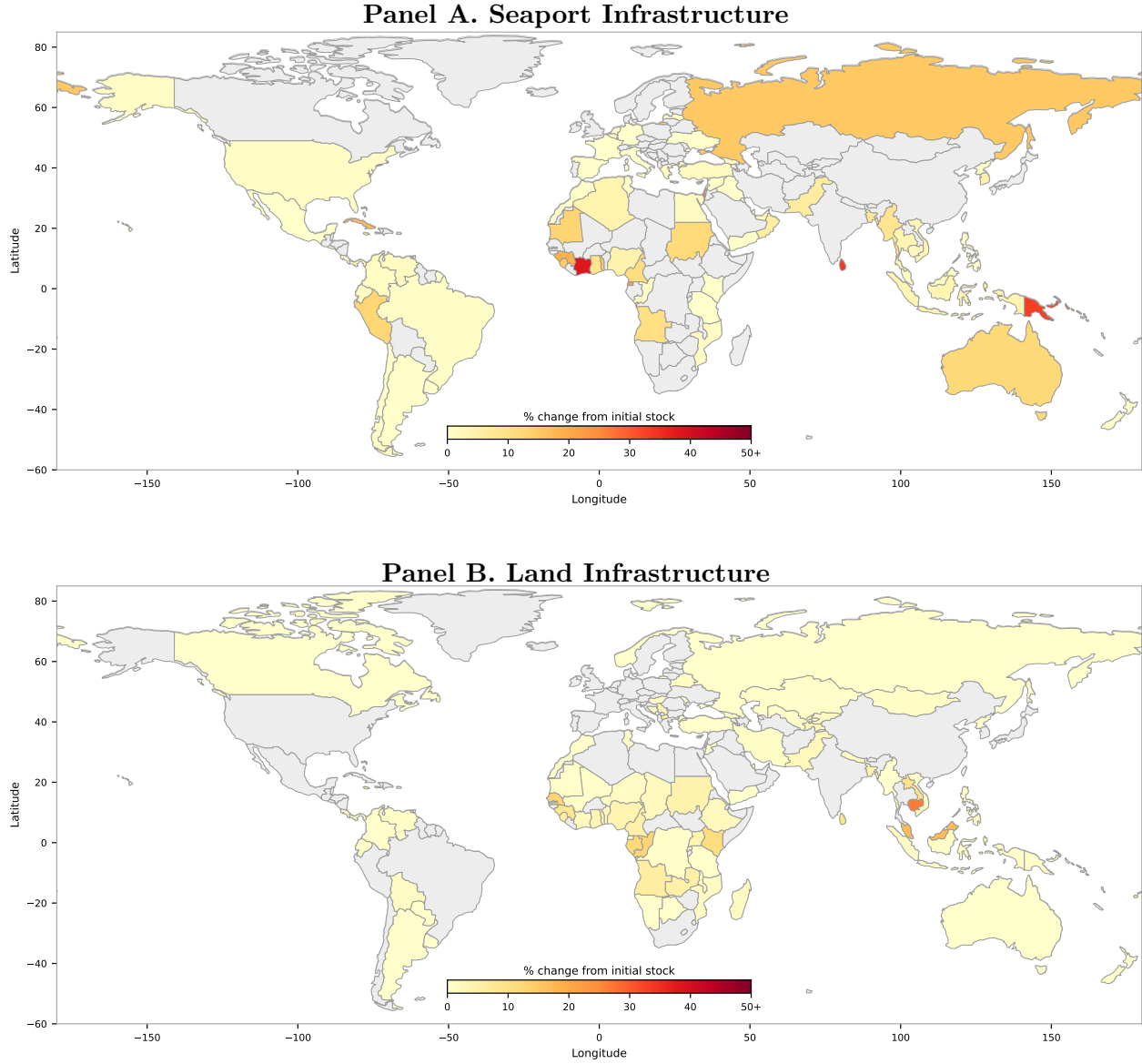
$$\Delta V_j = \underbrace{\Delta [U(C_j) - P_j^C C_j]}_{\text{consumer surplus}} + \underbrace{\Delta \Pi_{Y,j}}_{\text{prod profits}} + \underbrace{\Delta \sum_{s \in \mathcal{S}(j)} \pi_{Ts}}_{\text{port profits}} + \underbrace{\Delta \sum_{s \in \mathcal{S}} (r_s - \kappa_{js}) I_{js}}_{\text{infrastructure returns}}. \quad (58)$$

Panel (a) reports the distribution of consumer-surplus gains. As expected, all countries benefit, to

³⁷An alternative approach to measure the marginal cost of road construction is to use the World Bank's Road Costs Knowledge System (ROCKS) database. However, ROCKS does not necessarily reflect China's infrastructure costs, and we are unaware of analogous dataset for rail construction, so we opt to use information directly from the AIDData to recover the marginal cost of building land infrastructure.

³⁸A current limitation of our data is that the network used to measure global trade relies only of container shipments. However, some of the BRI investments, including in Russia, are geared towards gas terminals. Further including natural gas ships into the network may affect the trade reallocation and economics effects we measure in countries that are large traders of natural gas.

Figure 7: Implied Percentage Increase in Infrastructure from China's Investments

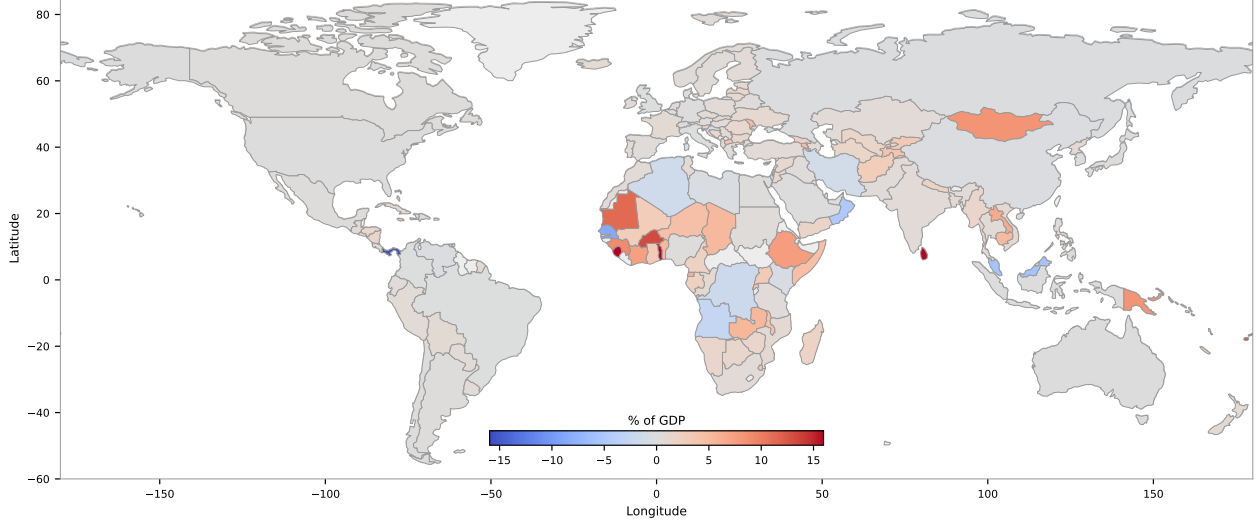


Notes: Maps plots the percentage increase from the initial stock of seaport and land infrastructure across countries due to China's investments. See Section 6.3 for details.

varying degrees, from improved access to cheaper foreign goods. Countries receiving the largest infrastructure investments relative to their initial infrastructure stock tend to experience the strongest gains.

Production profits, shown in panel (b), suggest the opposite, with most countries experience declines in production profits. This reflects changes in the composition of demand faced by domestic producers. On the one hand, domestic demand typically falls as expenditure is reallocated away from domestic goods toward imports. At the same time, foreign demand may increase as improved connectivity expands market access. Aside from Singapore and Malaysia in East Asia and Congo

Figure 8: Welfare Impact of Proposed BRI by Countries



Notes: Welfare impact $V_j(\{I_{ks}\}_{ks}, \{I_{CH,s}\}_{BRI}) - V_j(\{I_{ks}\}_{ks}, 0)$ reported as a percentage of the host country's GDP.

and Equatorial Guinea in Central Africa, the net effect is negative for countries.

Port profits, presented in panel (c), also exhibit substantial heterogeneity with no systematic relationship with the change in infrastructure from BRI investments. In the model, infrastructure improvements generate two offsetting forces. First, lower transportation costs reduce shipping fees and may depress revenues from shipping services, as in the case of Panama, Malta or Oman. Second, improved connectivity can redirect gross trade flows toward certain ports, increasing shipping activity and profits in countries such as Sri Lanka, Ivory Coast, and Ghana. Countries excluded from the BRI may also be adversely affected, such as Croatia and the Dominican Republic, or positively affected, like Finland and India who experience an increase in port profits as trade flows are diverted toward better-connected infrastructure hubs.

Finally, panel (d) reports changes in infrastructure returns. These are negative for many countries because Chinese investment increases the global supply of infrastructure capital and thereby dilutes returns to pre-BRI investments.

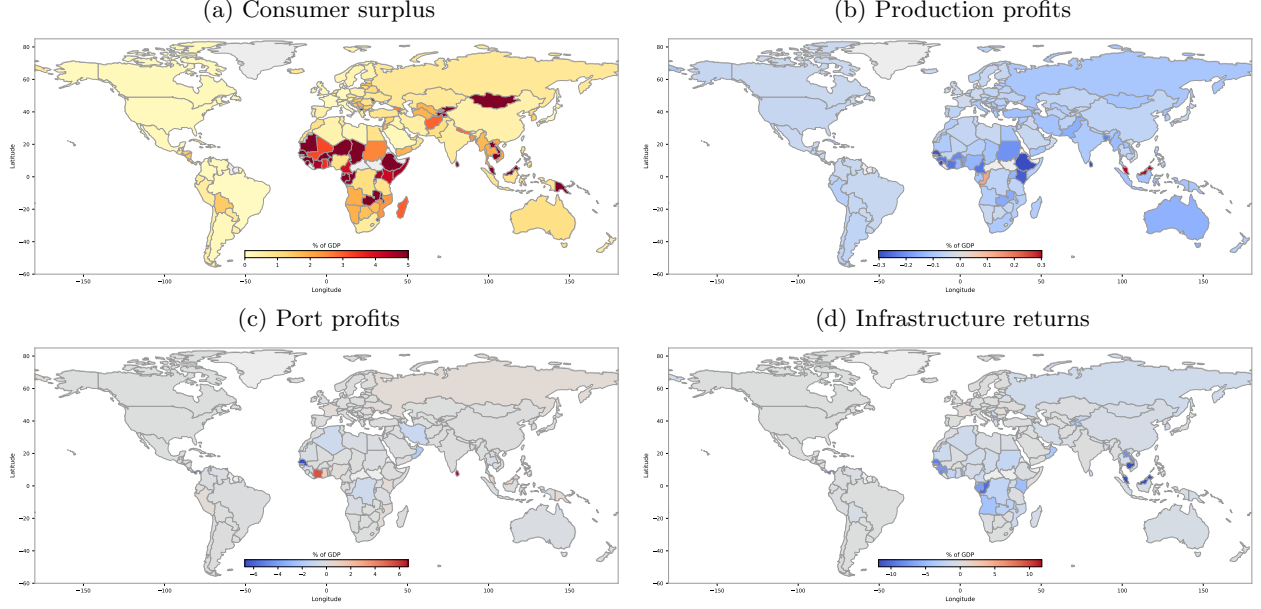
6.5 Rationalizing Observed Investments with Wedges

The BRI investments we have just described are not necessarily consistent with the optimization problem of a Chinese planner that maximizes V_{CH} defined in (33). If that was the case, then we should observe

$$\left. \frac{dV_{CH}}{dI_s} \right|_{post-BRI} = 0 \quad (59)$$

around the post-BRI investment equilibrium we have just simulated, for dV_j defined in (33). Such a condition would imply that the BRI investments are consistent with a Chinese planner concerned

Figure 9: Welfare Decomposition of Proposed BRI by Countries



Notes: The results are reported in % of the host country's GDP.

with economic benefits as specified in our model only. However, this condition is generally violated, suggesting that Chinese planners have other motivations besides maximizing purely economic.

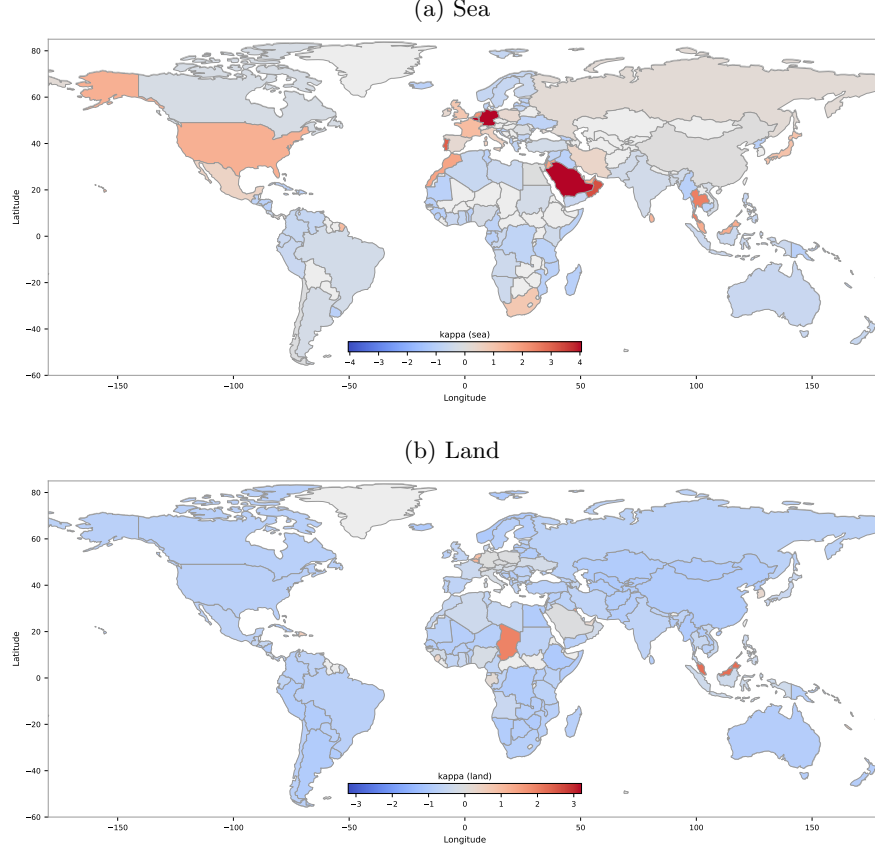
Therefore, to rationalize the observed BRI investments, we measure the “geopolitical” cost used by Chinese planners, $\kappa_{CH,s}^{GEO}$, such that, had Chinese planners used $\kappa_{CH,s}^{GEO}$ instead of the “true” economic costs $\kappa_{CH,s}^{ECO}$ we have measured in Section 6.3, condition (48) and (59) would have been satisfied. Values of the implied wedge $\kappa_{CH,s}^{GEO}$ above the benchmark $\kappa_{CH,s}^{ECO}$ suggest that large economic gains from investing in a given location, and therefore requires perceived investment costs by Chinese authorities to be higher than economic estimates to rationalize the observed allocation. Conversely, wedge values below $\kappa_{CH,s}^{ECO}$ suggest additional benefits associated with these investments that are not captured by the model. Such perceived costs or benefits may reflect several factors, including model misspecification, geopolitical objectives (for example, support for strategic partners), military considerations, transfers embedded in broader bilateral agreements, or other omitted forces such as default risk.

The estimated $\kappa_{CH,s}^{GEO}$, normalized to $\kappa_{CH,s}^{ECO}$, are reported in Figure 10. Land-investment wedges are generally low and relatively homogeneous across countries, suggesting a broad tendency towards over-investment in land infrastructure. By contrast, sea-investment wedges display substantial heterogeneity. In particular, China's major trading partners—including the United States, Europe, Japan—are associated with large wedges. According to the model, investments in these locations would generate sizable economic welfare gains, yet they have received little or no BRI investment, implying that China perceives additional costs associated with such projects. In contrast, many major BRI recipients in Africa and Central Asia are characterized by low wedges, consistent with

the presence of additional non-economic benefits attached to these investments.³⁹

Figure 11 correlates the implied $\kappa_{CH,s}^{GEO}$ for land and sea with countries' ideal point, again noting that higher values are indicative of support for U.S.-led international order and negative values opposing it. We observe a positive correlation between the $\kappa_{CH,sea}^{geo}$ and the ideal point, which is suggestive that China assigns a large cost to investing in US-aligned countries. The $\kappa_{CH,land}^{geo}$, on the other hand, do not exhibit meaningful variation across this dimension of geopolitics.

Figure 10: Wedges κ^{geo}



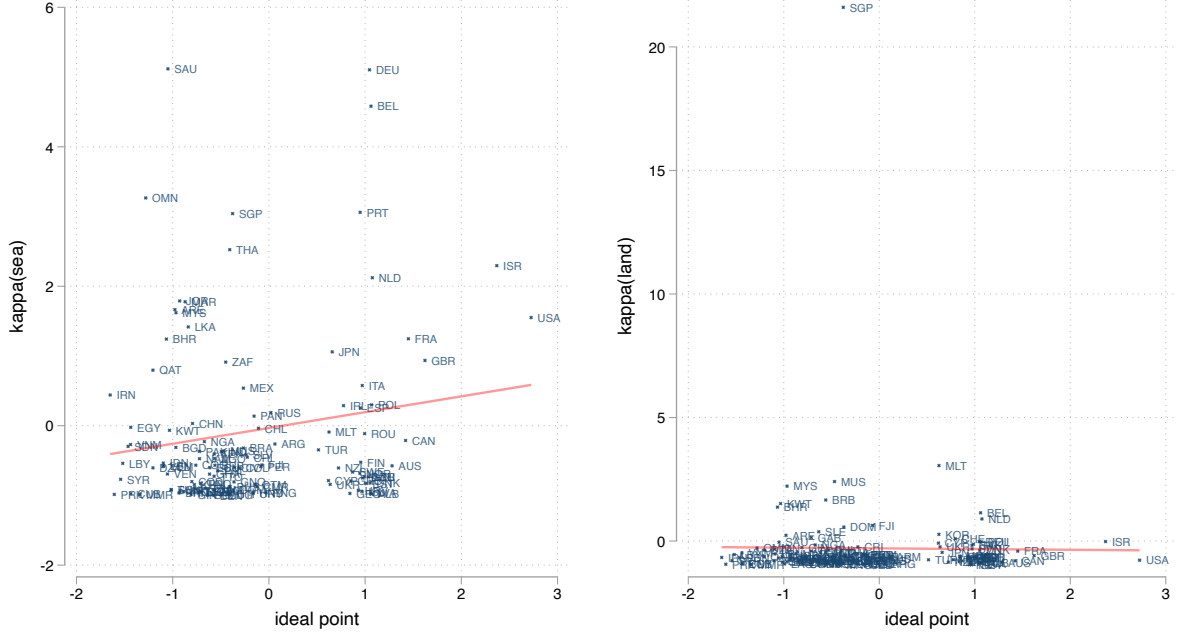
Notes: Wedges $\kappa_{CH,s}^{GEO}$ are reported relative to $\kappa_{CH,s}^{ECO}$.

6.6 Welfare Gradient and Optimal Network in the Absence of Political Preferences

Absent geopolitical motives, how should a Chinese planner optimally allocate foreign infrastructure investments? Figure 12 reports the welfare gradient $dV_{CH}/dI_{CH,s}$ associated with investments in each country's sea node (panel a) and land node (panel b), evaluated at the calibrated pre-BRI economy and setting the investment costs to 0 (so as to visualize only the gross benefit of the in-

³⁹The overall scale of these wedges is impacted by port-level shifters to returns to infrastructure. Our work in progress aims to control the overall level of these returns bringing in information about BRI loan rates to the calibration.

Figure 11: Wedges $\kappa_{CH,s}^{GEO}$ vs. Ideal Point



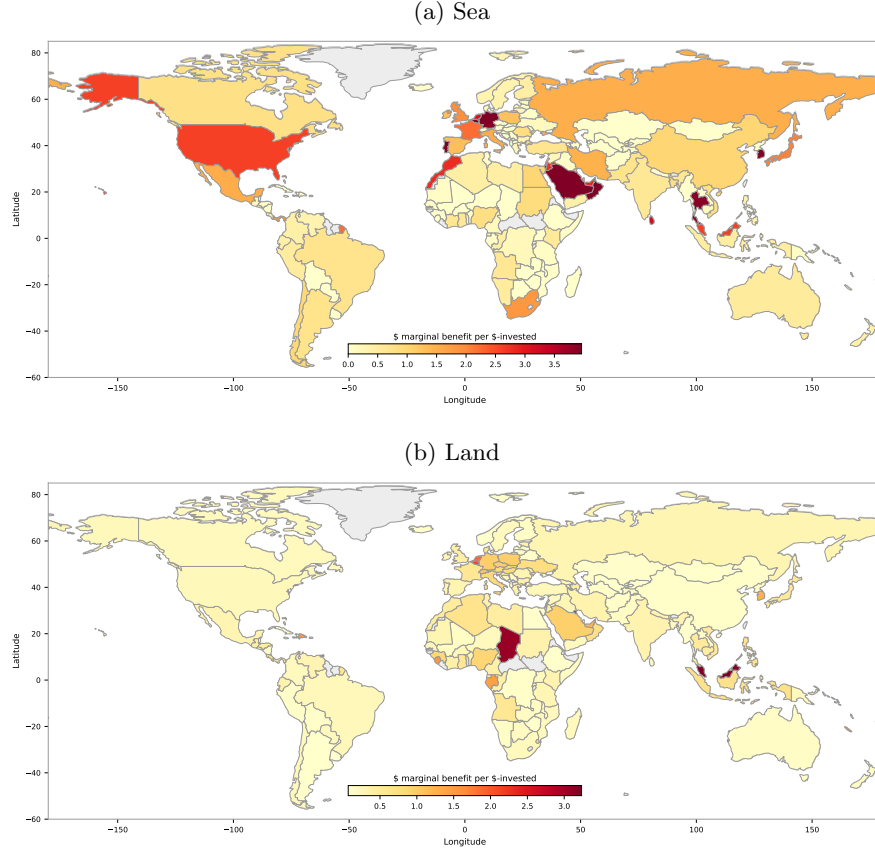
Notes: Wedges $\kappa_{CH,s}^{GEO}$ are reported relative to $\kappa_{CH,s}^{ECO}$.

vestment). Several large recipients of BRI investments do appear attractive on the map of seaports (Israel, Sri Lanka). However, our model predicts large welfare gains in several countries with little or no investment: Saudi Arabia, Germany, Belgium, Japan, and the U.S. Our model also predicts much larger gains from investing in maritime than in land infrastructure.

We examine these patterns more systematically in Figure 13. Countries that are important destinations for Chinese exports (panel a) have the highest welfare gradients while the relationships are flatter vis-a-vis imports and gross exports.

To conclude, we compare the proposed BRI allocation with the investment network a Chinese planner without geopolitical preferences (using $\kappa_{CH,s}^{ECO}$ rather than $\kappa_{CH,s}^{GEO}$) would optimally choose, assuming that the remaining countries remain at the pre-BRI equilibrium (Appendix B.3 discusses details of the optimization routine). The comparison reveals a markedly different allocation pattern. A purely economic maximizing planner would direct a larger share of investments toward China's most advanced trading partners. In contrast, developing countries that are major recipients of the current BRI—particularly West Africa and Latin America—would receive substantially less investment.

Figure 12: China's Infrastructure Gradient in Calibrated Economy



Notes: This figure represents $dV^{CH}/dI_{CH,s}$ gross of the investment cost at infrastructure autarky in dollars per invested dollar.

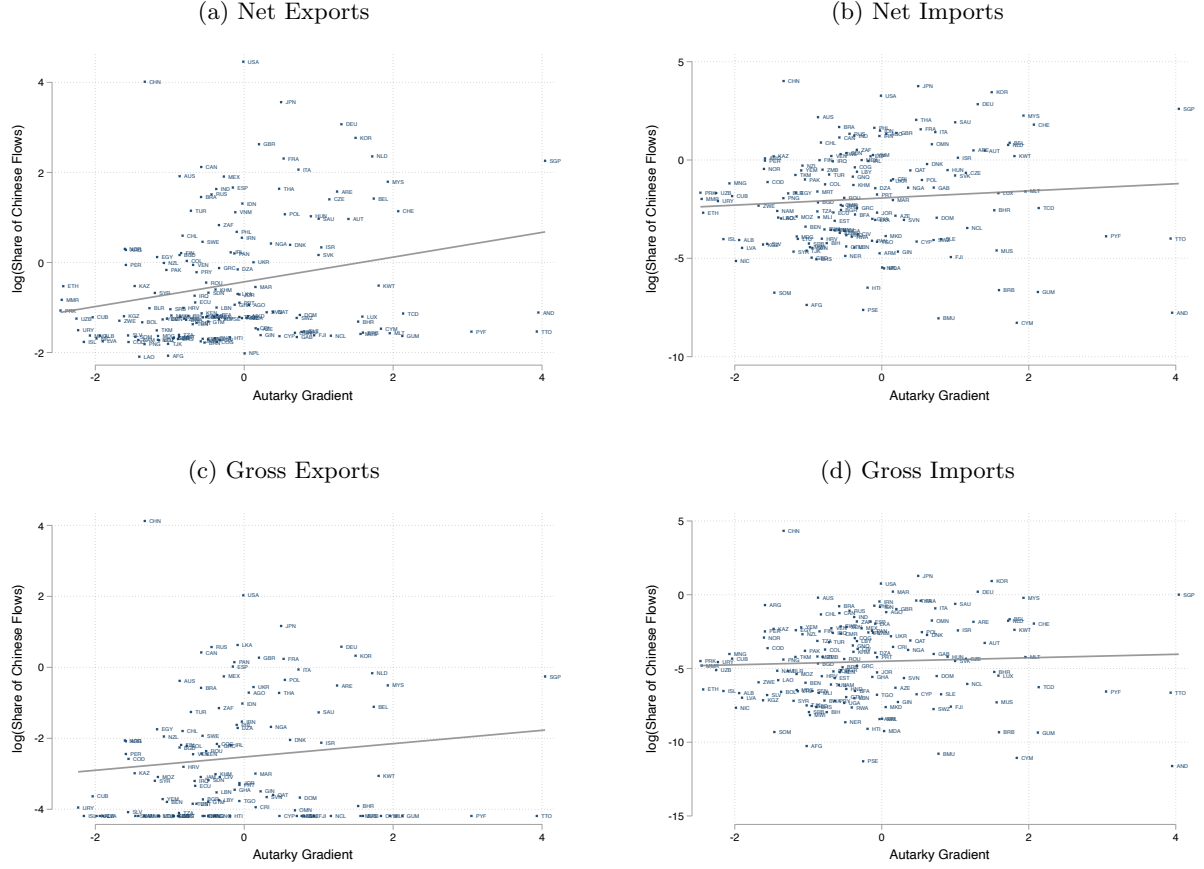
7 Conclusion

In this project we are studying China's Belt and Road Initiative (BRI). We combine several datasets, including geocoded investments from different major investors, detailed information on port infrastructure, and port-to-port shipment. Using these data, we establish that China's investments seem to respond to both economic and non-economic factors in ways that seem specific to China, and that these investments materially affect port infrastructure and trade.

We use a standard two-country trade model with an outside good to theoretically study the strategic incentives of countries to invest in trade infrastructure. We establish conditions under which a foreign country gains from investing in infrastructure, and we show that in equilibrium these investments are likely to arise due to the price movements implied by infrastructure investments.

We then develop a multi-country trade model with a transport network over different modes that incorporates congestion by node. Our quantification, as well as other features highlighted through the draft are still in progress. Among other things, we are working on bringing additional data to the analysis (such as current information on cranes and berths, which we lack), refining the network (including air transport, a more granular geography, and non-container shipments),

Figure 13: Determinants of the Infrastructure Gradient



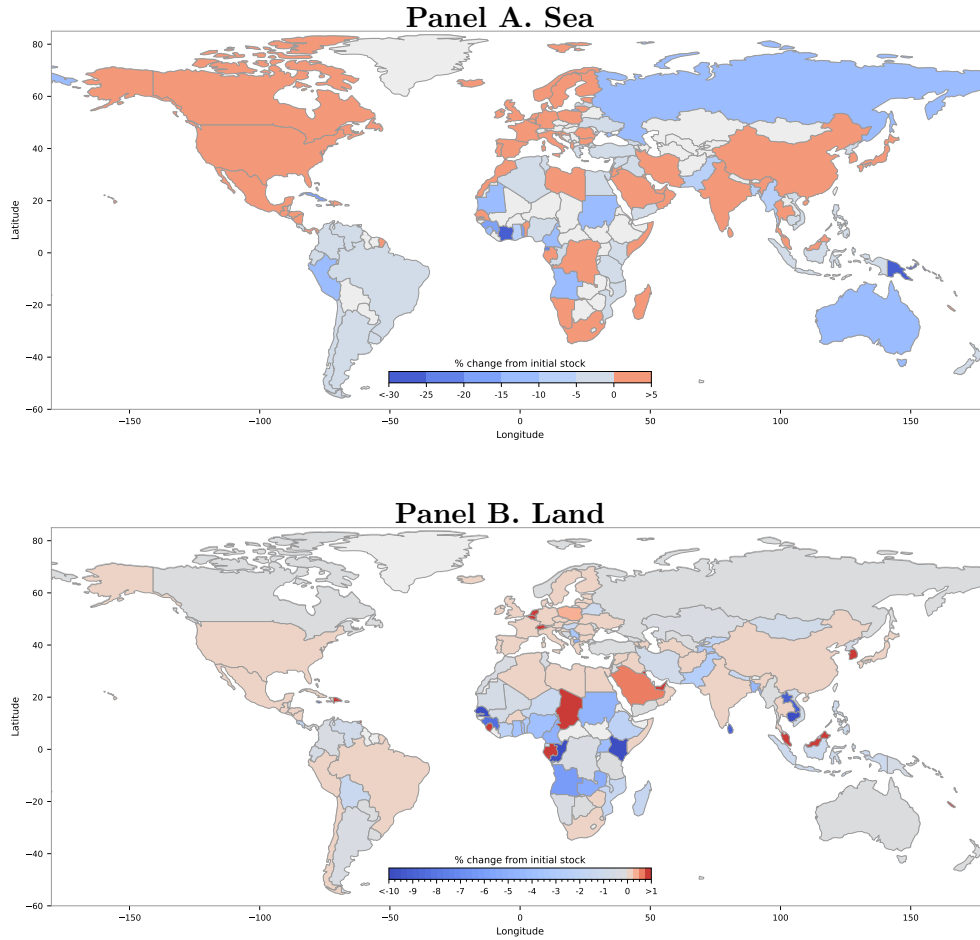
Notes: Net export are the FOB value of Chinese exports to each country, net import the FOB value of goods exported by each country to China. Gross exports represent the total FOB value $\sum_{s \in S(j)} p_{jCH}^{jCH} (Q_{ss'}^{jCH} + Q_{s's}^{jCH})$ of Chinese goods transiting through country j . Gross imports represent the total FOB value of foreign goods eventually consumed by China transiting through a country (attributed as pro-rata in each node).

incorporating additional moments to our calibration (such as the observed loan rates charged by China), refining model features (such as adding intermediate inputs or a more flexible demand system), and implementing additional counterfactuals such as assessing the optimal responses of the U.S. and other major organizations to the Chinese investments.

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Figure 14: Difference between Non-Geopolitically Motivated Optimum and Proposed BRI



Notes: The map shows the difference between the non-geopolitically motivated optimum and the actual.

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A Data Appendix

For all investment data sources, we employ the following set of common procedures to build the dataset of locode-level investments that have been implemented (but not necessarily completed):

- We assign each project a UN/LOCODE by matching its name, description, and geographic coordinates (where available) to the relevant code in the UN/LOCODE list.⁴⁰ For seaport investments made at terminals within a port, we assign the port-level locode after matching the terminal name to its Global Integrated Shipping Information System (GISIS) code.
- For each project, AIDData provides a year of commitment, implementation, and completion. The implementation year is assigned to the year the actual work on the infrastructure project began (or best estimate). The completion year is assigned to the year when the infrastructure project was completed and began operation (or best estimate). Additionally, AIDData also provides a status flag indicating if a project was cancelled or suspended or in the pipeline. For our analysis, we restrict the analysis to projects with with an implementation year and not cancelled/suspended. Details regarding the particulars of the cleaning procedure for individual dataset are discussed below.
- Investment value are total over the lifetime of projects and converted to constant 2023 USD values.

⁴⁰<https://unece.org/trade/cefact/unlocode-code-list-country-and-territory>.

A.1 AIDdata

LOCODE assignment AIDdata includes an infrastructure flag and identifies projects in the "transport and storage" sector, but we have found that this classification may miss infrastructure investments classified under other sector names. For example, the Silk Road Fund's investment in the construction of a greenfield seaport facility in Sabetta, Russia is classified as "industry, mining, and construction" due to its affiliation with the Yamal Liquefied Natural Gas (LNG) Project. We, therefore, supplement these categories with the text of the project descriptions to identify transportation infrastructure investments in seaports, airports, roads, and railroads.

We generate candidate loans by systematically seeking out infrastructure related words in the loan descriptions. Seaport indicators are based on "port", "seaport", "ports", "harbor", "dockyard", "crane(s)", "terminal", "LNG", "liquefied", "onshore", "canal", "wharf", "dredging", "bulk", and "container". Airport indicators are based on "airport", "airstrip", "airfield", "hangar". Rail indicators are based on "railroad", "railway", "railport", "dry port". Road indicators are based on "road", "expressway", "interchange", "bridge", "bypass", "tunnel". Ancillary port/airport terms include "scanners" and "scanning".

As a validation of our candidates, we merge our seaport candidates with AidData's "Chinese-Financed Ports Overseas and Related Terminals (CPORTS) Dataset, Version 2.0, 2000-2025" and confirm that AIDdata's classification of seaport investments represents a subset of our assignment. We manually exclude candidate projects that do not involve actual trade transportation infrastructure and classify the remaining projects as genuine investments in the respective type of infrastructure. For example, we exclude all of the syndicated loans provided for the Royal Albert Dock in the UK because they were funding a residential construction project at an old dockyard in London. We also drop projects like the the \$3.23 billion loan for the Mexico City–Querétaro High-Speed Railway because the infrastructure was for passenger transportation not freight transportation.

AIDdata includes regional or cross-border projects that we re-assign to individual countries based on the most prominent country within the AIDdata description. Examples include TAZARA-related projects assigned to Zambia or Tanzania, a Central Asia-China gas pipeline project assigned to Kazakhstan, a transcontinental railroad feasibility study assigned to Brazil, and selected acquisition projects reassigned to South Korea or Colombia. After these manual adjustments, we drop projects that cannot be assigned to a single country.

Road and rail investments span multiple locodes (e.g., an origin and a terminal point). At present, we have assigned a single locode to these investments, but the next iteration of this project anticipates assigning the full string of locodes to each land investment.

Projects Years As mentioned above, AIDdata provides three year variables which we will use in selecting our final set of investment projects. However, there are instances where the status variable and the years are inconsistent or missing. We implement the following adjustments. First, if the implementation year is missing but the completion year is not missing then we reassign the project as complete and assign the implementation year to the commitment year. Second, if both

implementation year and completion year are non-missing but implementation year is later than the completion year then we assign the implementation year to the completion year. Similarly, if the commitment year is later than the implementation year or the completion year, we assign the commitment year to the implementation or completion year, respectively.

Investment Values For a subset of transportation infrastructure projects, AIDData does not report a nominal value for the loan. We manually correct for these cases based on the a combination of AidData descriptions or external web/document sources.

We use AIDdata’s constant 2023 USD value of the loan as our preferred value for the loan. If the constant value is missing but nominal value non-missing, we apply a deflator from FRED to convert it to 2023 constant USD.

Contractual Terms AIDData provides some information on ownership based on the Ultimate Beneficial Owner (defined as a shareholding that exceeds 25%) of the loan’s Direct Receiving Agency (DRA). Ownership shares are collapsed by AidData record, activity country, and parent-owner nationality. Ownership percentage is constructed as the parent ownership percentage divided by the total DRA count affiliate with the project. We create three mutually exclusive ownership components: Chinese ownership, host ownership for owners from the activity country, and other ownership for owners that are neither Chinese nor host-country.

Infrastructure Details AIDdata also includes descriptive information on the infrastructure built as part of the loan. Of particular interest for our purpose is the total route length for road and rail projects. We extract this information as first remaining numeric distance attached to km, kilometer, or kilometer within the description of a road or rail project.

A.2 Council on Foreign Relations

In order to supplement the debt-based investments from AIDdata, we further employ data from the Council on Foreign Relations (CFR) covering China’s investments in seaports and airports. This data uses a broader definition of Chinese involvement including debt, equity investments by Chinese companies, as well as projects where Chinese companies were purely contractors for the project. We exclude all projects with debt components to maintain a mutual exclusivity between CFR and AIDdata.

LOCODE assignment Since the CFR datasets only include transportation infrastructure projects, we only need assign UN locodes to the respective projects. We therefore locate the seaport/airport information from the name and description fields then utilized our general procedure above to make the assignment.

Projects Years We generate an equivalent of AIDdata’s status variable by mapping the CFR’s status, construction timing, and cooperation mode variables into AIDdata’s status categories. We further map the construction dates in CFR to the commitment, implementation, and completion years framework from AIDdata. We assign the implementation year to the construction start year and the completion year to the construction end date. The commitment year is set to the implementation year except for a handful of projects whose commitment years can be validated from their descriptions.

Investment Values All investment values in CFR are recorded in nominal USD. However, because CFR records the equity acquisition of port terminals by Chinese companies, some of the values require manual correction. For example, China Ocean Shipping Company’s acquisition of Terminal Link involved the acquisition of terminals at several ports around Europe but CFR only assigns the value of the investment to the port of Marseille. In that case, we proportionally split the total value across the different ports involved for this type of Chinese investment activity. Following the procedure in AIDdata, we deflate to 2023 constant USD.

Contractual Terms CFR includes measures of Chinese ownership in their projects. We manually adjust all projects with the lease agreements to 100% ownership reflect the fact that these agreements allow China to collect all of the rents from the ports for the duration of the lease. In order to ensure mutual exclusivity with AIDdata, we drop cases where China is treated as only a contractor and limit the CFR projects in our analysis by dropping all turnkey investments, all greenfield investments in which China is only a contractor, and all leases in which China provides financial support through aid construction.

A.3 International Finance Corporation

The International Finance Corporation (IFC) is the private-sector arm of the World Bank Group. The IFC reports publicly data on all its projects going back to its founding.

LOCODE assignment To separate trade transportation infrastructure projects, we employ the same process of constructing candidate projects using relevant terminology in the project descriptions from AIDdata. We then perform the same manual checks to exclude unreasonable candidates and utilize our general procedure above to make the Locode assignment.

Projects Years The IFC project dataset only includes implemented/pipeline projects and does not include completion dates. As such, we assign the implementation year to the IFC investment date and the commitment year to the IFC approval/signed date. We compile investments from 2000-25.

Investment Values All investment values for the IFC are recorded in nominal terms. We follow the same procedure as employed for the nominal values in AIDdata to deflate these investment values to 2023 constant USD.

A.4 World Bank

We obtain project investments from World Bank Group: the International Bank for Reconstruction and Development (IBRD) and the International Development Association (IDA). The IBRD only finances sovereign governments directly, or projects backed by sovereign governments. The IDA complements the IBRD by lending to the world’s poorest countries who cannot afford the lending terms offered by the IBRD. Project data from these organizations is publicly available and goes back to the founding of the World Bank.

LOCODE assignment To separate trade transportation infrastructure projects, we employ the same process of constructing candidate projects using relevant terminology in the project descriptions from AIDdata. We then perform the same manual checks to exclude unreasonable candidates and utilize our general procedure above to make the locode assignment.

Projects Years To align with AIDdata’s timing variables, we further extract the completion year from the closing date, the implementation year from loan effective date, and the commitment year from the board approval date. If commitment year is missing, we use the disclosure date as an alternative. We further reclassify closed projects as complete when a closing date exists, active or closed-without-closing-date projects as implemented, pipeline as pipeline, and dropped projects as suspended or cancelled depending on whether the last stage reached was Bank Approved. We compile investments from 2000-25.

Investment Values All investment values for the IBRD/IDA projects are recorded in nominal terms. We follow the same procedure as employed for the nominal values in AIDdata to deflate to 2023 constant USD.

A.5 Japan International Cooperation Agency

The Japan International Cooperation Agency (JICA) is a governmental agency that delivers the bulk of Official Development Assistance (ODA) for Japan. JICA maintains a public project database going back to its founding in the 1950s.

LOCODE assignment To separate trade transportation infrastructure projects, we employ the same process of constructing candidate projects using relevant terminology in the project descriptions. We then perform the same manual checks to exclude unreasonable candidates and utilize our general procedure above to make the locode assignment.

Projects Years Like the IFC, JICA does not record completion dates. We assign the implementation year to the date of loan approval since this is the only date available to us in the project data. We compile investments from 2000-25.

Investment Values JICA separates its loans denominated in Yen from those denominated in USD. For the subset of loans denominated in Yen, we first convert JPY to USD using FRED’s nominal exchange rate from the respective year. We then follow the same procedure as employed for the nominal values in AIDdata to deflate these investment values to 2023 constant USD.

A.6 U.S. International Development Finance Corporation

The U.S. International Development Finance Corporation (DFC), and its predecessor the Overseas Private Investment Corporation (OPIC) is the U.S. government’s development finance institution that providing financing to overseas private-sector projects. While the DFC maintains a database of more recent projects, we rely on the Political Risk and Finance Dataset from [Arel-Bundock et al. \(2020\)](#) for historical OPIC project data.

LOCODE assignment To separate trade transportation infrastructure projects, we employ the same process of constructing candidate projects using relevant terminology in the project descriptions. While the DFC projects include the relevant fields, the OPIC data is more limited so we lean more heavily on the project name and sector categories to recover our candidates. We then perform the same manual checks to exclude unreasonable candidates and utilize our general procedure above to make the Locode assignment.

Projects Years Like IFC and JICA, this data does not include completion years so we set the implementation years to the date at which the loan was made. We compile DFC (or its predecessor Overseas Private Investment Corporation) from 2000-25.

Investment Values All investment values for the DFC/OPIC projects are recorded in nominal USD terms. We follow the same procedure as employed for the nominal values in AIDdata to deflate these investment values to 2023 constant USD.

A Appendix to Section 4

A.1 Effects of Domestic Taxes or Public Investment on Equilibrium Infrastructure

Consider an equilibrium $\{Q^*, r^*, \tau^*, p_M^*, p_X^*, I_{P,M}^*, I_{P,X}^*\}$ consistent with (17) and (18) given policies. We partition possible equilibria in three cases: (i) no private sector entry in ($I_{P,j}^* = I_{P,k}^* = 0$), (ii) country j entry ($I_{P,j}^* > 0$), and (iii) country k entry without country j entry ($I_{P,k}^* > 0 = I_{P,j}^*$). First, suppose we start without private investors in both countries. Then

$r^* - t_{P,j} - (1 - \chi_j)t_I < \kappa_j$ for $j = X, M$. Given any marginal shock, $I_{P,j} = dI_{P,j} = 0$ and public entry has a first order impact: $dI_{G,j} = dI_j = dI^*$. Second, assume country j investors are initial present. Then, $r^* - t_{P,j} - (1 - \chi_j)t_I = \kappa_j$, so $dt_{P,j} = dr^*$ and $dI_{P,j}^* = dI_j = dI^*$. From (17) and (13), $dr^* = -T_{IQ}dQ^* - T_{II}dI^* = dt_{P,j}$. Any desired dI_j can therefore be implemented with the change in tax $dt_{P,j} = \left(-T_{IQ}\frac{dQ^*}{dI^*} - T_{II}\right)dI_j$ (alternatively, the public sector in j can invest to crowd out the foreign private sector, $\Delta I_{G,j} = I_{P,k}^*$; around that point, further investments raise aggregate I). Third, if there is country k entry without country j entry, the initial equilibrium features $r^* - t_{P,k} - (1 - \chi_k)t_I = \kappa_k < \kappa_j = r^* - t_{P,j} - (1 - \chi_j)t_I$. Country j can offer a tax $t_{P,j} = \kappa_k - \kappa_j + t_{P,k} + (\chi_j - \chi_k)t_I$, at which point private firms in country j will be indifferent about entry, $r^* - t_{P,j} - (1 - \chi_j)t_I = \kappa_j$. Around that point, a reduction in the tax (a larger subsidy if negative) leads to country j private firms crowding out country k private firms with $dr^* = 0$. When foreign private firms are fully crowded out, further reductions in $t_{P,j}$ lead to domestic entry (where we use that $\frac{dr^*}{dI^*} < 0$ as shown in Part (ii) of Proposition 1). Therefore, around any initial equilibrium, and therefore any set of parameters and country k policies, there are country j policies such that $dI_j = dI^*$. Country j cannot implement any I^* below a level $\underline{I_{P,k}}$ that would prompt empty from country k private firms. Therefore, country j can only implement $I^* > \min \left\{ \underline{I_{P,k}}, I_{G,k} \right\}$.

A.2 Pareto Efficient Allocation

The Pareto frontier maximizes the utility of the exporting country subject to the feasibility constraint for each level of utility $\overline{V_M}$ of the importing country:

$$\max_{Q, I_M, I_X, C_M^O, C_X^O} V_X = C_X^O \quad (\text{A.1})$$

subject to

$$C_M^O + C_X^O + \Psi(Q) + T(Q, I) + \kappa_M I_M + \kappa_X I_X + \bar{\tau}Q \leq \overline{O_M} + \overline{O_X}. \quad (\text{A.2})$$

and

$$U(Q) + C_M^O \geq \overline{V_M}. \quad (\text{A.3})$$

The feasibility constraint says that the world endowment of outside goods must be used either for consumption, as an input to produce Q , as an input into port operations, to produce infrastructure or as an input into other variable costs. Because of the outside good, there is transferrable utility. Therefore, the problem can be written as maximizing the utility net of costs:

$$\max_{Q, I_M, I_X, C_M^O, C_X^O} V_X = \overline{O_M} + \overline{O_X} - \overline{V_M} + U(Q) - \Psi(Q) - T(Q, I_M + I_X) - \kappa_M I_M - \kappa_X I_X - \bar{\tau}Q. \quad (\text{A.4})$$

The planner's first order conditions are

$$U'(Q^{**}) = \Psi'(Q^{**}) + T_Q(Q^{**}, I^{**}) + \bar{\tau} \quad (\text{A.5})$$

and

$$-T_I(Q^{**}, I_X^{**} + I_M^{**}) \leq \kappa_j \text{ for } j = X, M. \quad (\text{A.6})$$

This condition are consistent with the market equilibrium 17 and 18 when $t_{P,j} = t_{Q,j} = 0$ and governments jointly invest at any level below the globally efficient allocation, $I_{G,X} + I_{G,M} \leq I_X^{**} + I_M^{**}$.

A.3 Proof of Proposition 1

Part (i) Totally differentiating (17) and solving for $\frac{dQ^*}{dI^*}$ we have:

$$\frac{dQ^*}{dI^*} = \frac{T_{QI}}{U_{QQ} - \Psi_{QQ} - T_{QQ}} > 0, \quad (\text{A.7})$$

and totally differentiating (12) and using (A.7):

$$\frac{d\tau^*}{dI^*} = \frac{U_{QQ} - \Psi_{QQ}}{U_{QQ} - \Psi_{QQ} - T_{QQ}} T_{QI} < 0$$

where to determine the signs we have used $T_{QI} < 0$, $U_{QQ} < 0$, $\Psi_{QQ} > 0$ and $T_{QQ} > 0$. Therefore, $\frac{dp_M}{dI} = U''(Q) \frac{dQ}{dI} < 0$ and $\frac{dp_X}{dI} = \Psi''(Q) \frac{dQ}{dI} > 0$. Totally differentiating (13) and using (A.7) we obtain:

$$\frac{dr^*}{dI^*} = -\frac{(T_{QI})^2}{U_{QQ} - \Psi_{QQ} - T_{QQ}} - T_{II} = -\frac{(T_{QI})^2 - T_{II}T_{QQ} + T_{II}(U_{QQ} - \Psi_{QQ})}{U_{QQ} - \Psi_{QQ} - T_{QQ}} < 0$$

where to determine the sign we have also used $T_{II} > 0$ and convexity of T , which implies $(T_{QI})^2 < T_{II}T_{QQ}$.

Part (ii) Totally differentiating (4) and (5) and using 6 and 7,

$$dV_j = (1_{j=X} - 1_{j=M}) Q^* dp_j^* + d\mathcal{T}_j. \quad (\text{A.8})$$

where $1_{j=J}$ equals 1 if $j = J$ and zero otherwise for $J = X, M$. Except in proposition Prop 3, we assume $dt_I = 0$. Consider next the variations $dI^* = dI_j$ or $dt_{Q,j}$. Totally differentiating (16) and using (12) and (13):

$$d\mathcal{T}_j = \chi_j \underbrace{(Q^* d\tau^* - I^* dr^*)}_{d\Pi_T} + I_j dr^* + (r^* - \kappa_j) dI_j + d(t_{Q,j} Q^*) - (1 - \chi_j) t_I dI_j \quad (\text{A.9})$$

We can re-write (A.9) as follows:

$$d\mathcal{T}_j = \chi_j (Q^* d\tau^* - I_j dr^*) + (1 - \chi_j) (I_j dr^* - t_I dI_j) + (r^* - \kappa_j) dI_j + d(t_{Q,j} Q^*) \quad (\text{A.10})$$

where either $dr = 0$ and $r = \min \kappa_j$ (if $\sum_j I_{P,j}^* > 0$ in initial equilibrium) or $dr < 0$ and $r < \min \kappa_j$ (if $\sum_j I_{P,j}^* = 0$). Combining (A.10) with (A.8) we obtain

$$dV_j = (1_{j=X} dp_X^* - 1_{j=M} dp_M^*) Q^* + \chi_j (Q^* d\tau^* - I_{-j} dr^*) + (1 - \chi_j) (I_j dr^* - t_I dI_j) + (r^* - \kappa_j) dI_j + d(t_{Q,j} Q^*). \quad (\text{A.11})$$

When $dI^* = dI_j$ and $dt_{Q,j} = 0$, using that in that case $dp_X > 0 > dp_M$ then (A.11) becomes:

$$dV_j = (r^* - \kappa_j) dI_j + Q^* |dp_j^*| + \chi_j (Q^* d\tau^* - I_{-j} dr^*) + (1 - \chi_j) (I_j dr^* - t_I dI_j) + t_{Q,j} dQ^*, \quad (\text{A.12})$$

which using $\chi_f = 0$ gives (19). Further letting h be the host country and f be the foreign country and using the fact that $|dp_h^*| + d\tau^* = -|dp_f^*|$ we obtain (20).

Part (iii) When both $dI_j = dI^* \neq 0$ and $dt_{Q,j} \neq 0$ then for the foreign country (A.11) implies

$$dV_f = (1_{f=X} dp_X^* - 1_{f=M} dp_M^*) Q^* + (I_f dr^* - t_I dI_f) + (r^* - \kappa_f) dI_f + d(t_{Q,f} Q^*), \quad (\text{A.13})$$

and for the host country

$$dV_h = (1_{h=X} dp_X^* - 1_{h=M} dp_M^*) Q^* + (Q^* d\tau^* - I_f dr^*) + (r^* - \kappa_h) dI_h + d(t_{Q,h} Q^*), \quad (\text{A.14})$$

where

$$dp_M^* = dp_X^* + d\tau^* + dt_{Q,M} + dt_{Q,X}. \quad (\text{A.15})$$

Combining (A.13) with (A.15), for the foreign country:

$$dV_f = \begin{cases} -(dp_X^* + d\tau^*) Q^* + I_f dr^* + (r^* - \kappa_f - t_I) dI_f + t_{Q,f} dQ^* & \text{if } f = M, \\ (dp_M^* - d\tau^*) Q^* + I_f dr^* + (r^* - \kappa_f - t_I) dI_f + t_{Q,f} dQ^* & \text{if } f = X. \end{cases} \quad (\text{A.16})$$

And further expressing $dr^* \equiv \frac{\partial r^*}{\partial Q} dQ^* + \frac{\partial r^*}{\partial I} dI^*$ and $d\tau^* \equiv \frac{\partial \tau^*}{\partial Q} dQ^* + \frac{\partial \tau^*}{\partial I} dI^*$ and collecting terms:

$$dV_f = \underbrace{\left[-1_{f=M} \left(\frac{dp_X}{dQ^*} + \frac{\partial \tau^*}{\partial Q} \right) Q^* + 1_{f=X} \left(\frac{dp_M}{dQ^*} - \frac{\partial \tau^*}{\partial Q} \right) Q^* + I_f \frac{\partial r^*}{\partial Q} + t_{Q,f} \right]}_{=0 \text{ at the optimal tariff}} dQ^* - Q^* \frac{\partial \tau^*}{\partial I} dI_f + I_f \frac{\partial r^*}{\partial I} dI_f + (r^* - \kappa_f) dI_f. \quad (\text{A.17})$$

Setting this expression equal to zero when $dI_f = 0$ and letting dQ^* respond to $dt_{Q,f}$ gives the optimal tariff of the foreign country:

$$t_{Q,f}^* = \left(1_{f=M} \frac{dp_X}{dQ^*} - 1_{f=X} \frac{dp_M}{dQ^*} - \frac{\partial \tau^*}{\partial Q} \right) Q^* - \frac{\partial r^*}{\partial Q} I_f. \quad (\text{A.18})$$

Using $t_{Q,f}^*$ in (A.17) gives (21). In turn, combining (A.14) with (A.15) gives:

$$dV_h = \begin{cases} Q^* dp_M - I_f dr^* + (r^* - \kappa_h) dI_h + t_{Q,h} dQ^* & \text{if } h = X, \\ -Q^* dp_X^* - I_f dr^* + (r^* - \kappa_h) dI_h + t_{Q,h} dQ^* & \text{if } h = M. \end{cases} \quad (\text{A.19})$$

And further expressing $dr^* \equiv \frac{\partial r}{\partial Q} dQ^* + \frac{\partial r}{\partial I} dI^*$ and collecting terms:

$$dV_h = \underbrace{\left(1_{h=X} Q^* \frac{dp_M}{dQ^*} - 1_{h=M} Q^* \frac{dp_X}{dQ^*} - I_f \frac{\partial r^*}{\partial Q} + t_{Q,h} \right)}_{=0 \text{ at the optimal tariff}} dQ^* - I_f \frac{\partial r^*}{\partial I^*} dI_h + (r^* - \kappa_h) dI_h, \quad (\text{A.20})$$

Setting this expression equal to zero when $dI_h = 0$ and letting dQ^* respond to $dt_{Q,h}$ gives the optimal tariff of the host country:

$$t_{Q,h} = \left(-1_{h=X} \frac{dp_M}{dQ^*} + 1_{h=M} \frac{dp_X}{dQ^*} \right) Q^* + I_f \frac{\partial r^*}{\partial Q}. \quad (\text{A.21})$$

which gives (22).

A.4 Proof of Proposition 2

In a Nash equilibrium, (19) and (20) imply:

$$\frac{dV_f}{dI_f} = \left| \frac{dp_f^*}{dI^*} \right| Q^* + I_f^* \frac{dr^*}{dI^*} + (r^* - \kappa) \leq 0, = \text{if } I_f^* > 0 \quad (\text{A.22})$$

$$\frac{dV_h}{dI_h} = - \left(\left| \frac{dp_f^*}{dI^*} \right| Q^* + I_f^* \frac{dr^*}{dI^*} \right) + (r^* - \kappa) \leq 0, = \text{if } I_h^* > 0. \quad (\text{A.23})$$

Suppose $I_f = 0$. Then, $\frac{dV_h}{dI_h} = 0$, implying $r^* = \kappa + \left| \frac{dp_f^*}{dI^*} \right| Q^* \equiv r^{*a}$. However, at $I_f = 0$, $\frac{dV_f}{dI_f} = \left| \frac{dp_f^*}{dI^*} \right| Q^* + (r^{*a} - \kappa) = 2 \left| \frac{dp_f^*}{dI^*} \right| Q^* > 0$, which violates (A.22). Therefore, $I_f^* > 0$ and (A.22) implies $r^* - \kappa = - \left(\left| \frac{dp_f^*}{dI^*} \right| Q^* + I_f^* \frac{dr^*}{dI^*} \right)$. Suppose $I_h^* > 0$; then (A.22) and (A.22) and (A.23) give:

$$r^* = \kappa \quad (\text{A.24})$$

$$\left| \frac{dp_f^*}{dI^*} \right| Q^* + I_f^* \frac{dr^*}{dI^*} = 0 \quad (\text{A.25})$$

and the equilibrium is given by $\{Q^*, I_f^*, I^*, r^*\}$ such that (A.24), (A.25), 17 and 13 hold. Alternatively, suppose $I_h^* = 0$. Then, (A.22) implies $r^* - \kappa = - \left(\left| \frac{dp_f^*}{dI^*} \right| Q^* + I_f^* \frac{dr^*}{dI^*} \right)$ and $\frac{dV_h}{dI_h} = -2 \left(\left| \frac{dp_f^*}{dI^*} \right| Q^* + I_f^* \frac{dr^*}{dI^*} \right)$. If $\left| \frac{dp_f^*}{dI^*} \right| Q^* + I_f^* \frac{dr^*}{dI^*} < 0$ at the $\{I_f^*, Q^*\}$ that solves 17 and 26, then $\frac{dV_h}{dI_h} > 0$, and the equilibrium necessarily features both countries investing. On the other hand, if given the I_f^* that solves 17 and 26 we have $\left| \frac{dp_f^*}{dI^*} \right| Q^* + I_f^* \frac{dr^*}{dI^*} > 0$ for the Q^* , $\left| \frac{dp_f^*}{dI^*} \right|$, and $\frac{dr^*}{dI^*}$ that solve 17

for any $I^* \geq I_f^*$, then $\frac{dV_h}{dI_h} < 0$ for any $I_h^* > 0$, implying $I_h^* = 0$ at the equilibrium. Such an equilibrium can be ensured assuming $\frac{dr^*}{dI^*}$ is sufficiently small or $\left|\frac{dp_f^*}{dI^*}\right|$ sufficiently large. In this case, $r^* < \kappa$ implies that in an implementation via taxes the foreign government sets a subsidy $t_{P,j}^* = -\left(\left|\frac{dp_f^*}{dI^*}\right|Q^* + I_f^*\frac{dr^*}{dI^*}\right) < 0$ to its private developers.

A.5 Proof of Proposition 3

In a Nash equilibrium around an arbitrary t_I , (19) and (20) imply:

$$\frac{dV_f}{dI_f} = \left(Q^*\left|\frac{dp_f^*}{dI^*}\right| + I_f^*\frac{dr^*}{dI^*}\right) + (r^* - \kappa - t_I) \leq 0, = \text{if } I_f^* > 0 \quad (\text{A.26})$$

and

$$\frac{dV_h}{dI_h} = -\left(Q^*\left|\frac{dp_f^*}{dI^*}\right| + I_f^*\frac{dr^*}{dI^*}\right) + (r^* - \kappa) \leq 0, = \text{if } I_h^* > 0. \quad (\text{A.27})$$

In turn, the welfare effect on the host from modifying its foreign tax around any equilibrium is

$$\frac{dV_h}{dt_I} = \left(t_I - \left(Q^*\left|\frac{dp_f^*}{dI^*}\right| + I_f^*\frac{dr^*}{dI^*}\right)\right) \frac{dI_f^*}{dt_I} + I_f^*. \quad (\text{A.28})$$

Suppose $I_f^* = 0$. Then, (A.27) implies $r^* = \kappa + \left|\frac{dp_f^*}{dI^*}\right|Q^* \equiv r^{*a}$. A necessary condition for this to be an equilibrium is $\frac{dV_f}{dI_f} \leq 0$ around this point, implying from (A.26) that $t_I \geq \bar{t}_I \equiv 2Q^*\left|\frac{dp_f^*}{dI^*}\right|$. At $t_I = \bar{t}_I$ and $I_f^* = 0$ then $\frac{dV_f}{dI_f} = 0$ and $\frac{dV_h}{dt_I} = Q^*\left|\frac{dp_f^*}{dI^*}\right|\frac{dI_f^*}{dt_I}$. Assuming $\frac{dI_f^*}{dt_I} < 0$ around this point, $\frac{dV_h}{dt_I} < 0$ and the host would gain from lowering t_I . Hence, $I_f^* > 0$ and the optimal tax solves

$$t_I^* = -\frac{I_f^*}{dI_f^*/dt_I} + \left(Q^*\left|\frac{dp_f^*}{dI^*}\right| + I_f^*\frac{dr^*}{dI^*}\right). \quad (\text{A.29})$$

If both countries enter, then (A.26), (A.27) and (A.29) imply $r^* = \kappa - 2\frac{I_f^*}{dI_f^*/dt_I} > \kappa$ and $t_I^* = 2(r^* - \kappa) > 0$. If the foreign country implements via taxes to domestic investors, entry of investors abroad requires $r^* - \kappa - t_{P,f} - \tau_I^* = 0$, giving the subsidy $t_{P,f} = -\frac{t_I^*}{2} < 0$. If only foreign invests, then (A.26) and (A.29) yield $r^* = \kappa - \frac{I_f^*}{dI_f^*/dt_I} > 0$. If this is an equilibrium, then (A.27) implies

$$Q^*\left|\frac{dp_f^*}{dI^*}\right| + I_f^*\frac{dr^*}{dI^*} > r^* - \kappa > 0 \quad (\text{A.30})$$

at the solution for (A.26), (A.29), and 17. If (A.30) fails, then both countries entering is the only equilibrium. If foreign implements with domestic taxes it must give the subsidy $t_{P,f} = r^* - \kappa - \tau_I^* = -\left(Q^*\left|\frac{dp_f^*}{dI^*}\right| + I_f^*\frac{dr^*}{dI^*}\right) < 0$ where the last inequality follows from (A.30), implying again a subsidy to private investors in the foreign country.

A.6 Proof of Proposition 4

In a Nash equilibrium with equal κ across countries and no foreign investment taxes, (21) and (22) imply:

$$\frac{dV_f}{dI_f} = -Q^* \frac{\partial \tau^*}{\partial I} + I_f^* \frac{\partial r^*}{\partial I} + (r - \kappa) \leq 0, = \text{if } I_f^* > 0 \quad (\text{A.31})$$

$$\frac{dV_h}{dI_h} = -I_f \frac{\partial r^*}{\partial I^*} + (r - \kappa) \leq 0, = \text{if } I_h^* > 0. \quad (\text{A.32})$$

Suppose $I_f = 0$. Then, $\frac{dV_h}{dI_h} = 0$, implying $r^* = \kappa \equiv r^{*a}$. However, at $I_f = 0$, $\frac{dV_f}{dI_f} = -Q^* \frac{\partial \tau^*}{\partial I} > 0$, which violates (A.31). Therefore, $I_f^* > 0$. The host country must enter in equilibrium if, at the (I^*, Q^*) solving free entry of foreign of foreign (A.31),

$$r^* - \kappa = Q^* \frac{\partial \tau^*}{\partial I} - I^* \frac{\partial r^*}{\partial I^*} \quad (\text{A.33})$$

and (17) we have $\frac{dV_h}{dI_h} = -2I^* \frac{\partial r^*}{\partial I^*} + Q^* \frac{\partial \tau^*}{\partial I} > 0$. Otherwise, the host country not entering may be an equilibrium. If the host country does not enter, $\frac{dV_h}{dI_h} \leq 0$ implies $r^* - \kappa \leq I_f \frac{\partial r^*}{\partial I^*} < 0$. If it enters, $r^* - \kappa = I_f \frac{\partial r^*}{\partial I^*} < 0$. In either case, $r^* < \kappa$, implying private investors are subsidized in the foreign country, $t_{P,f}^* = r^* - \kappa < 0$.

B Appendix to the Quantitative Sections

B.1 Summary of Model Equations

Denote by $N_s = \#\mathcal{S}$ the number of nodes and $N_e = \#\mathcal{E}$ the number of edges. The quantitative model given infrastructure and tariffs is summarized by the following system of equations in the unknowns: p_{ns} (price of good n at node s , $J \times N_s$ unknowns), p_{nj}^C (consumer price of good n in country j , J^2 unknowns), P_j^C (price index in country j , J unknowns), C_{nj} (consumption of good n in country j , J^2 unknowns), C_j (aggregate consumption in country j , J unknowns), Y_n (production of good n , J unknowns), Q_s (total flows through infrastructure node s , N_s unknowns), $Q_{nss'}$ (gross flows of good n through edge ss' , $J \times N_e$ unknowns), $\mu_{nss'}$ (Lagrange multiplier on the constraint $Q_{nss'} \geq 0$, $J \times N_e$ unknowns). This amounts to a system in $3J + 2J^2 + N_s + 3JN_s$ unknowns with the following $3J + 2J^2 + N_s + 3JN_s$ equations:

1. Price indices (J equations):

$$P_j^C - \left(\sum_{n=1}^J d_{nj} (p_{nj}^C)^{1-\sigma} \right)^{\frac{1}{1-\sigma}} = 0;$$

2. Consumer prices (J^2 equations):

$$p_{nj}^C - p_{nj} - 1_{n \neq j} (t_{nj}^M + t_{nj}^X) = 0;$$

3. Aggregate consumptions (J equations):

$$C_j - (U'_j)^{-1} (P_j^C) = 0;$$

4. Consumption matrix (J^2 equations):

$$C_{nj} - C_j d_{nj} \left(\frac{p_{nj}^C}{P_j^C} \right)^{-\sigma} = 0;$$

5. Productions (J equations):

$$Y_n - A_n^{\frac{1}{1-\alpha_n}} (\alpha_n p_{nn})^{\frac{\alpha_n}{1-\alpha_n}} = 0;$$

6. Resource constraints ($J \times N_s$ equations):

$$1_{s=s_0(j)} C_{nj} + \sum_{s' \in \mathcal{N}(s)} Q_{nss'} - 1_{s=s_0(j)} 1_{\{n=j\}} Y_j - \sum_{s' \in \mathcal{N}(s)} Q_{ns's} = 0;$$

7. Flows through infrastructure nodes (N_s equations):

$$Q_s - \sum_n \sum_{s' \in \mathcal{N}(s)} \omega_n (Q_{nss'} + Q_{ns's}) = 0.$$

8. No-arbitrage conditions ($J \times N_e$ equations):

$$p_{ns'} + \mu_{nss'} - p_{ns} - \omega_n (1 + \beta) \bar{T}_s Q_s^\beta I_s^{-\gamma} - \omega_n (1 + \beta) \bar{T}_{s'} Q_{s'}^\beta I_{s'}^{-\gamma} = 0,$$

where $\mu_{nss'}$ is the Lagrange multiplier on the constraint $Q_{nss'} \geq 0$;

9. Complementary slackness conditions ($J \times N_e$ equations):

$$\mu_{ss'}^n Q_{ss'}^n = 0;$$

B.2 Equilibrium given infrastructure $(I_{js})_{j \in \mathcal{J}, s \in \mathcal{S}}$ and tariffs $\{t_{nj}^M, t_{nj}^X\}_{(n,j) \in \mathcal{J}^2}$

Approximation of the Convex-cost Flow Problem Given an allocation of consumption $\{C_{nj}\}_{(n,j) \in \mathcal{J}^2}$, production $\{Y_n\}_{n \in \mathcal{J}}$ and infrastructure $\{I_s\}_{s \in \mathcal{S}}$, the solution of the problem solved by infrastructure operators and shipping companies is the solution of the convex-cost flow problem

$$\min_{\{Q_s, Q_{nss'} \geq 0\}} \sum_{n \in \mathcal{J}} \sum_{(s,s') \in \mathcal{E}} \bar{\tau}_{nss'} Q_{nss'} + \sum_{s \in \mathcal{S}} T_s(Q_s, I_s) \quad (\text{B.1})$$

subject to

$$Q_s = \sum_{n \in \mathcal{J}} \sum_{s' \in \mathcal{S}} \omega_n (Q_{nss'} + Q_{ns's})$$

$$1_{s=s_0(j)} C_{nj} + \sum_{s' \in \mathcal{N}(s)} Q_{nss'} \leq 1_{s=s_0(j)} 1_{n=j} Y_n + \sum_{s' \in \mathcal{N}(s)} Q_{ns's}, \forall n \in \mathcal{J}, s \in \mathcal{S}.$$

Under the convexity assumptions on T stated in Section 4, this problem is a well-defined convex optimization problem that satisfies the same first-order conditions (43) and no-arbitrage conditions (40) as the full equilibrium problem in Definition 1. The solution of the optimal flow problem $\{Q_{nss'}\}_{n \in \mathcal{J}, (s,s') \in \mathcal{E}}$ and $\{Q_s\}_{s \in \mathcal{S}}$ given $\{C_{nj}, Y_n, I_s\}$ thus coincides with the equilibrium allocation.

Being convex, problem B.1 can be solved using standard optimization approaches such as, for instance, an interior-point method. This approach is nonetheless costly to implement because it requires the computation of potentially very large Hessians. To speed up this step, we approximate the functions T_s with piecewise linear interpolants $\hat{T}_s$ (Croxtton et al., 2007; Crainic and Kim, 2007). Specifically, for an approximation using K segments over the breakpoints $(\hat{Q}_0 = 0, \dots, \hat{Q}_K = Q_{max})$, each function T_s is approximated by the affine interpolant

$$T_s(Q_s, I_s) \simeq \hat{T}_s^{k,0} + \hat{T}_s^{k,1} (Q_s - \hat{Q}_s) \text{ for } \hat{Q}_k \leq Q_s < \hat{Q}_{k+1}.$$

Problem B.1 is then approximated by the linear programming problem with capacity constraints:

$$\min_{\{Q_{s,k}\}_{s \in \mathcal{S}, k=1 \dots K}, \{Q_{nss'} \geq 0\}, Q_s} \sum_{n \in \mathcal{J}} \sum_{(s,s') \in \mathcal{E}} \bar{\tau}_{nss'} Q_{nss'} + \sum_{s \in \mathcal{S}} \sum_{k=1}^K \hat{T}_s^{k,1} Q_{s,k} \quad (\text{B.2})$$

subject to

$$0 \leq Q_{s,k} < \hat{Q}_k - \hat{Q}_{k-1}$$

$$Q_s = \sum_{k=1}^K Q_{s,k}$$

$$Q_s = \sum_{n \in \mathcal{J}} \sum_{s' \in \mathcal{S}} \omega_n (Q_{nss'} + Q_{ns's})$$

$$1_{s=s_0(j)} C_{nj} + \sum_{s' \in \mathcal{N}(s)} Q_{nss'} \leq 1_{s=s_0(j)} 1_{n=j} Y_n + \sum_{s' \in \mathcal{N}(s)} Q_{ns's}, \forall n \in \mathcal{J}, s \in \mathcal{S}.$$

In our implementation, we choose an approximation with $K = 16$ segments and solve problem B.2 with the LP solver Gurobi 12.0.3. While this reformulation introduces new variables (i.e., the flows in each subsegment $Q_{s,k}$) and new constraints (i.e., the capacity constraints $Q_{s,k} < \hat{Q}_k - \hat{Q}_{k-1}$), the resulting LP problem substantially improves the numerical tractability of the problem by avoiding the repeated calculation of large Hessians in every iteration.

Resolution algorithm Given infrastructure $(I_{js})_{j \in \mathcal{J}, s \in \mathcal{S}}$ and tariffs $\{t_{nj}^M, t_{nj}^X\}_{(n,j) \in \mathcal{J}^2}$, we solve the model using a simple iterative procedure over the producer prices $\{p_{nn}\}_{n=1 \dots J}$. The algorithm is described in pseudo-code below:

1. **Initialization.** Set the iteration counter $k := 0$. Guess a $J \times 1$ vector of prices $p_{nn}^{(k)}$. Get a vector of production $\hat{Y}_n^{(k)} := A_n^{\frac{1}{1-\alpha_n}} (\alpha_n p_{nn})^{\frac{\alpha_n}{1-\alpha_n}} L_n$ from the firm's FOC and guess a feasible matrix of consumption $\hat{C}_{nj}^{(k)}$, for instance: $\hat{C}_{nj}^{(k)} := \hat{Y}_n / J$;
2. **Flow problem.** Given $(\hat{C}_{nj}^{(k)}, \hat{Y}_n^{(k)})$, solve for the flows $Q_{nss'}^{(k)}$ and $Q_s^{(k)}$ using the piecewise linear approximation B.2. Retrieve a matrix of prices $p_{ns}^{(k)} = p_{nn}^{(k)} + \lambda_{ns}^{(k)}$ where $\lambda_{ns}^{(k)}$ is the resulting Lagrange multiplier on the resource constraint;
3. **Equilibrium variables.** From $p_{ns}^{(k)}$, extract $p_{nj}^{(k)} := p_{ns_0(j)}^{(k)}$, compute the consumer price $p_{nj}^{C,(k)} := p_{nj}^{(k)} + t_{nj}^M + t_{nj}^X$, the consumer price index $P_j^{C,(k)} := \left(\sum_{n=1}^J d_{nj} \left(p_{nj}^{C,(k)} \right)^{1-\sigma} \right)^{\frac{1}{1-\sigma}}$, consumption $C_j^{(k)}$ from (36), consumption $C_{nj}^{(k)}$ from (37), total demand $C_n^{(k)} := \sum_{j=1}^J C_{nj}^{(k)}$ and production $Y_n^{(k)}$ from (53);
4. **Convergence test.** Check convergence by evaluating

$$D = \frac{1}{J} \sum_n \left(Y_n^{(k)} - C_n^{(k)} \right)^2 + \frac{1}{J^2} \sum_{n,j} \left(\hat{C}_{nj}^{(k)} - C_{nj}^{(k)} \right)^2.$$

If D is small enough, stop: the market for each good is cleared and our guess for the allocation of consumption has converged. Otherwise, we continue with new guesses. For the producer prices, we compute

$$\hat{p}_{nn}^{(k)} = \frac{1}{\alpha_N} \left(\frac{C_n^{(k)}}{L_n} \right)^{\frac{1-\alpha_n}{\alpha_n}} A_n^{-\frac{1}{\alpha_n}},$$

which corresponds to the prices that would make producers accommodate current aggregate demand. We set our guess

$$p_{nn}^{(k+1)} := (1 - \delta) p_{nn}^{(k)} + \delta \hat{p}_{nn}^{(k)},$$

for some $\delta \in (0, 1)$. To construct our new guesses for the allocation $(\hat{C}_{nj}^{(k+1)}, \hat{Y}_n^{(k+1)})$ to be fed in the LP solver, we construct the matrix of prices $\hat{p}_{ns}^{(k+1)} = p_{nn}^{(k+1)} + \lambda_{ns}^{(k)}$ consistent with our producer prices. We set our new guess for consumption as the matrix of consumption $\hat{C}_{nj}^{(k+1)}$ consistent with these prices and our guess for production to total demand $\hat{Y}_n^{(k+1)} := \sum_j \hat{C}_{nj}^{(k+1)}$ (which guarantees feasibility). Set $k := k + 1$ and go back to step 2.

B.3 Optimization

An equilibrium given $(I_{js})_{j \in \mathcal{J}, s \in \mathcal{S}}$ and tariffs $\{t_{nj}^M, t_{nj}^X\}_{(n,j) \in \mathcal{J}^2}$ satisfies the system of equations

$$\mathbf{F}(\mathbf{x}; \mathbf{I}) = 0,$$

where $\mathbf{F}$ is the vector-valued function of all the equilibrium conditions provided in Section B.1, $\mathbf{x}$ is the vector of endogenous variables

$$\mathbf{x} = (\{p_{ns}\}, \{p_{nj}^C\}, \{P_j^C\}, \{C_{nj}\}, \{C_j\}, \{Y_n\}, \{Q_s\}, \{Q_{nss'}\}, \{\mu_{nss'}\})$$

and $\mathbf{I}$ the $N_s \times 1$ vector of total infrastructure supply, such that $I_s = \sum_j I_{js}$. We compute the Jacobian of the endogenous variables with respect to changes in infrastructure using the implicit function theorem:

$$Jac_{\mathbf{I}}\mathbf{x} = -(Jac_{\mathbf{x}}\mathbf{F})^{-1} Jac_{\mathbf{I}}\mathbf{F}.$$

The gradient of welfare for any country j is computed using equation (48) and substituting the corresponding derivatives from $Jac_{\mathbf{I}}\mathbf{x}$. We search for the optimal infrastructure network using a gradient descent algorithm and stop when local optimality conditions are satisfied. Our game-theoretic setup with multiple planners does not allow us to offer conditions to guarantee global optimality without additional parametric restrictions.

B.4 Extensions

Congestion externalities

The main text specification assumes that congestion is internalized by infrastructure operators. We show in this subsection how to extend our numerical approach to deal with congestion externalities of a certain form. Suppose the cost function faced by port operators is given by

$$T_s(Q_s, \bar{Q}_s, I_s) = \bar{T}_s \bar{Q}_s^{\bar{\beta}} Q_s^{1+\beta} I_s^{-\gamma},$$

where $Q_s = \sum_n \omega_n Q_{ns}$ is the throughput chosen by the port operator and $\bar{Q}_s$ is the source of a negative congestion spillover, taken as given by the operator but that must satisfy $\bar{Q}_s = Q_s$ in equilibrium. The problem of the port operator is

$$\max_{Q_{ns}, Q_s, I_s} \sum_n \tau_{ns} Q_{ns} - \bar{T}_s \bar{Q}_s^{\bar{\beta}} Q_s^{1+\beta} I_s^{-\gamma} - r_s I_s.$$

The first-order condition over Q_s yields

$$\tau_{ns} = (1 + \beta) \bar{T}_s Q_s^{\beta + \bar{\beta}} I_s^{-\gamma}. \quad (\text{B.3})$$

Define the following convex-cost flow problem:

$$\min_{Q_s, Q_{nss'}, Q_{ns's}} \sum_{n \in \mathcal{J}} \sum_{(s, s') \in \mathcal{E}} \bar{\tau}_{nss'} Q_{nss'} + \frac{1 + \beta}{1 + \beta + \bar{\beta}} \sum_{s \in \mathcal{S}} T_s Q_s^{1 + \beta + \bar{\beta}} I_s^{-\gamma} \quad (\text{B.4})$$

subject to

$$Q_s = \sum_{n \in \mathcal{J}} \sum_{s' \in \mathcal{S}} \omega_n (Q_{nss'} + Q_{ns's})$$

$$1_{s=s_0(j)} C_{nj} + \sum_{s' \in \mathcal{N}(s)} Q_{nss'} \leq 1_{s=s_0(j)} 1_{n=j} Y_n + \sum_{s' \in \mathcal{N}(s)} Q_{ns's}, \forall n \in \mathcal{J}, s \in \mathcal{S}.$$

Problem B.4 is a well-defined convex-cost flow problem that satisfies the exact same optimality conditions as the first-order condition (B.3), the non-arbitrage equation (40) and the resource constraint (47). It thus defines the same allocation of flows $\{Q_{nss'}\}$, $\{Q_s\}$ and $\{Q_{ns}\}$. Problem B.4 can be solved using a piecewise-linear approximation, like the one described in the previous sections.

Iceberg trade costs

The model in the main text assumes that transport costs are specified in terms of the outside good. Our model can also accommodate iceberg costs. Suppose, for simplicity, that infrastructure operators only charge a fee on exporting goods, instead of both on importing and exporting. Suppose further that exporting good n from location s requires the payment of an iceberg cost $\hat{\tau}_{ns}(Q_s, I_s)$ per unit of the good itself. Denote $Q_{ns}^{Exp} = \sum_{s' \in \mathcal{N}(s)} Q_{nss'}$ and $Q_s^{Exp} = \sum \omega_n Q_{ns}^{Exp}$. The infrastructure operator solves the problem

$$\max_{Q_{ns}, Q_s, I_s} \sum_n \tau_{ns} Q_{ns}^{Exp} - \sum_n p_{ns} \hat{\tau}_{ns}(Q_s^{Exp}, I_s) Q_{ns}^{Exp} - r_s I_s.$$

The first-order condition over Q_{ns}^{Exp} yields

$$\tau_{ns} = p_{ns} \hat{\tau}_{ns} + \omega_n \sum_{n'} p_{n's} \frac{\partial \hat{\tau}_{n's}}{\partial Q_s}.$$

Suppose shipping companies have to pay the exogenous iceberg cost $\bar{\tau}_{nss'}$ in location s in addition to paying the port fee τ_{ns} . The no-arbitrage condition for shipping companies gives

$$p_{ns'} \leq p_{ns} (1 + \bar{\tau}_{nss'}) + \tau_{ns} = p_{ns} (1 + \bar{\tau}_{nss'} + \hat{\tau}_{ns}) + \omega_n \sum_{n'} p_{n's} \frac{\partial \hat{\tau}_{n's}}{\partial Q_s}.$$

Define the cost flow problem:

$$\min_{Q_{nss'} \geq 0, Y_n} \sum_{n \in \mathcal{J}} \Psi_n(Y_n) \quad (\text{B.5})$$

subject to

$$\begin{aligned}
& 1_{s=s_0(j)} C_{nj} + \sum_{s' \in \mathcal{N}(s)} (1 + \bar{\tau}_{nss'}) Q_{nss'} + \hat{\tau}_{ns} (Q_s^{Exp}, I_s) Q_{ns}^{Exp} \\
& \leq 1_{s=s_0(j)} 1_{n=j} Y_n + \sum_{s' \in \mathcal{N}(s)} Q_{ns's}, \forall n \in \mathcal{J}, s \in \mathcal{S}.
\end{aligned} \tag{B.6}$$

For $\hat{\tau} (Q_s^{Exp}, I_s) Q_{ns}^{Exp}$ quasi-convex, this problem is a well-defined quasi-convex optimization problem. Labeling p_{ns} the Lagrange multiplier on the resource constraint, the first-order condition over $Q_{nss'}$ is

$$p_{ns'} \leq p_{ns} (1 + \bar{\tau}_{nss'} + \hat{\tau}_{ns} (Q_s, I_s)) + \omega_n \sum_{n'} p_{n's} \frac{\partial \hat{\tau}_{n's}}{\partial Q_s},$$

with equality if $Q_{nss'} > 0$. This first-order conditions, along with the optimality condition over production Y_n and the resource constraint (B.6), are identical to the equilibrium conditions that characterize the model with iceberg costs. The two problems are thus equivalent. Problem B.5 can be solved with a piecewise-linear approximation similar to the one described in the previous sections.

B.5 Growth in Seaport Infrastructure due to BRI

Ideally, we would like to directly measure changes in the total number of cranes, but we do not observe cranes in the most recent period in our data. Therefore we proceed in two steps. First, we estimate the response of pier depth to investment over time, and then we translate this growth to cranes using the cross-sectional relationship between depth and cranes before the BRI. The first step mimics the regressions from Section 3.4, but in levels.

$$\Delta \text{pierdepth}_p = \alpha + \beta_1 \text{value}_{CHN,p} + \beta_2 \text{dredgecost}_p + \beta_3 \text{value}_{CHN,p} \times \text{dredgecost}_p + \epsilon_p \tag{B.7}$$

where we allow for potential heterogeneity in the response of pier depth to investment by dredging costs. We then construct the average pier depth response to financing across all ports p within each country i : $\beta_i = \hat{\beta}_1 + \hat{\beta}_3 \text{dredgecost}_p$. Then, we use our data on cranes and total berth length from Lloyd's Ports of the World to estimate the mapping between pier depth and cranes/berths.⁴¹ We aggregate the total number of cranes and berth length for each country along with its average pier depth, and estimate:

$$\text{cranes}_i = \alpha + \gamma_1 \text{pierdepth}_i + \nu_i. \tag{B.8}$$

This implies that the response of cranes to pier depth is given by γ_1 . The product $\beta_i \times \gamma_1$ is number of cranes (or total berth length) per USD 1 million financing in country i . In our model, $\beta_i \gamma_1$ is the inverse of the dollar cost per unit of infrastructure $\kappa_{CH,s}$ for the sea node s of country i . We can then convert the total dollars invested by China into a country's ports to the total country-level

⁴¹We can only do this as a cross-sectional regression at the moment since we do not observe a long time series of port-level cranes and berths.

increase in cranes (or berths) as: $\beta_i \times \gamma_1 \times \text{value}_{CHN,i}$. Scaling this increase by the initial cranes yields the implied percentage increase in seaport infrastructure.

Appendix Tables and Figures

Table C.1: China's Largest Overseas Infrastructure Investments, by Transport Mode

mode	country	locode	name	year implement	year complete	value (\$b)
seaport	RUS	RUSAB	Sabetta	2012	2018	14.8
land	MYS	MYKBR	Kota Bharu	2017		9.3
seaport	AUS	AUGLT	Gladstone	2009	2013	7.7
seaport	CUB	CUCFG	Cienfuegos	2011	2017	6.6
land	LAO	LAVTE	Vientiane	2008	2015	5.4
land	BGD	BDDAC	Dhaka	2018	2024	4.8
land	KHM	KHPNH	Phnom Penh	2001	2002	4.7
seaport	GIN	GNBRP	Port de Boke	2015	2015	4.6
land	ETH	ETADD	Addis Ababa	2001	2004	3.8
land	KEN	KEMBA	Mombasa	2014	2017	3.6
airport	TUR	TRISL	Istanbul New Airport	2014		3.3
seaport	LKA	LKHBA	Hambantota	2008	2011	3.2
seaport	SGP	SGPPT	Singapore Pasir Panjang New Berth Project	2017	2018	3.2
seaport	SLE	SLPEP	Pepel	2012	2012	3.1
land	PAK	PKKHI	Karachi	2016	2019	3.1

Notes: Table reports the largest overseas investments by China in the AidData.

Table C.2: Investment and Changes in Port Characteristics and Trade: Details

Panel A: Changes in Port Characteristics							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Δ harbor	Δ breakwater	Δ shelter	Δ tidal	Δ pier depth	Δ chan depth	Δ anchor depth
asinh(chn investment)	0.006 (0.005)	0.002 (0.005)	0.010 (0.006)	-0.001 (0.008)	0.053** (0.022)	0.025* (0.013)	0.017 (0.016)
R2	0.00	0.00	0.00	0.00	0.01	0.00	0.00
Obs	2,739	2,790	2,790	1,795	2,410	2,419	2,490

Panel B: Changes in TEU Imports and Exports				
Δ Trade				
	(1)	(2)	(3)	(4)
	Δ TEU in	Δ TEU in CHN	Δ TEU out	Δ TEU out CHN
asinh(chn investment)	0.065*** (0.024)	0.041** (0.018)	0.012 (0.009)	0.025 (0.023)
R2	0.04	0.01	0.00	0.00
Obs	369	369	364	364

Notes: The left panel regresses the underlying changes in port characteristics on (inverse hyperbolic sine) Chinese investment. The right panel examines correlations with TEU imports and exports (and their shares from/to China). Standard errors clustered by country.

Table C.3: Non-China Investment and Changes in Port Characteristics and Trade

	Δ Characteristics		Δ Trade			
	(1)	(2)	(3)	(4)	(5)	(6)
	Δ port idx	Δ depth idx	Δ capacity	Δ vessels	Δ TEU tpt	Δ TEU tpt CHN share
asinh(nchn investment)	0.012*** (0.004)	0.021 (0.013)	0.024 (0.020)	0.017 (0.012)	0.013 (0.015)	0.015 (0.020)
R2	0.00	0.00	0.00	0.00	0.00	0.00
Obs	2,790	2,699	655	655	374	374

Notes: The left panel regresses the port characteristics index and depth index on (inverse hyperbolic sine) non-Chinese investment. The right panel examines correlations with changes in capacity, vessels, TEU throughput and the share of TEU throughput to/from China. Standard errors clustered by country.

Table C.4: Confirmed Dredging and Changes in Port Characteristics

	Δ Port Chars				Δ Depth Chars		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Δ harbor	Δ breakwater	Δ shelter	Δ tidal	Δ pier depth	Δ chan depth	Δ anchor depth
dredging (0/1)	0.098 (0.083)	-0.027 (0.061)	0.061 (0.079)	-0.029 (0.095)	0.487** (0.213)	-0.069 (0.139)	0.430** (0.202)
R2	0.02	0.00	0.00	0.00	0.05	0.00	0.04
Obs	115	116	116	88	112	107	110

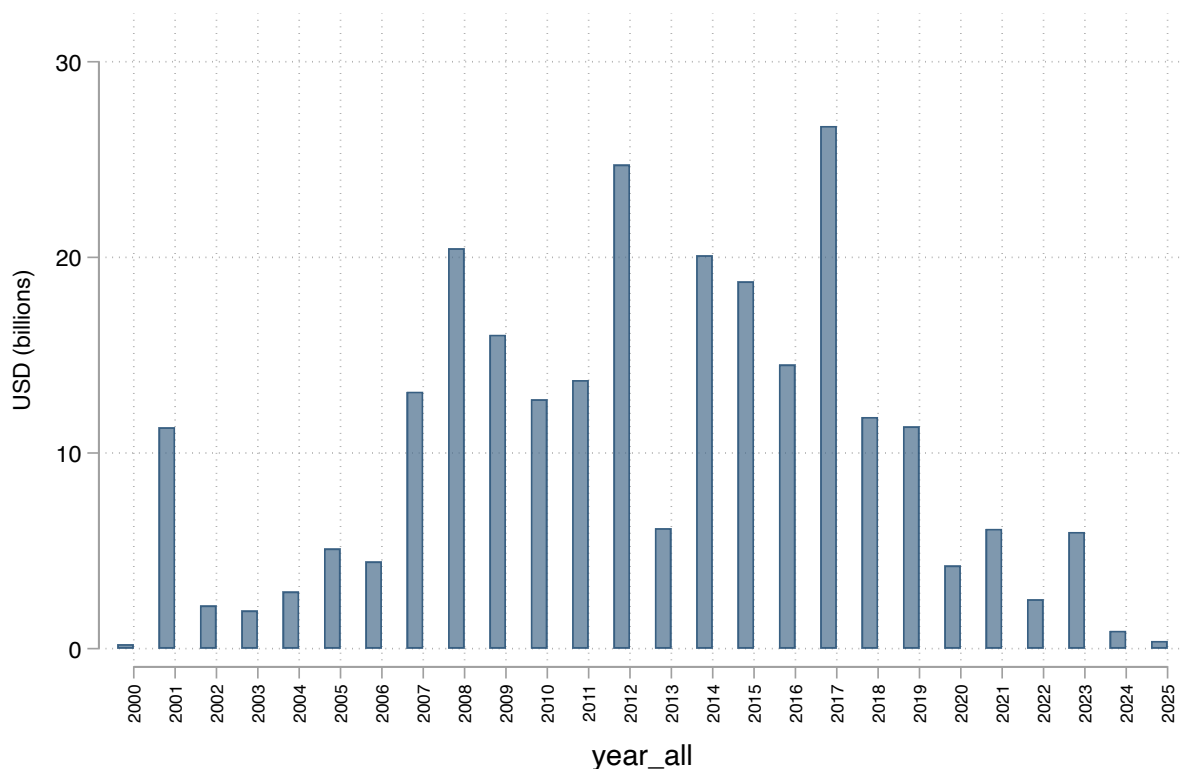
Notes: Table reports regression coefficients of changes in China's investment (row 1) and changes in port-level characteristics (subsequent rows) on an indicator for confirmed dredging at ports. Standard errors clustered by country.

Table C.5: China's Investment and Changes in Port Characteristics and Trade: Larger Ports

	Δ Characteristics		Δ Trade			
	(1)	(2)	(3)	(4)	(5)	(6)
	Δ port idx	Δ depth idx	Δ capacity	Δ vessels	Δ TEU tpt	Δ TEU tpt CHN share
asinh(chn investment)	0.007* (0.004)	0.031* (0.016)	0.048 (0.039)	0.005 (0.016)	0.049** (0.024)	0.030 (0.028)
R2	0.00	0.01	0.01	0.00	0.04	0.01
Obs	440	437	259	259	176	176

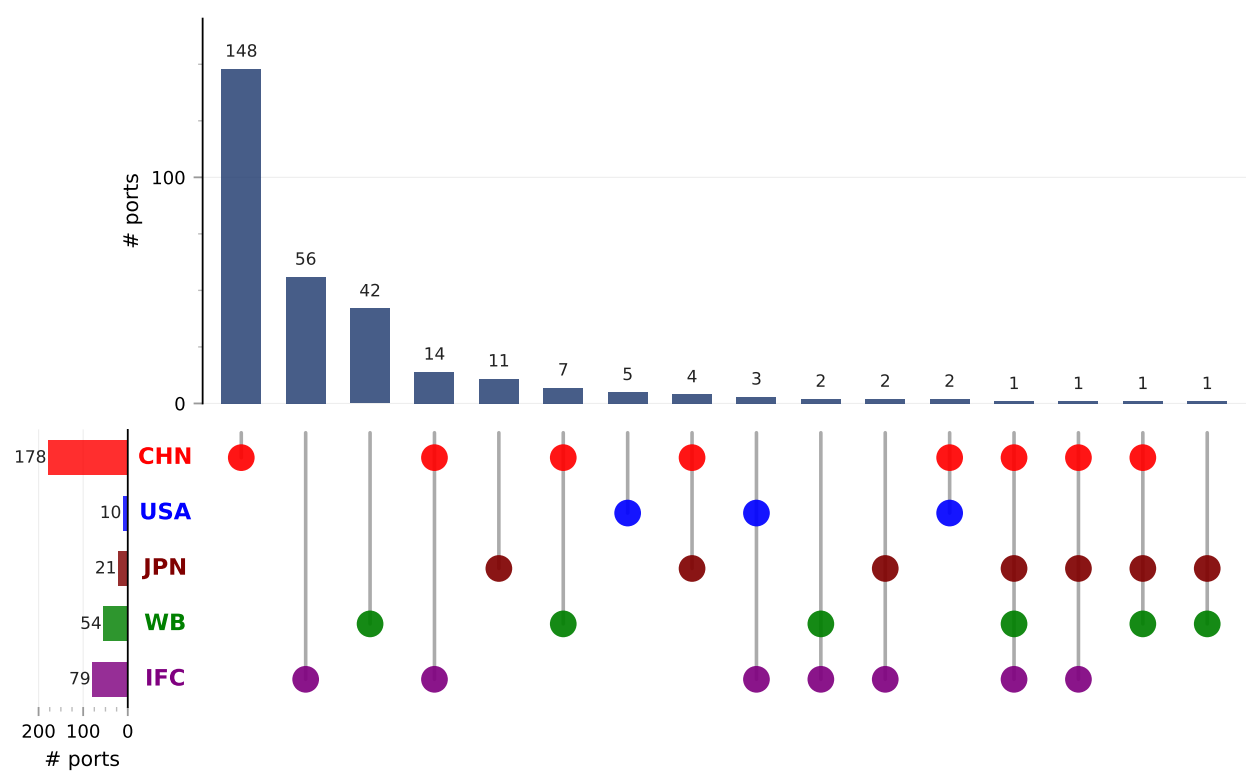
Notes: The left panel regresses the port characteristics index and depth index on (inverse hyperbolic sine) Chinese investment among WPI ports initially classified as medium/large. The right panel examines correlations with changes in capacity, vessels, TEU throughput and the share of TEU throughput to/from China. Standard errors clustered by country.

Figure C.1: Timeline of China's Overseas Investments, 2000-2025



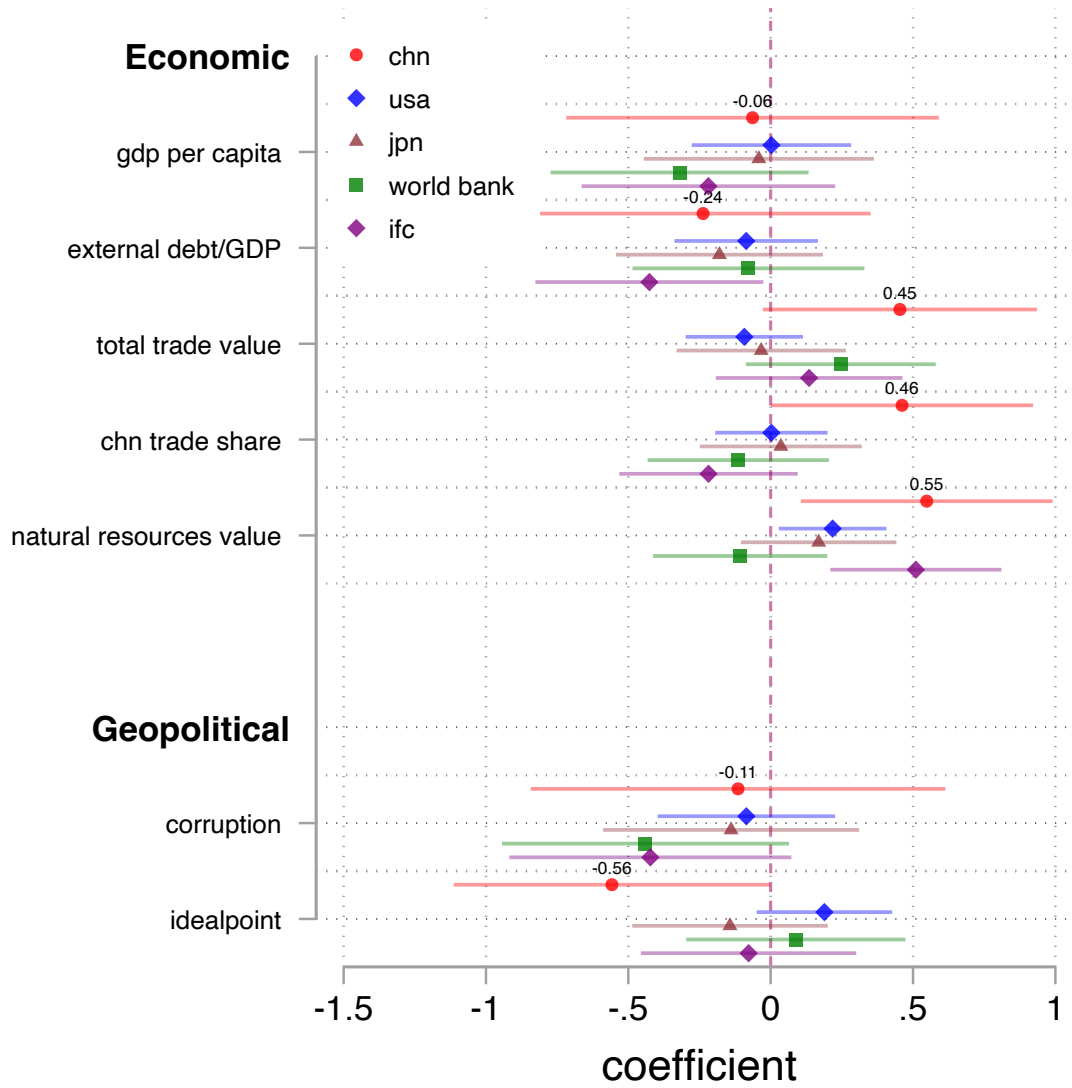
Notes: Figure plots committed value (USD billions, 2023 base) of Chinese overseas infrastructure investments by implementation year across all modes, and for seaports only.

Figure C.2: Overlap in Seaport Investments



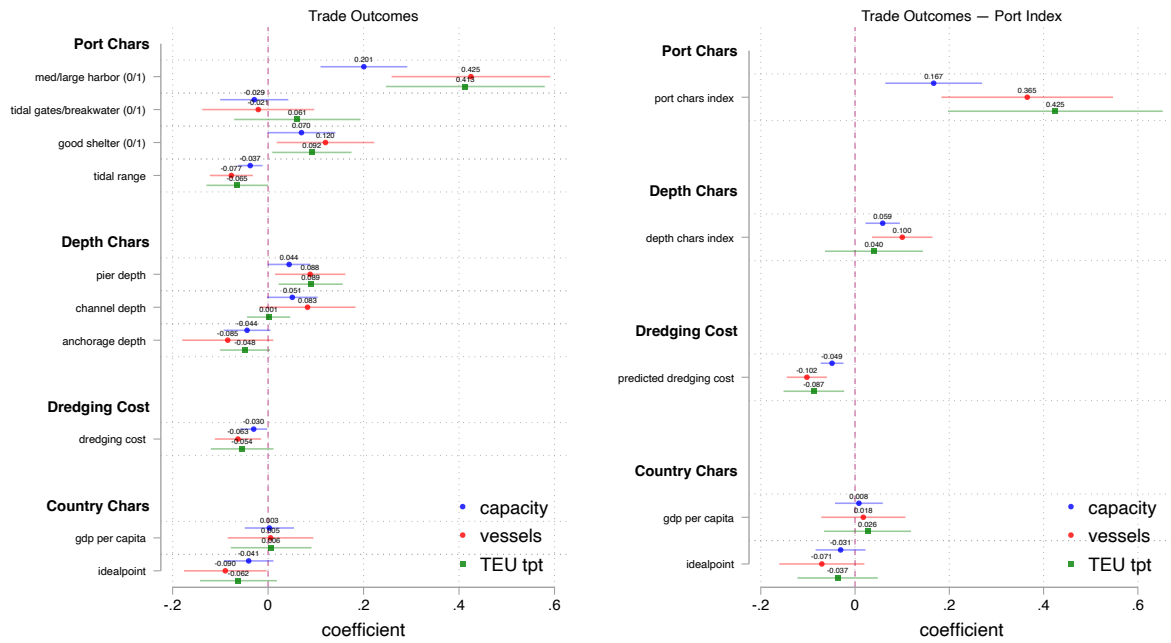
Notes: Figure plots an UpSet plot of seaport investments across the five sources.

Figure C.3: Investments and Country Correlates, All Investors



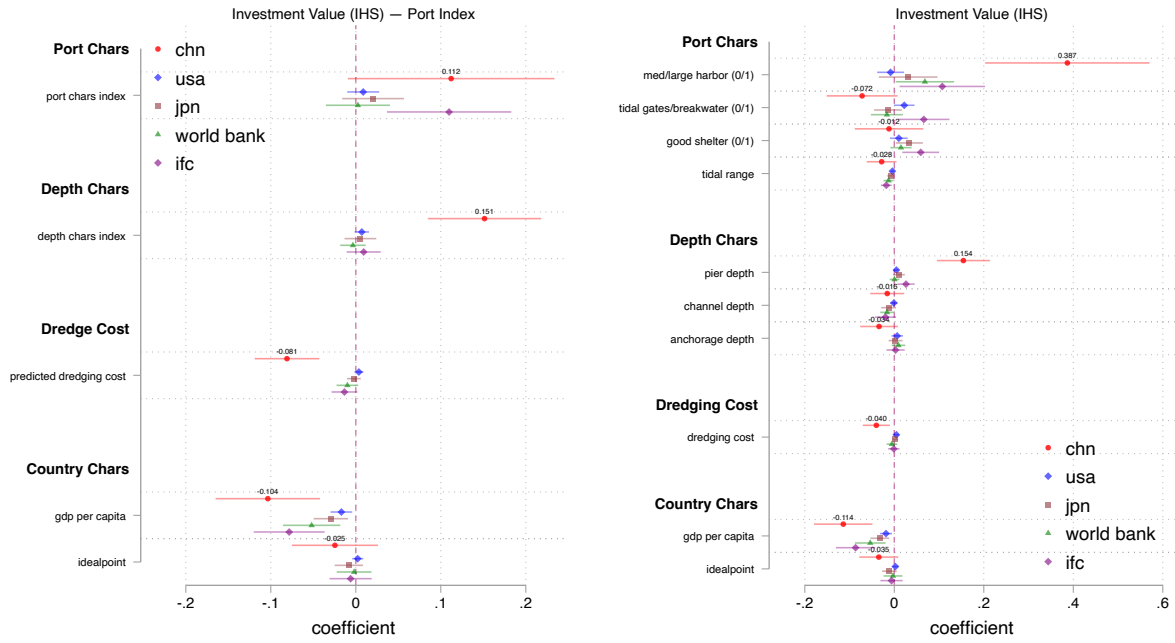
Notes: Figure plots the coefficients on the country correlates in (1) for China, U.S., Japan, World Bank, and IFC. All variables are standardized; if a covariate is missing, the standardized value is set to zero and a missing-value indicator is included. Total number of observations in each regression is 209. The number of treated countries are: CHN 89; USA 10; JPN 17; World Bank 34; IFC 38. 90/10 errors bars.

Figure C.4: Trade, Port Characteristics and Port Indexes



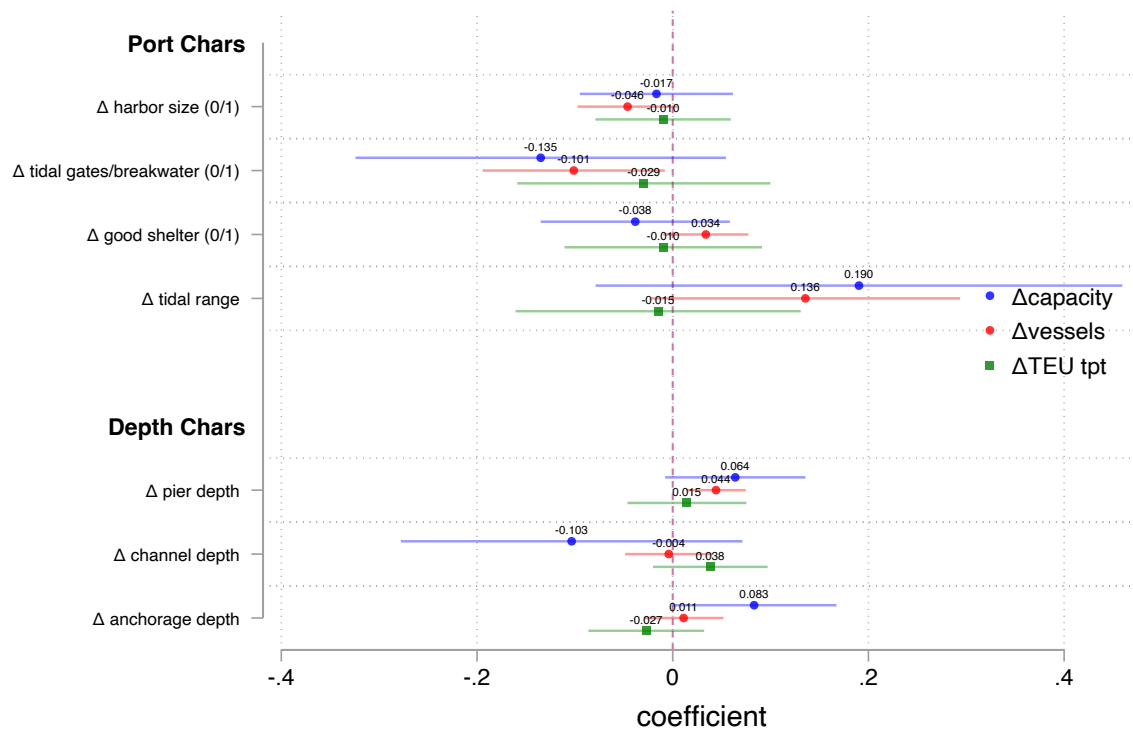
Notes: The left panel plots coefficients from three regressions of port-level trade measures—number of vessels departing the port (N=823), outbound shipping capacity (N=823), and total TEU throughput (N=397)—on initial port-level characteristics and two initial country covariates (GDPPC and ideal point). Continuous variables are standardized. The right panel re-runs these three regressions, but now averages the four natural port characteristics into a port characteristics index and averages the three depth characteristics into a depth index. Standard errors in all regressions clustered by country, 90/10 bars shown.

Figure C.5: Initial Port Characteristics and Investments, All Sources



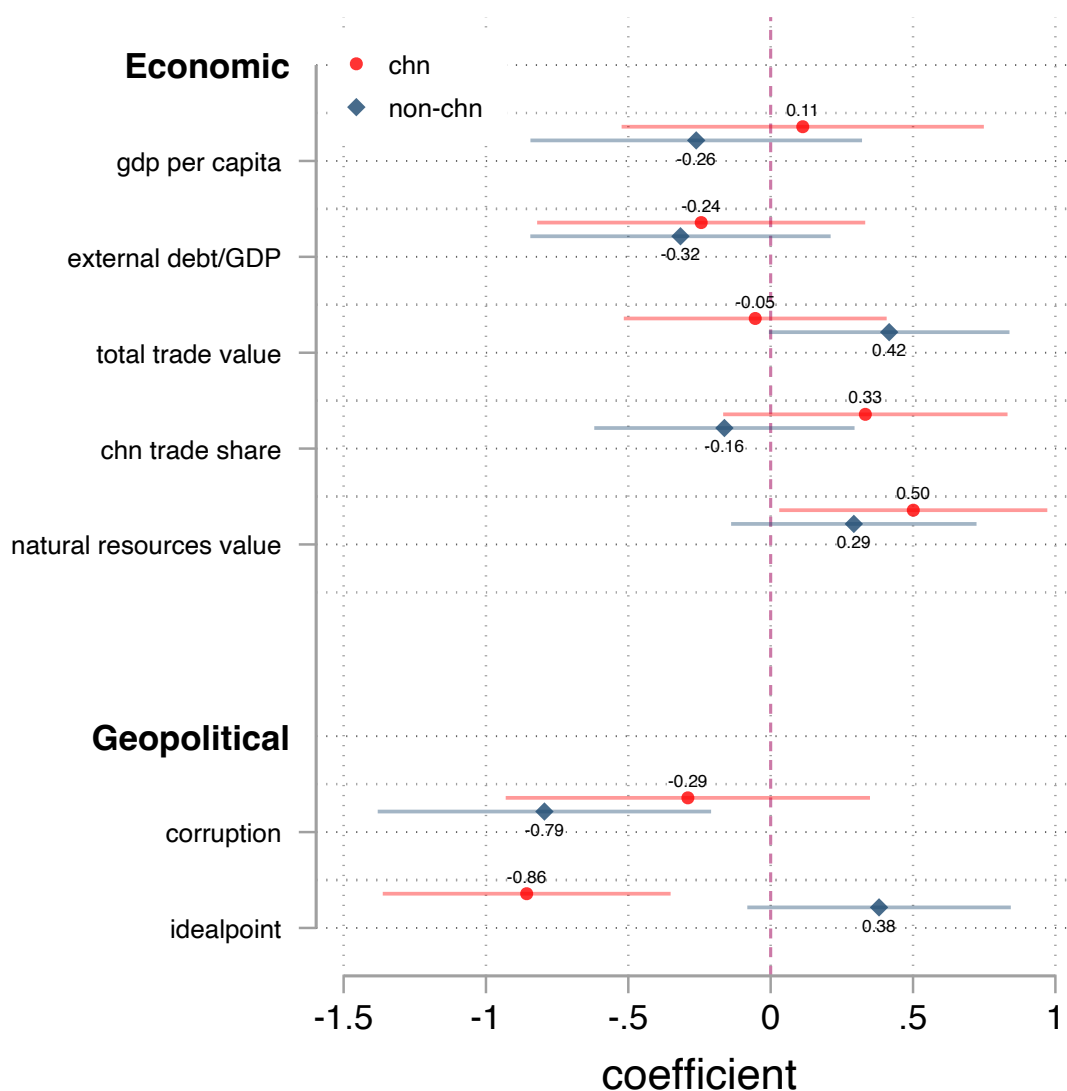
Notes: Figure plots the coefficients on the port characteristics in (2) for China and non-China's investments. "Port chars index" is the average of harbor size (0/1), breakwater/tidal gates (0/1), shelter (0/1), and standardized tidal range. "Depth chars index" is the average of standardized depths of cargo pier, channel, and anchorage. Total number of observations in each regression is 3,531. The number of treated ports by source are: CHN 152; USA 7; JPN 16; WB 31; IFC 53. Standard errors clustered by country; 90/10 error bars shown.

Figure C.6: Changes in Trade Regressed on Changes in Port Characteristics



Notes: Figure reports regression coefficients of changes port-level trade on changes in all seven characteristics (a multivariate regression). Standard errors in all regressions clustered by country.

Figure C.7: Investments and Country Correlates, China vs Non-China; Non-OECD Countries



Notes: Figure plots the coefficients on the country correlates in (1) for China and non-China's investments amongst non-OECD countries. All variables are standardized; if a covariate is missing, the standardized value is set to zero and a missing-value indicator is included. Total number of observations in each regression is 175. The number of treated non-OECD countries are: CHN 69; non-CHN 62. 90/10 errors bars.